

Mathematics

Year 1

Student Manual



Mathematics

Year 1 · Student Book

Count it. Check it. Then say how you know.

INSIDE THIS BOOK

- Sixteen units, six projects, a counting fair to finish
- Every page meant to be written in
- Count it, check it, say how you know
- A second method whenever there is one
- Original watercolour illustration throughout

IMPRINT

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 For teachers

This book is meant to be written in. Pupils will draw rings, tick boxes, join pairs with a line, colour squares and write numerals on these pages, and that is the point.

There are **no printed answers** anywhere in this book except inside the **Watch me do one** panels, where one worked example is solved in full before pupils try their own. Let pupils say what they think out loud first, then write.

Every topic ends by asking **How do you know?** Accept an answer in a pupil's own words, including pointing, counting again out loud or moving the objects. Explaining the method matters more here than the number.

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WELCOME

Maths is already happening

Under the roots of an old oak tree there is a room called the Burrow Schoolroom. There are acorns in baskets, pebbles in a jar, wooden blocks on the rug and a small pair of brass scales on the bench.

Nobody in the burrow says the word **mathematics**. They just do it all morning long.

♣ The one big idea in this book

Count it. Check it. Then say how you know.

Bramble counts the acorns into a bag. Then he counts them again, to be sure. If you asked him how he knew there were six, he would not say **because I am older than you**. He would say **because I counted them, look**, and count them again in front of you.

That is all this book asks you to do. Count it. Check it. Say how you know.

★ **Pip's find**

A tiny wren called Pip lives on the window ledge of the burrow. Pip hops along the number line all through this book, and every single hop is worth one. When you see Pip, count the hops.



Otto counts the acorns one at a time, pointing at each one. One point, one number, one acorn.

Meet the six



Otto, Tessa, Hattie, Milo, Bramble and Pip. You will meet all six of them in every unit.

Otto the owl

Otto points at each thing as he counts it. One point, one number, one thing.

Hattie the hedgehog

Hattie guesses first, out loud, then counts to check. Sometimes she is close. Sometimes she is not. Both are useful.

WHO WORKS IN THE BURROW • CONTINUED

Tessa the tortoise

Tessa draws it, or lines it up, so she can be sure. She invents a way to write things down.

Milo the mouse

Milo asks **how do you know?** about everything, including his own answers.

Bramble and Pip

Bramble the badger keeps the store, the brass scales and the till. He asks the real questions, because he has to weigh things and give change. **Pip** is the burrow wren. Pip hops along the number line, and every hop is one.

What is inside?



Sixteen units, six places in the burrow, and a counting fair of your own at the end.

Unit 1	Numbers to 10 Counting, saying, reading and writing numbers. Odd and even	1
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The second half of the year

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Fair	Counting fair Run your own stall: the final project	221
Words	Burrow words Every new word, with its own picture	227

Along the way there are six **Along the way** projects, one after Unit 4, Unit 5, Unit 9, Unit 10, Unit 11 and Unit 16.

Signs to look for

The same signs come back on every page. Once you know them, you always know what to do next.

Open the door

What you already know. Do this first, before anything new.

✓ By the end you will

- know what this unit is for.

Do you remember?

One question from before, to warm up your thinking.

☀ In the burrow today

A little story.

Something is happening in the burrow. Listen, then help.

WAKE UP
YOUR
COUNTING

A quick game with your voice, your hands and your feet.

♣ Let's find out

The teaching part. Read it out loud together.

 Watch me do one

How many acorns?

- 1** One step at a time, in order.
-

Answer: the answer is printed here, and **only** here. This is the one place in the book where you can see how it is done before you try.

- 1 Your turn.** A numbered job for you. A dashed box or line is always yours to write, draw or tick in.

No answers are printed for these. Say what you think first, then write it.

How do you know?

Every topic asks this. Say it out loud, or point, or count again. Explaining how you know IS the maths.

Six more signs



★ Pip's find

Something true and surprising, left on a torn ticket by the wren.

→ Try a bigger one

The same idea, with bigger numbers. Have a go.

⚖ Burrow challenge

A harder job. Take your time, and work with a partner if you like.

✂ The material you need

The things to fetch before you start: counters, ribbon, a jug.

♂ Find out together

Work with a partner or in a group. Talk about it, try it, then tell the class what you found. There is often more than one good way.

▣ BURROW WORDS

The new words of the unit, each one with its own picture. Every word in this book has a picture of exactly what it means.

At the end of every unit

▣ Check what you can do

- 1** New questions about the whole unit. These are never the same as the ones you have already done.

☐ **Now I can** is a list of what you have learned. Tick each one you can do.

	confident	almost there	needs more work
How did it go asks you to be honest. Tick one box for each thing.	☐	☐	☐

A map of the burrow



Every unit in this book happens in one of six places. The colour at the bottom of the page tells you where you are.



The Burrow Schoolroom and Fern Meadow, with the six places you will work in this year.

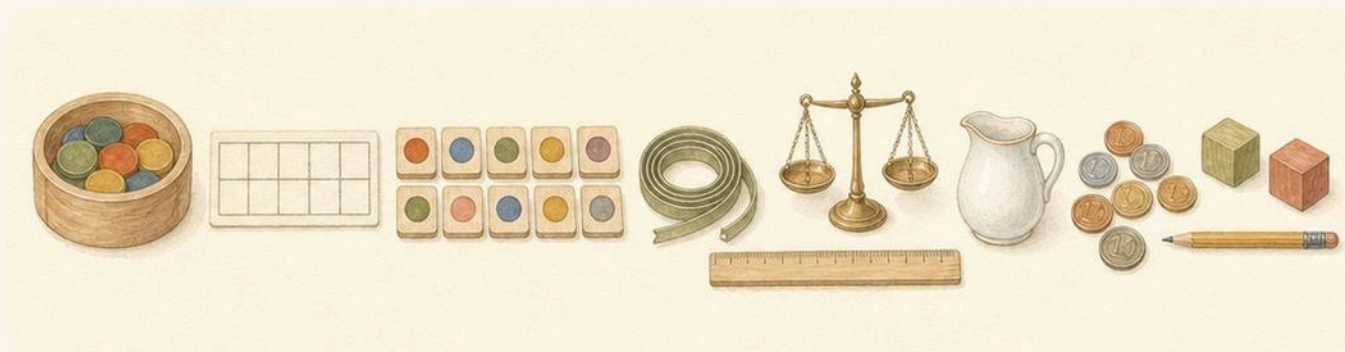
 The Counting Table	Counting, adding, taking away, and numbers to 20
 The Woven Rug	Solid shapes, flat shapes, position and patterns
 The Halving Board	Halves of shapes and halves of sets
 The Measuring Bench	How long, how tall, how heavy, how full
 The Sorting Baskets	Sets, rings and charts that tell you something
 The Cuckoo Clock	Days, o'clock, half past and the months of the year

Before you start

✂ The material you need

Keep these together in a box or a tray. You will reach for them all year.

- a tub of round counters, and a handful of small pebbles or acorns
- a ten-frame card and some number tiles from 0 to 20
- a length of ribbon, a short ruler and a tape measure
- a small pair of balance scales, a jug and a cup
- a few euro coins, real or paper
- wooden blocks: a cube, a cuboid, a cylinder, a cone and a ball
- a pencil, coloured pencils and a rubber



Everything you need for a whole year of maths, laid out on the table.

♣ **How to work in this book**

1. **Say it before you write it.** Tell a partner what you think first.
2. **Count out loud** and point at each thing as you say its number.
3. **Check it.** Count a second time, a different way if you can.
4. **Say how you know.** That part is the maths.

★ **Pip's find**

A dashed line or a dashed box always means the space is yours. Write, draw, tick or colour in it. This book is meant to be written in.

UNIT 1

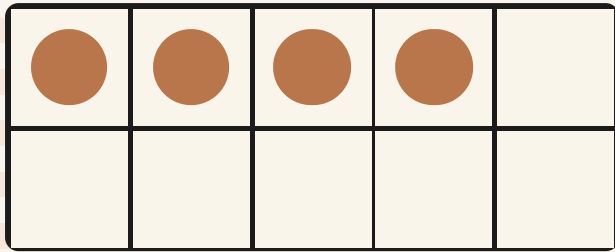
Numbers to 10

Counting acorns at the table, saying the number, writing it down, and finding out which numbers come in pairs.

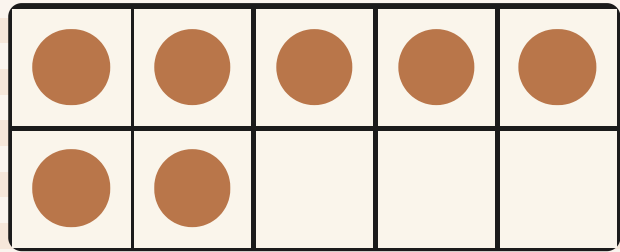


Open the door

Before you begin, show what you know already.



How many?



How many?

Say each number out loud. Then write it in the box.

☀ In the burrow today

It is the first morning of the year in the burrow.

Otto has tipped a basket of acorns onto the table. Nobody knows how many there are. Hattie says **about ten**. Milo says **how do you know?**

So they count them. That is how this book starts, and it is how every unit starts: count it, check it, then say how you know.

• CONTINUED

✓ By the end you will

- count a set of things and say how many there are
- say, read and write every number from 0 to 10
- say which of two sets has more, and which has fewer
- match a number word to its numeral
- sort numbers to 10 into odd and even



One point, one number, one acorn. Otto never counts the same acorn twice.

Counting sets of objects

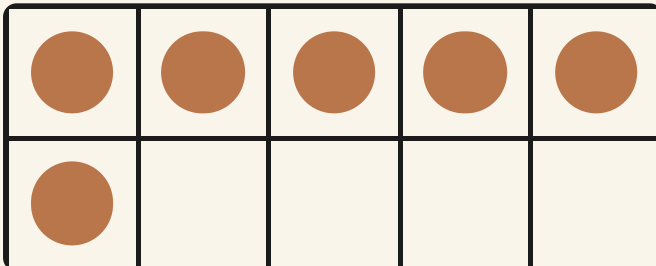
WAKE UP
YOUR
COUNTING

Clap and count with me. Clap once for every number: one, two, three, four, five. Now stamp your feet and count back down to one.

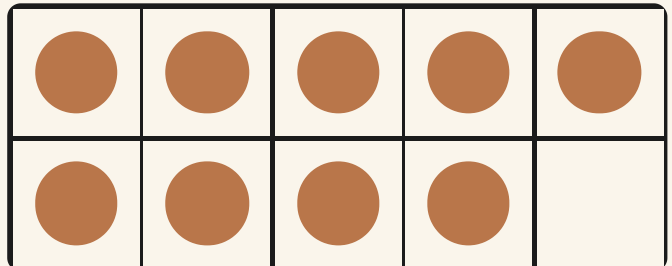
✦ Counting is touching, not guessing

When you count a set, you say one number for one thing. Point at it as you say it. Then move it away, so you cannot count it twice.

The last number you say is **how many there are**.



six acorns



nine acorns

 Watch me do one 1

How many acorns are on the table?

- 1** Point at the first acorn and say **one**.
 - 2** Keep going along the row: two, three, four, five, six.
 - 3** The last number you said was six, so there are six acorns.
-

Answer: there are **6** acorns

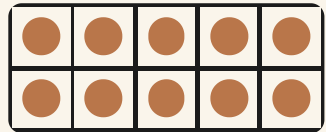
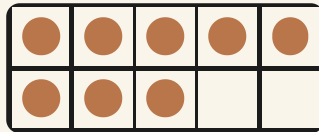
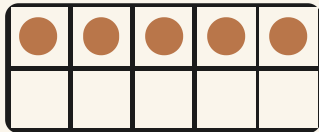
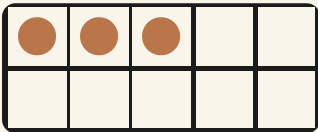
How do you know?

Count the six acorns again, but start from the other end. Do you still get six? Tell your partner why that happens.

Count and write



1 Count the counters in each frame. Write how many.



2 Draw counters in this frame to show seven.

TOPIC 1.1 · YOUR TURN · CONTINUED

3 Count these shells. Ring the number that matches.

Point at each shell as you count it.

6

7

8

4 Count out loud from one to ten. Then count back to one.

The material you need: nothing at all, just your voice

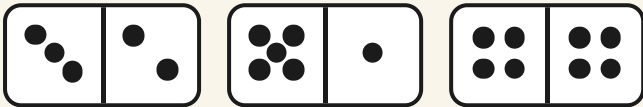
★ **Pip's find**

Your fingers are the oldest counting machine there is. Ten fingers is why we count in tens all over the world.

How many are there?



5 Count the dots on each die. Write the number.



6 Count the walnuts. Write how many.



Count carefully. Move each one as you count it.

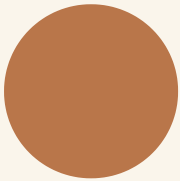
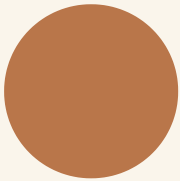
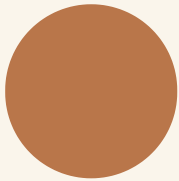
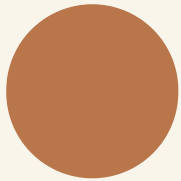
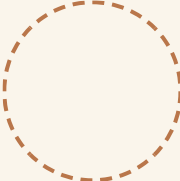
TOPIC 1.1 · YOUR TURN · CONTINUED

7 Draw five acorns in the box.

--	--	--	--	--	--	--	--	--	--

8 Count the counters. Then draw one more.

The material you need: a pencil and a rubber

9 Count backwards from ten to zero out loud.



count

10 Fill this frame to show ten counters.

11 Count the shells in the ring. Write the number.



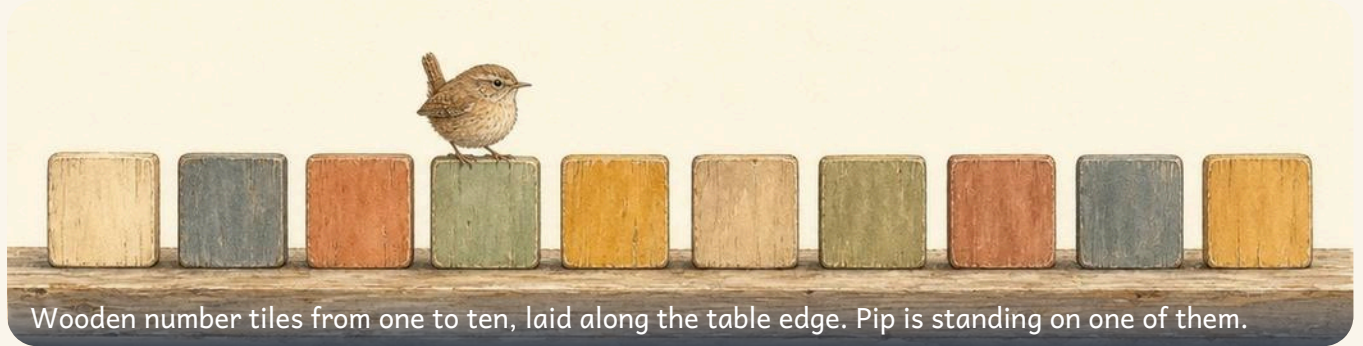
How many shells?

12 Count your own fingers. Say the last number out loud.

→ **Try a bigger one**

Count a set of twelve pebbles. Then count them again in pairs. Do you still get twelve?

Say, read and write numbers to 10



Wooden number tiles from one to ten, laid along the table edge. Pip is standing on one of them.

✿ Every number has a shape you can draw

A number has three parts: the **set** you can count, the **word** you say, and the **numeral** you write.

Trace the numeral with your finger first. Start where the dot is.

3 3 3 3

8 8 8 8

 Watch me do one 1

Write the numeral for this set.

- 1 Count the acorns: one, two, three, four, five.
 - 2 The word you say is **five**.
 - 3 The numeral you write for five looks like this.
-

Answer: the numeral is **5**

How do you know?

Point at a numeral on the tiles. Say its name. How would you show that many on your fingers?

Write the numbers



numeral

1 Trace each numeral. Then write it once on your own.

1 1 1 1

2 2 2 2

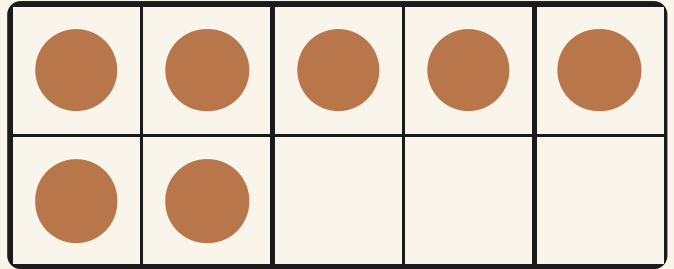
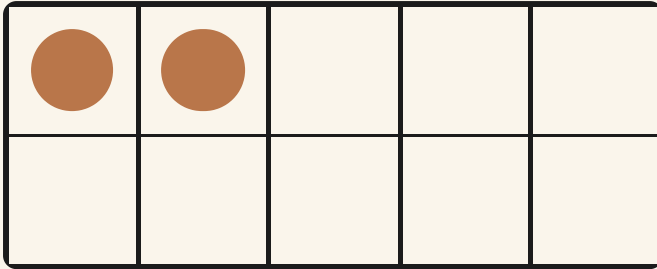
4 4 4 4

2 Write the missing numbers on the track.

0	1	2		4	5		7	8		10
---	---	---	--	---	---	--	---	---	--	----

3 Count the counters. Write the numeral in the box.

TOPIC 1.2 • YOUR TURN • CONTINUED



4 Trace the numeral for six. Then trace nine.

6 6 6 6

9 9 9 9

5 Write every number from zero to ten in order.

TOPIC 1.2 · YOUR TURN · CONTINUED



numeral

6

Write the numeral that comes after four.

A large rectangular box with a dashed brown border, intended for the student to write the answer.

Numbers and sets



numeral

7 Draw a ring around the numeral eight.

3

8

5

8 Write the numeral that comes before seven.

9 Draw counters to match the numeral ten.

TOPIC 1.2 • YOUR TURN • CONTINUED

10 Write the missing numbers on this track.

0		2	3	4		6	7		9	10
---	--	---	---	---	--	---	---	--	---	----

11 Trace the numeral zero. Then trace ten.

0 0 0 0



numeral

12 Say each number on the track out loud, then write it.

--	--	--	--

★ **Pip's find**

The numeral 0 was the last one to be invented. For a long time people had no way at all to write **none**.

Comparing numbers



WAKE UP
YOUR
COUNTING

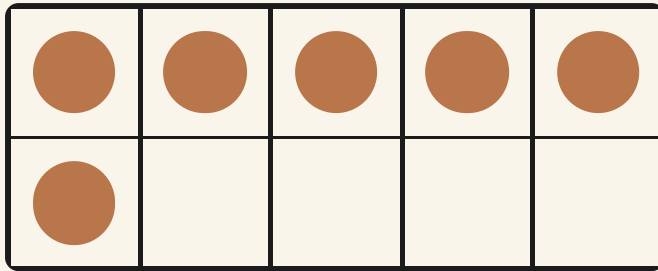
Show me four fingers. Now show me six. Which hand has more? Say it out loud.

♣ More, fewer, or the same

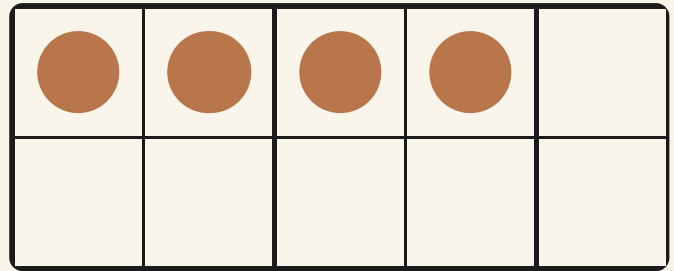
To compare two sets, line them up in pairs. The set that runs out first has **fewer**. The set left over has **more**.

If nothing is left over, the sets are the **same**.

TOPIC 1.3 • CONTINUED



six



four

Watch me do one 1

Which is more, six or four?

- 1 Put six counters in one frame and four in the other.
- 2 Pair them up: one with one, two with two, and so on.
- 3 Six has two counters left over, so six is more.

Answer: 6 is more than 4

How do you know?

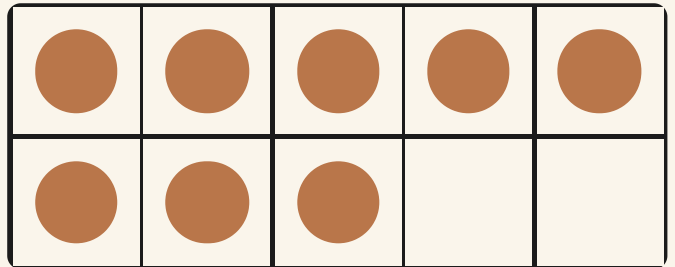
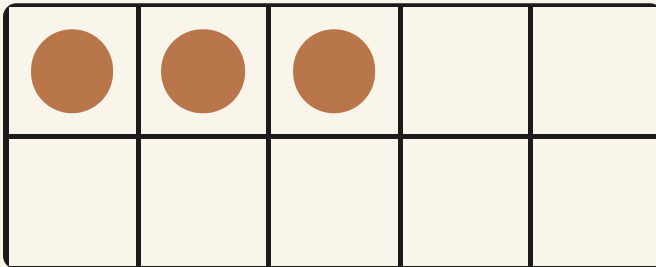
Hattie says nine is more than ten, because nine sounds longer. Is she right? How would you show her?

Which is more?



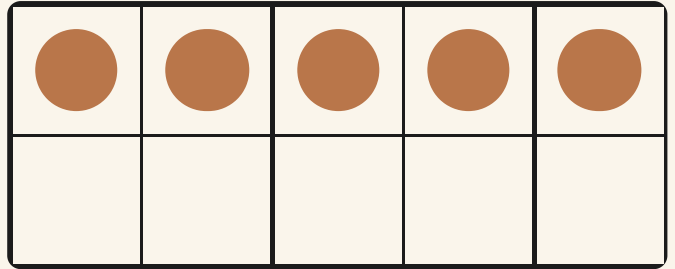
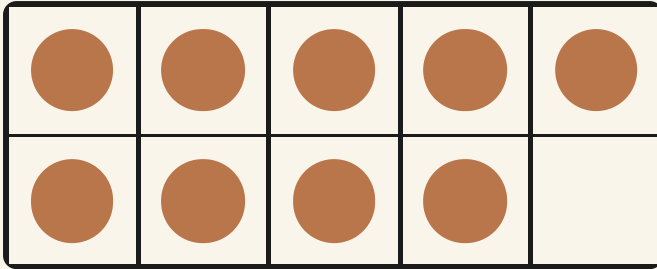
more

1 Ring the frame that shows more.



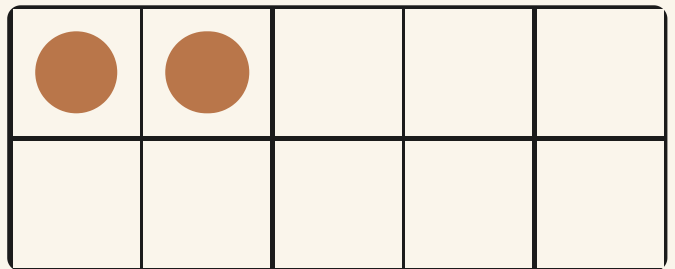
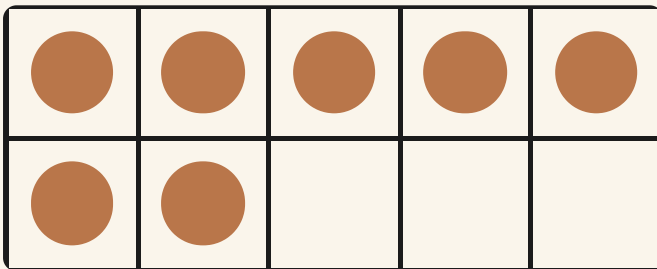
2 Ring the frame that shows fewer.

TOPIC 1.3 • YOUR TURN • CONTINUED



3 Write the missing number to make the sets the same.

4 Count both sets. Write which number is bigger.



TOPIC 1.3 · YOUR TURN · CONTINUED

5 Order these numbers from smallest to largest.

5

2

9

smallest to largest

--	--	--



more

6 Order these numbers from largest to smallest.

4

10

7

largest to smallest

--	--	--

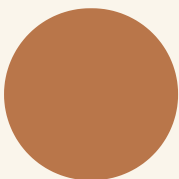
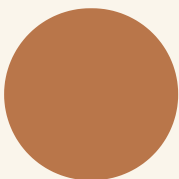
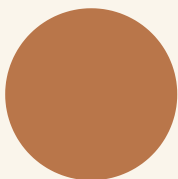
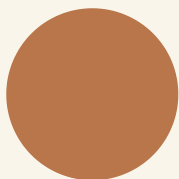
Same, more, fewer



more

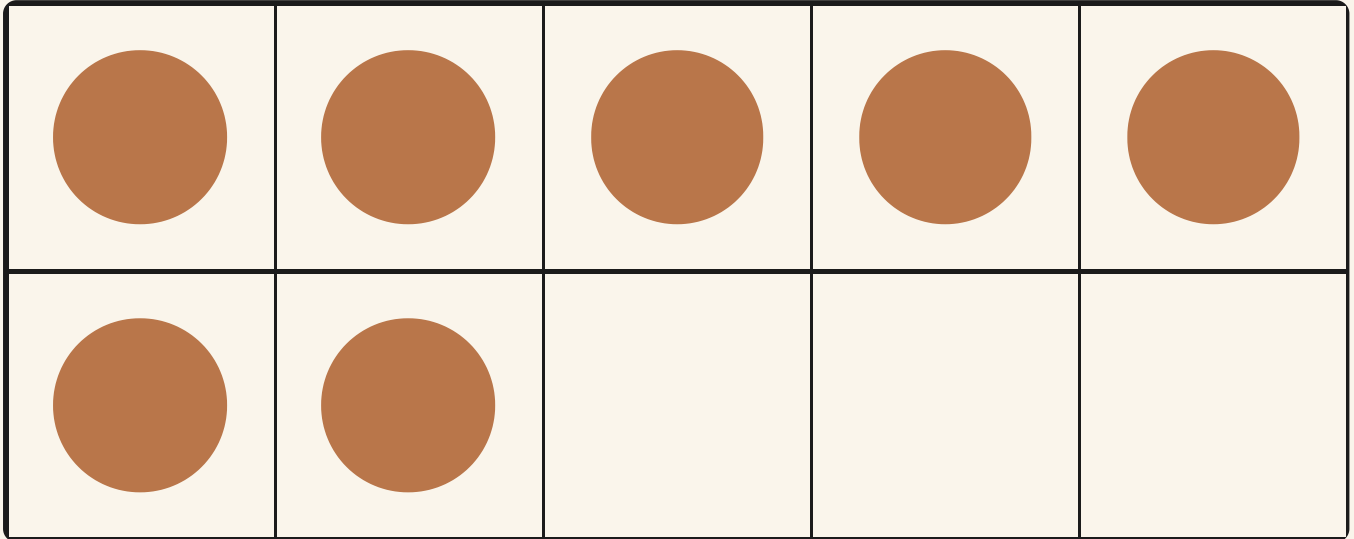
7

Draw a set with more counters than this one.

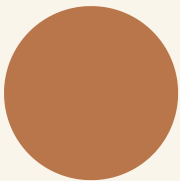
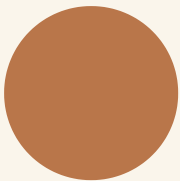
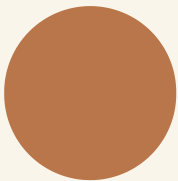
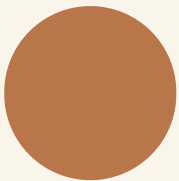
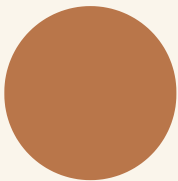
TOPIC 1.3 · YOUR TURN · CONTINUED

8 Draw a set with fewer counters than this one.



9 Draw a set that is the same as this one.

TOPIC 1.3 · YOUR TURN · CONTINUED

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10 Count each set. Write both numbers in the boxes.

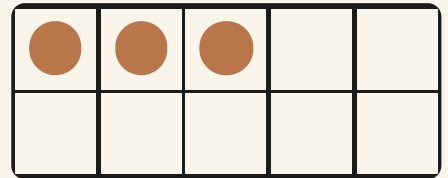
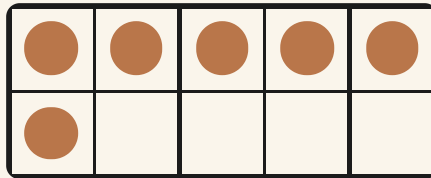
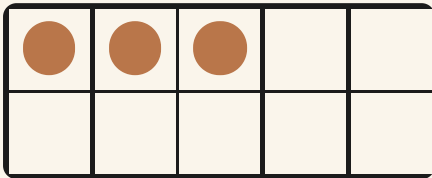
				
				

TOPIC 1.3 · YOUR TURN · CONTINUED

--	--

11 Ring the two sets that are the same.



more

12 Order these four numbers from smallest to largest.

6

1

10

4

smallest to largest

--	--	--	--

→ Try a bigger one

Take two handfuls of pebbles. Guess which handful has more, then pair them up to check.

Number words



Blank cream cards pegged to a line of string. Hattie reaches up towards one.

✚ The word and the numeral mean the same thing

Tessa writes labels for the store. She can write **4** or she can write **four**. Both say the same amount.

A numeral is quick to write. A word is easy to read out loud.

one



5

three



1

five



3

 Watch me do one 1

Match the word two to its numeral.

- 1** Read the word out loud: two.
 - 2** Show two on your fingers.
 - 3** Find the numeral that means the same amount.
-

Answer: two matches **2**

How do you know?

Bramble wrote **six** on a label and **6** on another. Are they different amounts? Say how you know.

Words and numerals



number word

1 Join each number word to its numeral.

two



9

seven



2

nine



7

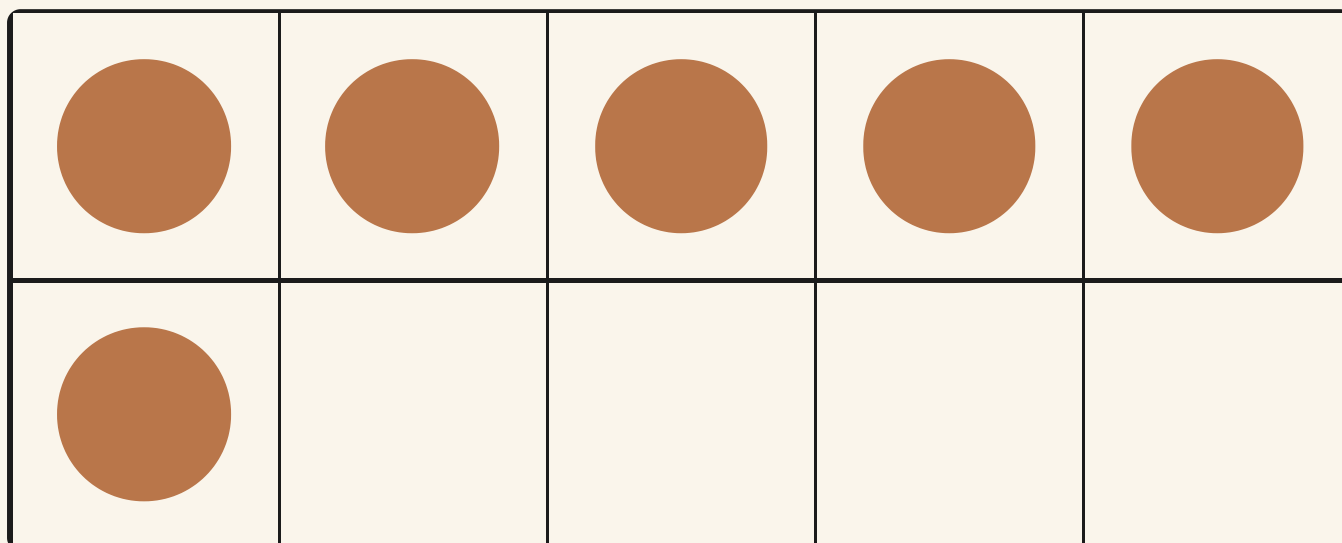
2 Write 3 and 8 in words.

3

8

TOPIC 1.4 · YOUR TURN · CONTINUED

3 Write the number word for this set in figures.



4 Read the word ten out loud. Draw ten counters.

TOPIC 1.4 · YOUR TURN · CONTINUED

5 Join each word to its set.

four



one



number word

6 Write the word for the numeral 5.

Reading the labels



7 Ring the word that says the same as 8.

six

eight

ten

8 Write the missing number words in order.

one

two

four

five

9 Draw a set to match the word seven.

TOPIC 1.4 · YOUR TURN · CONTINUED

10 Join each word to its numeral.

zero



10

six



0

ten



6

11 Write the word nine in your own hand.



number word

12 Ring the numeral that says the same as three.

2

3

5

★ **Pip's find**

The words **one** to **ten** are some of the oldest words in English. They have barely changed in a thousand years.

Odd and even numbers



Walnuts arranged in neat pairs on a slate tray, with one single walnut left over at the end.

WAKE UP
YOUR
COUNTING

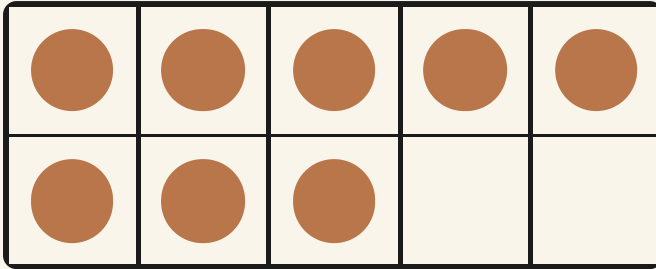
Stand with a partner. Hold up two hands between you.
Can you both make pairs, or is one hand left on its own?

♣ Pair them up and see what is left

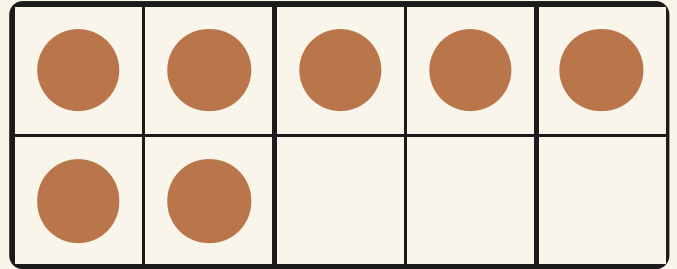
Put the things into **pairs**, two at a time. If every one has a partner, the number is **even**.

If one is left on its own, the number is **odd**.

TOPIC 1.5 • CONTINUED



eight pairs up



seven has one left

 Watch me do one 1

Is six odd or even?

- 1 Take six counters.
- 2 Push them together two at a time: that makes three pairs.
- 3 Nothing is left over, so six is even.

Answer: 6 is even

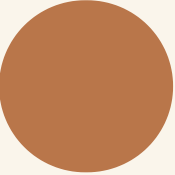
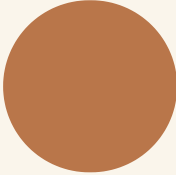
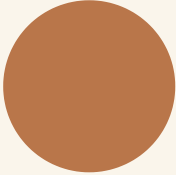
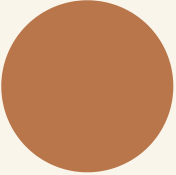
How do you know?

Milo says seven must be even because it is a big number. Pair seven counters up and tell him what you find.

Pairs and odd ones



1 Pair up the counters. Ring odd or even.

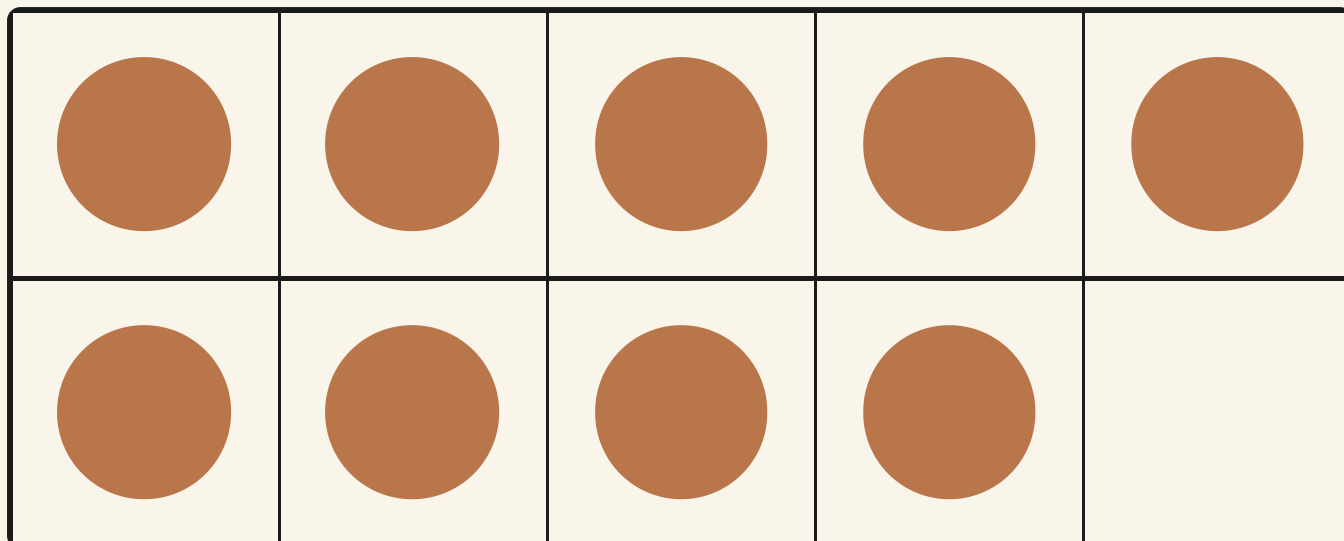
				

odd

even

2 Pair up the counters. Ring odd or even.

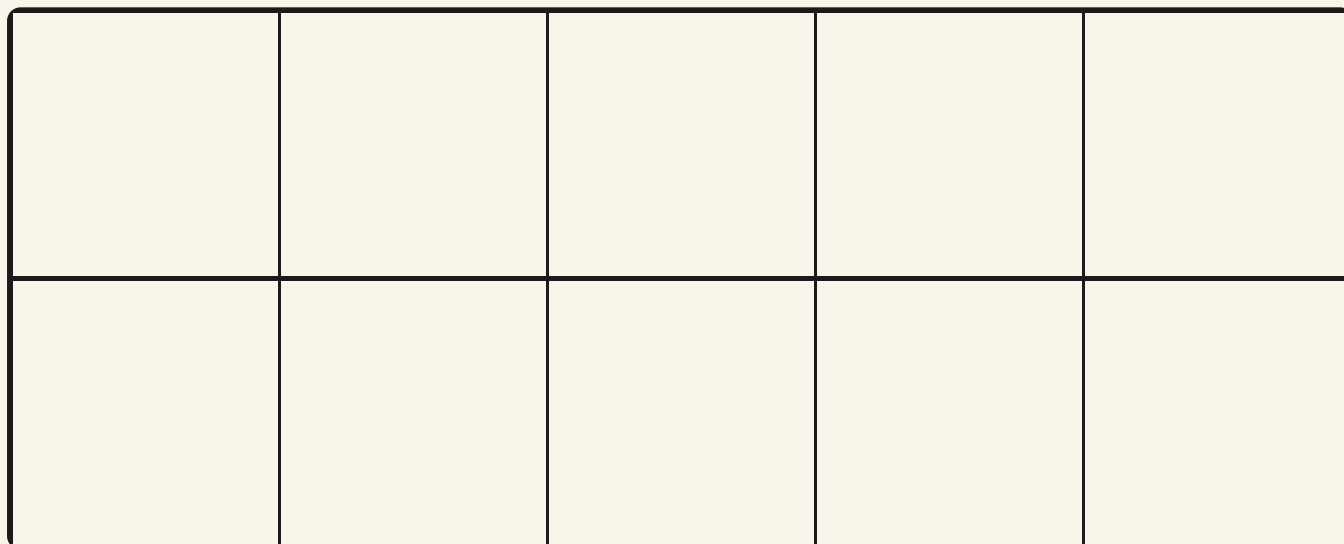
TOPIC 1.5 • YOUR TURN • CONTINUED



odd

even

3 Draw an even set of counters.



4 Draw an odd set of counters.

5 Ring all the even numbers.

2

3

6

9

10



even

6 Ring all the odd numbers.

1

4

5

7

8

Odd or even?



7 Count the walnuts. Write odd or even.

The material you need: ten walnuts or ten counters



Pair them up before you decide.

8 Write the even numbers from two to ten.

TOPIC 1.5 • YOUR TURN • CONTINUED

--	--	--	--	--

9 Write the odd numbers from one to nine.

--	--	--	--	--

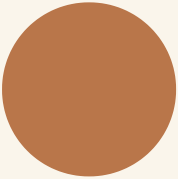
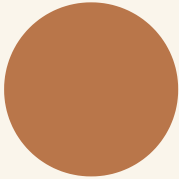
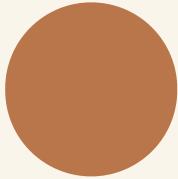
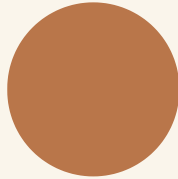
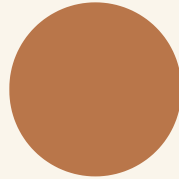
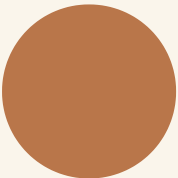
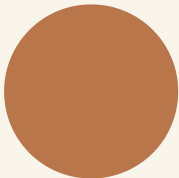
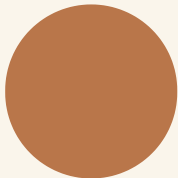
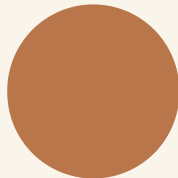
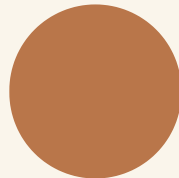
10 Ring the number that is odd.

4

7

8

11 Pair up ten counters. Write odd or even.



even

12 Ring the number that is even.

3

5

6

⚖️ Burrow challenge

Every child in your class stands in pairs. If one child is left on their own, is the number of children odd or even?

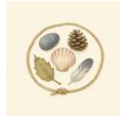
The words of this unit

▣ BURROW WORDS



count

one number for one thing



set

things grouped together



more

the bigger amount



fewer

the smaller amount



same

no difference at all



pair

two that go together



odd

one is left over



even

everything pairs up

▣ Check what you can do

- 1 Count the acorns in the picture and write how many.
- 2 Write the numeral that comes after eight.
- 3 Ring the set that has fewer counters.
- 4 Join the word five to its numeral.
- 5 Pair up nine counters. Write odd or even.

Now I can



- ☐ count a set of things and say how many there are
- ☐ say, read and write every number from zero to ten
- ☐ say which set has more and which has fewer
- ☐ match a number word to its numeral
- ☐ sort numbers to ten into odd and even

	confident	almost there	needs more work
counting a set carefully	☐	☐	☐
writing the numerals from zero to ten	☐	☐	☐
comparing two sets	☐	☐	☐
matching words and numerals	☐	☐	☐
finding odd and even numbers	☐	☐	☐

★ Pip's find

You have counted, compared and paired every number up to ten. In Unit 9 the same tricks will carry you all the way to twenty.

UNIT 2

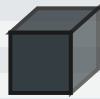
Solid shapes and flat shapes

Holding solid shapes, feeling their flat faces and their curved ones, and finding the flat shapes hiding in the burrow rug.



Open the door

Before you begin, show what you know already.



Can it
stack?



Can it
roll?

Say yes or no for each one. Then find something in the room that stacks, and something that rolls.

☀ In the burrow today

Tessa has rolled out the old woven rug.

The rug is covered in shapes. Some of them have three sides. Some have four. One has no straight sides at all, and Pip keeps hopping round and round it.

On the shelf above the rug sit the wooden solids: boxes, tins, a ball and a cone. Otto holds one up. **This one you can hold**, he says. **The ones on the rug are flat. You cannot pick a rug shape up.**

• CONTINUED

✓ **By the end you will**

- name a cube, a cuboid, a sphere, a cylinder, a cone and a pyramid
- count the flat faces on a solid shape
- say which solids roll and which solids stack
- name a circle, a triangle, a square, a rectangle and a hexagon
- count the sides of a flat shape



Otto turns the cube over and over. Every time it lands, one flat face is on the table.

Solid shapes you can hold

WAKE UP
YOUR
SHAPE
EYES

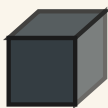
Make a shape with your body. Curl up small and round like a ball. Now stand up straight and tall like a tin. Now be flat like the rug.

✦ A solid shape takes up room

A **solid** shape is one you can pick up and hold. It has an inside. A flat shape drawn on the rug has no inside to hold.

The parts you touch are called **faces**. Some faces are **flat**, like the side of a box. Some are **curved**, like the skin of a ball.

Flat faces let a shape sit still and stack. Curved faces let a shape roll away.



cube



cuboid



sphere



cylinder



cone



pyramid

 Watch me do one 1

How many flat faces does a cube have?

- 1** Put the cube down. One flat face is on the table.
 - 2** Turn it over onto a new face. Count as you go: one, two.
 - 3** Keep turning until every face has touched the table.
 - 4** The top and the bottom, and the four faces round the side.
-

Answer: a cube has **6** flat faces

How do you know?

Otto says a ball has one curved face and no flat ones. Put a ball on the table and let go. What happens, and why?

Hold it, name it, count it



face

1 Name each solid shape below.

Use a word from the word box.



cube

cuboid

sphere

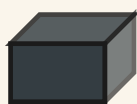
cylinder

cone

pyramid

2 Count the flat faces on this cuboid. Write how many.

TOPIC 2.1 · YOUR TURN · CONTINUED



3 Ring the solid shape that can roll.

cube

sphere

pyramid

4 Draw a ring around every solid with a curved face.



5 Match each solid to its name.



cube



cone



sphere



TOPIC 2.1 · YOUR TURN · CONTINUED



face

6 Sort these solids into rolls and does not roll.

The material you need: a ball, a tin, a wooden box and a cone

rolls

does not roll

7 Find a tin in the room. Say which solid it is like.

Faces, edges and towers

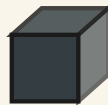


face

- 8** Hold a cube. Count its edges with your finger.

An edge is the line where two faces meet.

The material you need: one wooden cube

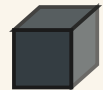


- 9** Build a tower with three cubes. Draw your tower.

The material you need: three wooden cubes

TOPIC 2.1 · YOUR TURN · CONTINUED

10 Tick the solid with one curved face and two flat faces.



11 Feel a solid in the bag. Say its name before you look.

The material you need: a cloth bag with four solids inside

12 Order these three solids, fewest faces to most faces.



fewest faces to most faces



face

13 Draw one solid you can see in Bramble's store.

A large dashed rectangular box for drawing.

★ Pip's find

A cone has one flat face and one curved face. That is why an acorn cup will spin on the table but never stack.

🦉 Burrow challenge

Otto hides a solid behind his wing. Ask him three questions he can answer with yes or no, then say which solid it is.



TOPIC 2.2

Flat shapes on the rug

WAKE UP
YOUR
DRAWING
HAND

Draw shapes in the air with one finger. A big circle. Now a triangle, and stop at each corner. Now a square: along, down, along, up.

Do you remember?

A cube has flat faces. What shape is one face of a cube?

✿ A flat shape has sides and corners

A **flat** shape lies down on the paper. It has no inside to hold. We name it by counting its **sides**.

A side is a straight line along the edge of the shape. A **corner** is where two sides meet.

A circle is the odd one out. It has no straight sides at all, so it has no corners either. Its edge just keeps going round.

TOPIC 2.2 · CONTINUED



circle



triangle



square



rectangle



pentagon



hexagon

Watch me do one 2

How many sides does this hexagon have?

- 1 Put your finger on one corner of the hexagon.
- 2 Run it along the straight side to the next corner, and say one.
- 3 Keep going the same way round: two, three, four, five, six.
- 4 You are back where you began, so stop counting.

Answer: a hexagon has **6** sides

How do you know?

Milo says a square and a rectangle are the same shape. Count the sides of each one. Then tell him what is the same, and what is not.

Count the sides



- 1** Name each flat shape.
Use a word from the word box.



circle

triangle

square

rectangle

pentagon

hexagon

- 2** Count the sides of this pentagon. Write how many.



TOPIC 2.2 · YOUR TURN · CONTINUED

3 Ring every shape with four sides.



4 Draw a triangle and a square in the box.

5 Match each flat shape to its number of sides.



4 sides



6 sides



3 sides

TOPIC 2.2 · YOUR TURN · CONTINUED



6 Sort these shapes into three sides and four sides.

three sides

four sides

7 Find the flat shape with no straight sides. Name it.



Shapes in the rug



- 8** Draw round a wooden block. Name the flat shape you made.

The material you need: a wooden block and a pencil

- 9** Count the sides of this octagon. Write the number.



10 Colour every hexagon in the rug pattern.

the woven rug

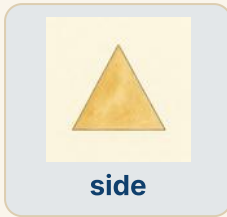


11 Draw a shape with five sides. Name it.

12 Order these shapes, fewest sides to most sides.



fewest sides to most sides



13 Tick the shape that is a rectangle, not a square.



→ Try a bigger one

Tessa turns a square round so it stands on one corner. Is it still a square? Count its sides and say why.

🔗 Find out together

Work with a partner. Walk round the room and find one circle, one triangle, one square and one rectangle. Draw each one, and write where you found it.

What you can do now



☐ Check what you can do

- 1** Hold a solid shape. Tell a friend two things about it.
- 2** Say which solid will not stack, and say why not.
- 3** Point to a curved face on something in the room.
- 4** Say what is the same about a square and a rectangle.
- 5** Say what a circle does not have.

- ☐ name a cube, a cuboid, a sphere, a cylinder, a cone and a pyramid
- ☐ count the flat faces on a solid shape
- ☐ say which solids roll and which solids stack
- ☐ name a circle, a triangle, a square, a rectangle and a hexagon
- ☐ count the sides of a flat shape

END OF UNIT 2 · CONTINUED

	confident	almost there	needs more work
Naming solid shapes	☐	☐	☐
Counting faces on a solid	☐	☐	☐
Naming flat shapes	☐	☐	☐
Counting the sides of a flat shape	☐	☐	☐
Saying how I know	☐	☐	☐

▣ BURROW WORDS



face



edge



side



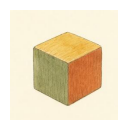
curved



straight



same



block



crate

UNIT 3

Halves of shapes

Folding, cutting and sharing, until the two parts match exactly and nobody in the burrow can say it was unfair.



Open the door

Before you begin, show what you know already.



One of these is cut fairly. Point to it. Then say out loud how you can tell.

☀ In the burrow today

Bramble has one flat bread and two hungry friends.

He puts his knife down the middle and cuts. Milo looks at the two pieces for a long time. **How do you know they are the same?** he asks.

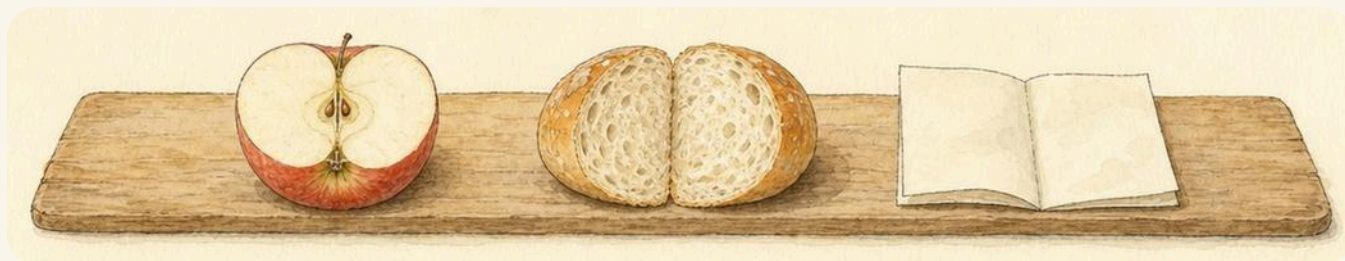
So Tessa lifts one piece and lays it straight on top of the other. The edges match all the way round. Nothing sticks out. Now everybody knows.

That is what a **half** means in this burrow: not just two pieces, but two pieces that match.

• CONTINUED

✓ **By the end you will**

- say what one half of a shape means
- fold or cut a shape into two equal halves
- tell two halves from two parts that are not equal
- find half of a small set of things
- say that two halves make one whole



Tessa lays one half on top of the other. The edges match, so the cut was fair.

Two halves that match

WAKE UP
YOUR
FOLDING
HANDS

Hold your hands out flat. Put them palm to palm. Do your two hands match? Now clap ten times, and count in twos as you go.

Do you remember?

You have counted six acorns before. If you share them fairly with a friend, how many will each of you hold?

♣ A half is one of two equal parts

Cut a shape into two parts. If the two parts are the same size, each one is a **half**. If one part is bigger, they are just parts, and neither one is a half.

The best way to check is to fold. Fold the shape over on itself. If the two parts land exactly on top of each other, you have two halves.

Put the two halves back together and you have the **whole** shape again, with nothing left over.



 Watch me do one 1

Is this circle cut into two halves?

- 1** Look at the line. It goes straight through the middle.
- 2** Fold the circle along the line, in your head or with paper.
- 3** The two parts land on top of each other with nothing over.
- 4** So the parts are equal, and each part is a half.

Answer: yes: there are **2** equal parts, so each part is one half

How do you know?

Hattie cuts a square near one edge, not down the middle. She says she has made two halves. Is she right? Say how you know.

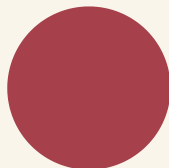
Fair or not fair



- 1 Ring every shape cut into two equal halves.
Look for a line straight through the middle.



- 2 Draw one line to cut this circle into two halves.



3 Colour one half of each shape.



4 Share ten pebbles between two friends. Draw what each gets.

The material you need: ten pebbles and two small trays



5 Fold a paper square to make two halves. Then cut it.

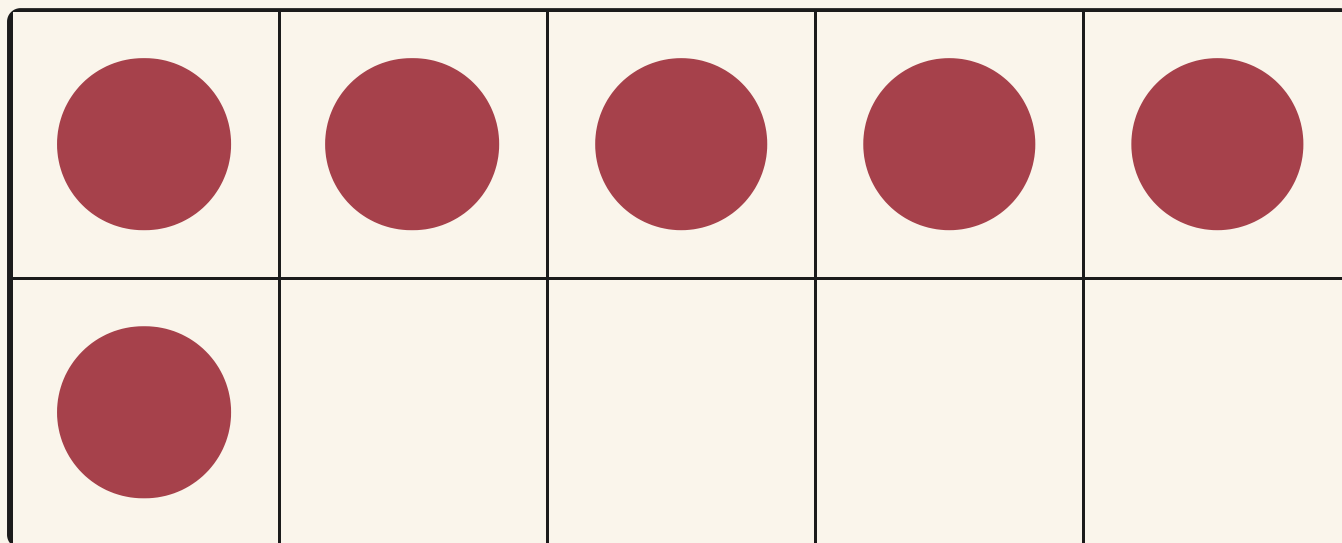
Check the two pieces match before you cut.

The material you need: a paper square and safe scissors



half

6 Find half of six counters. Write how many.



six counters

7 Say why these two parts are not halves.



Half of a set



Watch me do one 2

Bramble shares eight walnuts between two friends. How many each?

- 1 Put the walnuts in a line on the halving board.
- 2 Give one to Otto, then one to Pip, then one to Otto again.
- 3 Keep going until the line has gone. Nothing is left over.
- 4 Now count what Otto is holding.

Answer: each friend gets **4** walnuts



eight walnuts, shared fairly



half

TOPIC 3.1 · YOUR TURN · CONTINUED

8 Tick the shape that is not cut into two halves.



9 Match each set of walnuts to half of that set.

4



3

10



2

6



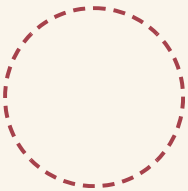
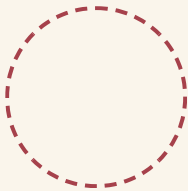
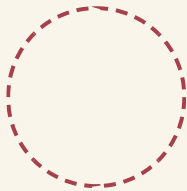
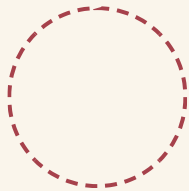
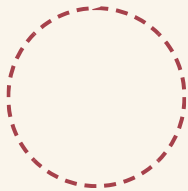
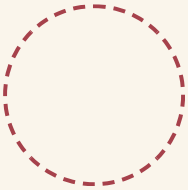
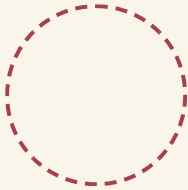
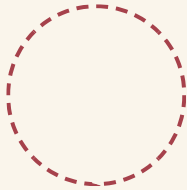
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10 Fold a ribbon in two. Check both parts are equal.

The material you need: a long ribbon

11 Colour half of these eight counters.

TOPIC 3.1 · YOUR TURN · CONTINUED

eight counters to colour

12 Find half of four walnuts, then half of ten walnuts.

The material you need: ten walnuts

--	--



half

13 Draw a shape. Draw one line to make two halves.

★ **Pip's find**

Fold a leaf down its middle vein and the two halves nearly match. That is how a leaf grows: the same on both sides of the line.

🐹 **Burrow challenge**

Tessa cuts a square from corner to corner instead of down the middle. Has she made two halves? Draw it, then say why.

♂ **Find out together**

Work with a partner. Take twelve counters and share them fairly. Then take seven and try again. Tell the class what happens with seven, and why.

What you can do now



☐ Check what you can do

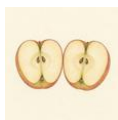
- 1** Show a friend one half of a piece of paper.
- 2** Say what makes two parts halves and not just parts.
- 3** Point to something in the room with two matching halves.
- 4** Say how many halves make one whole shape.
- 5** Say half of eight, then say half of two.

- ☐ say what one half of a shape means
- ☐ fold or cut a shape into two equal halves
- ☐ tell two halves from two parts that are not equal
- ☐ find half of a small set of things
- ☐ say that two halves make one whole

END OF UNIT 3 · CONTINUED

	confident	almost there	needs more work
Making two equal halves	☐	☐	☐
Spotting parts that are not halves	☐	☐	☐
Finding half of a set of things	☐	☐	☐
Putting two halves back into a whole	☐	☐	☐
Saying how I know	☐	☐	☐

▣ BURROW WORDS



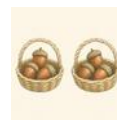
half



whole



equal



same



pair



walnut



ribbon

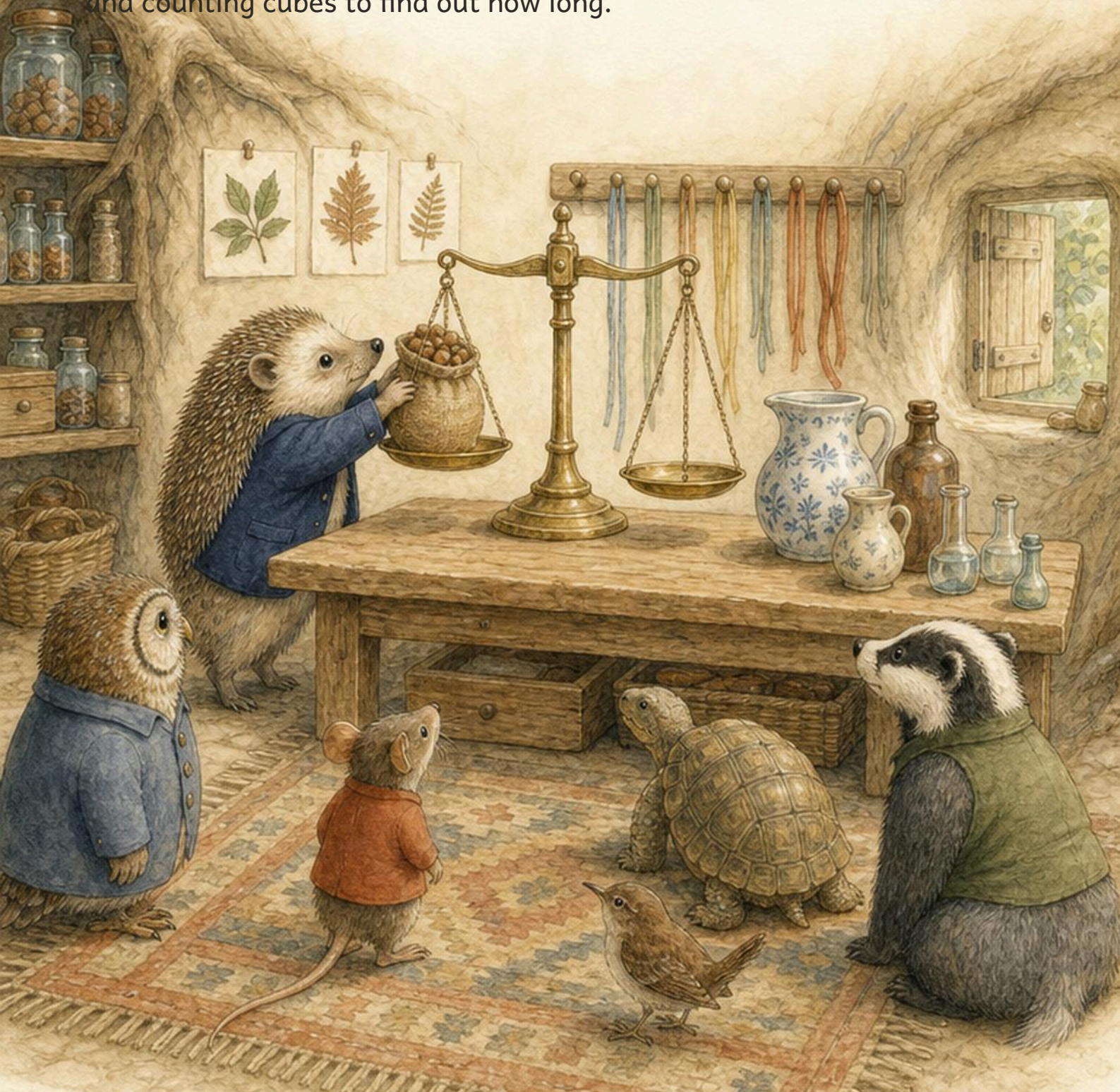


apple

UNIT 4

How long, how tall

Laying things side by side on the measuring bench, starting at the same line every time, and counting cubes to find out how long.



Open the door

Before you begin, show what you know already.

this ribbon

that ribbon

Point to the longer ribbon. Then find something on your table that is longer than your hand.

☀ In the burrow today

Bramble has cleared the long bench by the store.

Two ribbons are lying on it, and the friends cannot agree. Hattie holds them up in the air and guesses. Otto says the red one, then changes his mind.

Tessa says nothing. She lays both ribbons flat on the bench, and pushes both their ends against the wooden lip at the end. Now everybody can see.

You have to start them in the same place, she says. That is the whole trick of measuring, and it is the rule for this unit.

• CONTINUED

✓ **By the end you will**

- say which of two things is longer, and which is shorter
- say which of two things is taller
- line things up at the same start before you compare them
- measure a length with cubes and count how many
- put three or four things in order from shortest to longest



Tessa pushes both ribbons against the end of the bench, so they begin at the same line.

Longer, shorter, taller



Stretch your arms out as wide as they will go. Now make them as short as you can. Stand tall on your toes. Now be very short.

✦ Start both things at the same line

To find out which of two things is **longer**, lay them side by side. Push both their ends against the same line. Then look at the other end.

The one that reaches further is longer. The other one is **shorter**. For something standing up, we say **taller** instead of longer.

You can also measure with cubes. Lay cubes end to end along the thing, with no gaps, and count them. Then you can say how long it is even when the other thing is in another room.

✂ The material you need

a handful of cubes that join together, two ribbons, a pencil and a ruler with centimetre marks

TOPIC 4.1 · CONTINUED

a ruler, marked in centimetres



 **Watch me do one 1**

How many cubes long is this twig?

- 1** Lay the twig on the bench and put one cube at its end.
- 2** Add cubes end to end along the twig, with no gaps at all.
- 3** Stop when the cubes reach the far end of the twig.
- 4** Now count the cubes: one, two, three, four, five.

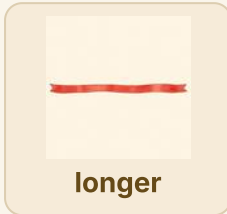
Answer: the twig is **5** cubes long

the twig, measured in cubes

How do you know?

Milo measures the same twig again, but he leaves a gap between two of his cubes. He counts six. Say why he is wrong.

Longer or shorter



1 Ring the longer stick.

Pip's stick

Otto's stick

2 Order these three ribbons, shortest to longest.

Write the names in the boxes.

Otto's ribbon

Hattie's ribbon

Milo's ribbon

TOPIC 4.1 · YOUR TURN · CONTINUED

shortest to longest

3 Find something in the room taller than you. Name it.

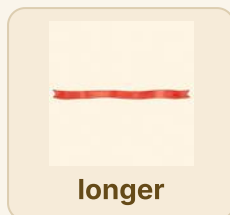
4 Count the centimetre marks on this ruler.



5 Measure the pencil with cubes. Write how many cubes.

The material you need: a pencil and ten cubes that join together

TOPIC 4.1 · YOUR TURN · CONTINUED



6 Draw a line longer than this one.

draw a longer line under this



7 Tick the shortest bar.

bar one

bar two

bar three

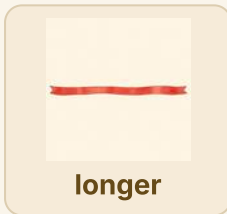
On the measuring bench

Watch me do one 2

Which stick is longer, Pip's or Otto's?

- 1 Lay both sticks on the bench, one under the other.
- 2 Push both ends against the lip at the end of the bench.
- 3 Look at the far ends. One stick reaches further along.
- 4 The stick that reaches further along is the longer one.

Answer: Otto's stick is longer



- 8 Compare two ribbons. Say which one is shorter.

The material you need: two ribbons of different lengths

- 9 Stand back to back with a friend. Say who is taller.

TOPIC 4.1 · YOUR TURN · CONTINUED

- 10** Measure the bench with your hands. Count the hand spans.
Do not leave a gap between your hands.



- 11** Draw a bar shorter than this one.

draw a shorter bar under this



- 12** Order these four pencils, longest to shortest.

the grey pencil

the red pencil

the green pencil

the blue pencil

longest to shortest





13 Find two things the same length. Show how you checked.

The material you need: a ribbon, a pencil and a strip of paper

*** Pip's find**

Your own hand span is a real measure, and people used it for hundreds of years. It has one problem: every hand is a different size, so two people never agree.

→ Try a bigger one

Otto's stick is longer than Pip's. Hattie's stick is longer than Otto's. Draw all three, shortest to longest.

♣ Find out together

Work with a partner. Cut a strip of paper exactly as long as your arm. Then find three things in the room that are shorter than your strip, and one thing that is longer.

What you can do now



☐ Check what you can do

- 1** Hold two pencils together. Say which one is longer.
- 2** Say what you must line up before you begin measuring.
- 3** Measure your foot with cubes. Say how many.
- 4** Name something in the burrow shorter than your arm.
- 5** Say which is taller, Bramble the badger or Pip the wren.

- ☐ say which of two things is longer, and which is shorter
- ☐ say which of two things is taller
- ☐ line things up at the same start before you compare them
- ☐ measure a length with cubes and count how many
- ☐ put three or four things in order from shortest to longest

END OF UNIT 4 • CONTINUED

	confident	almost there	needs more work
Comparing two lengths	☐	☐	☐
Lining things up at the same start	☐	☐	☐
Measuring with cubes and counting them	☐	☐	☐
Ordering things from shortest to longest	☐	☐	☐
Saying how I know	☐	☐	☐

▣ BURROW WORDS



long



short



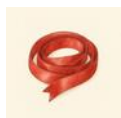
tall



order



line



ribbon



pencil



counter

Three ribbons



Otto, Hattie and Tessa kneel on the woven rug and lay the three ribbons out side by side.

Bramble keeps three ribbons in the store: a green one for parcels, a red one for hats and a thin brown one for bookmarks. This morning nobody can agree which ribbon is the longest. The brown one is rolled up in a curl, so Hattie is quite sure it is the shortest one of all. Otto is not so sure. He says a curled up ribbon only **looks** short.

The material you need

three ribbons of different lengths, a table with a straight edge, a pencil and paper

- 1 Lay all three ribbons flat on the table.
- 2 Line up one end of every ribbon against the same edge.
- 3 Point at the ribbon that reaches the furthest.
- 4 Put your three ribbons in order, shortest to longest.
- 5 Draw them in that order, one under the other.

How do you know?

Hattie thought the curled ribbon was the shortest. Straighten it out and lay it down again. Was she right? Tell the class how you found out.

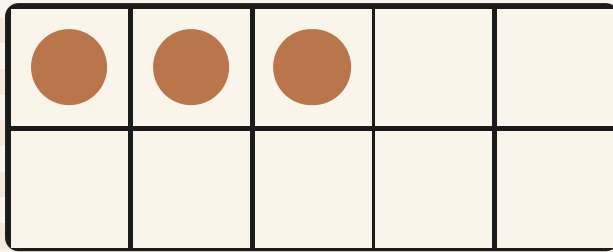
Adding and taking away

Tipping two baskets together to find a total,
then lifting some out again to see what is left.

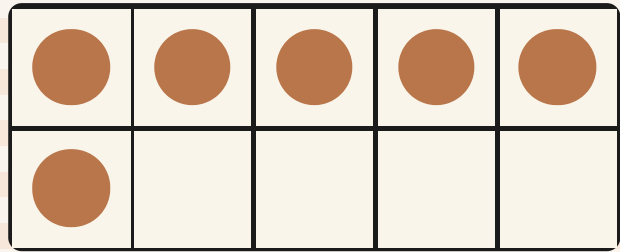


Open the door

Before you begin, show what you know already.



How many?



How many?

Count each set out loud. Then write the number in the box.

--	--

☀ In the burrow today

Two baskets are standing side by side on the counting table.

One basket holds acorns. The other holds acorns too. Otto tilts the first basket until every acorn slides into the second one.

Hattie says there will be **about ten**. Milo asks **how do you know?**

So Otto points at each acorn and counts the new heap. Later Bramble lifts some acorns out again for the store, and the heap gets smaller.

• CONTINUED

✓ By the end you will

- add two sets together and count the total
- count on from the bigger number to add
- write a number sentence with a plus sign
- take some away from a set and count what is left
- write a number sentence with a take away sign



Otto tilts one basket into the other. Nothing falls on the floor, so the two sets become one set.

Adding two sets together



WAKE UP
YOUR
COUNTING

Show me four fingers on one hand and three on the other. Now count them all, without starting again at one.

♣ Two sets, one total

When you put two sets together you get one bigger set. The number in the bigger set is the **total**.

Count the first set. Then keep counting straight into the second set. You do **not** go back to one.

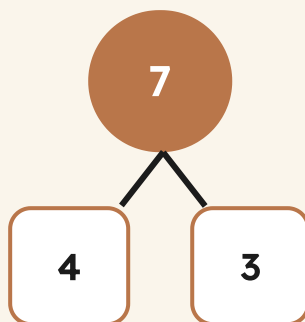
				

four acorns

three acorns

four and three make the whole



Watch me do one 1

How many acorns are there altogether?

- 1 Count the first basket, pointing as you go: one, two, three, four.
- 2 Do not start again. Keep counting into the second basket: five, six, seven.
- 3 Seven was the last number you said, so seven is the total.

Answer: the total is 7, so $4 + 3 = 7$

How do you know?

Hattie says four and three makes seven, but three and four makes six. Show her with counters whether she is right.

Find the total



total

1 Count both sets. Write the total in the box.

Point at every counter once, and only once.

●	●			

●	●	●	●	●

2 Count on from the bigger set. Write the total.

●	●	●	●	●
●				

●	●	●		

TOPIC 5.1 • YOUR TURN • CONTINUED

3 Finish each number sentence.

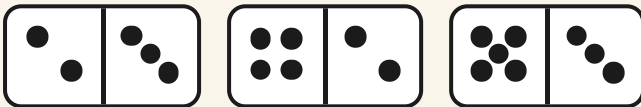
$3 + 2 = \square$

$5 + 4 = \square$

$1 + 6 = \square$

$7 + 2 = \square$

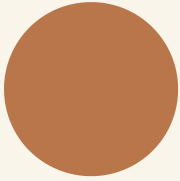
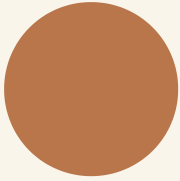
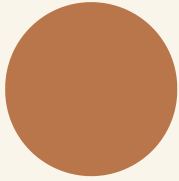
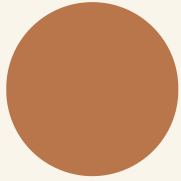
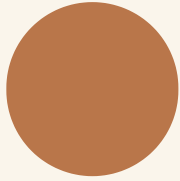
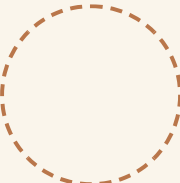
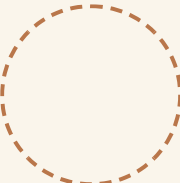
4 Add the spots on both halves of each domino.



5 Draw two more counters. Then write the new total.

The material you need: counters and a pencil

TOPIC 5.1 • YOUR TURN • CONTINUED



total

6 Fill in the missing part of each number bond.

$$0 + \square = 6$$

$$1 + \square = 6$$

$$2 + \square = 6$$

$$3 + \square = 6$$

$$4 + \square = 6$$

$$5 + \square = 6$$

$$6 + \square = 6$$

Counting on with Pip



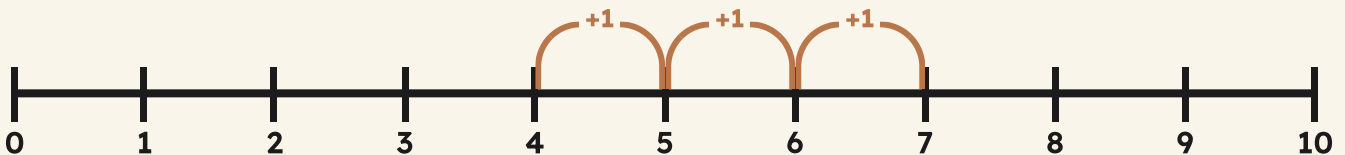
Pip hops along the numbers. One hop is one number, and Pip never skips a tick.



total

7 Hop along the number line to add three. Write where you land.

start at four



8 Add these pairs of numbers. Write each total.

TOPIC 5.1 · YOUR TURN · CONTINUED

$2 + 2 = \square$

$4 + 4 = \square$

$5 + 5 = \square$

- 9** Count the acorns. Then add two more acorns.



A row of acorns waiting on the counting table.

- 10** Write the total of six and three.

- 11** Draw counters to show four and five together.

TOPIC 5.1 • YOUR TURN • CONTINUED



total

12 Match each number sentence to its total.

$3 + 4 =$



9

$2 + 4 =$



7

$5 + 4 =$



6

★ Pip's find

Your two hands make adding easy. Four fingers and three fingers are always seven fingers, in every burrow in the world.

Taking away from a set



WAKE UP
YOUR
COUNTING

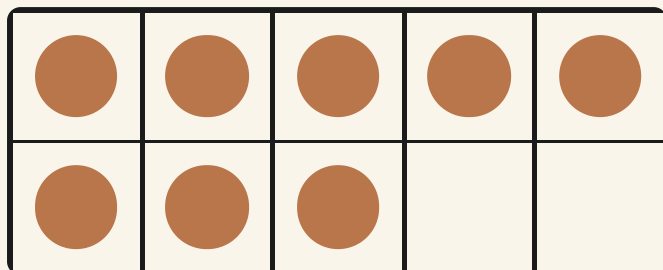
Hold up eight fingers. Fold two of them down. How many fingers are still standing up?

✦ Take some away, then count what is left

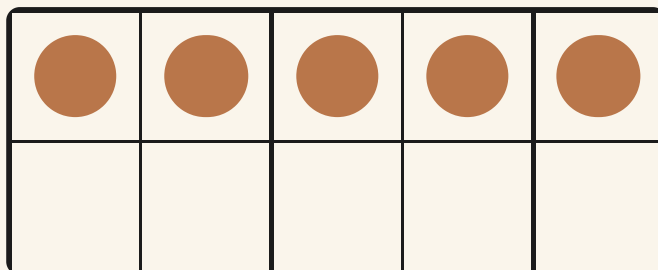
When you take things away from a set, the set gets smaller. You count what is **left**.

The take away sign is one short line. It tells you to move some things right out of the set, so they cannot be counted again.

TOPIC 5.2 · CONTINUED



eight shells on the tray



five shells left

Watch me do one 1

Bramble had eight shells and sold three. What is left?

- 1 Count eight shells onto the tray, one at a time.
- 2 Slide three shells *away*, right off the tray, into the bag.
- 3 Now count only what is left: one, two, three, four, five.

Answer: 5 shells are left, so $8 - 3 = 5$

How do you know?

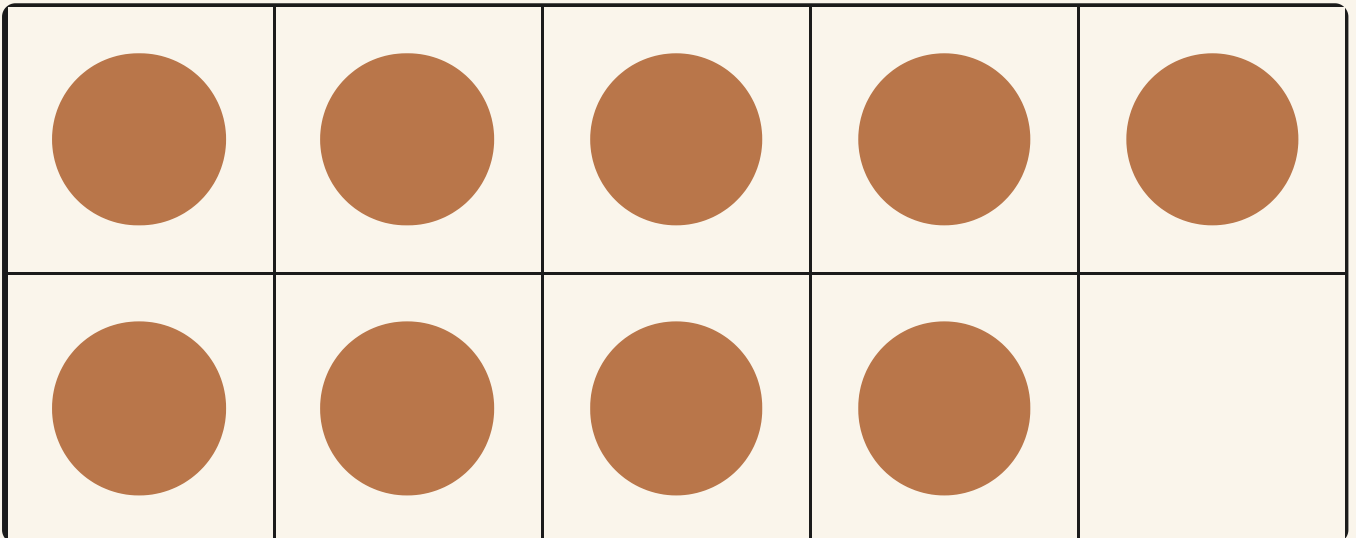
Tessa says taking away always leaves fewer things. What happens if you take away nothing at all?

What is left?



fewer

1 Cross out three counters. Write how many are left.



2 Finish each take away sentence.

TOPIC 5.2 · YOUR TURN · CONTINUED

$6 - 2 = \square$

$9 - 4 = \square$

$7 - 5 = \square$

$10 - 3 = \square$

- 3** Count the eggs. Cross two of them out.
Then write how many eggs are left in the straw.

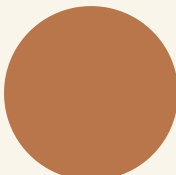
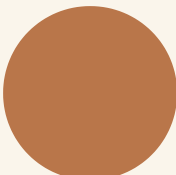
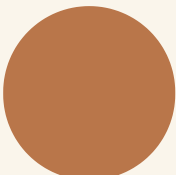
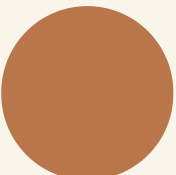
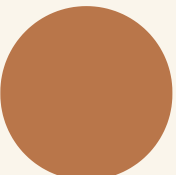
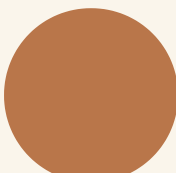
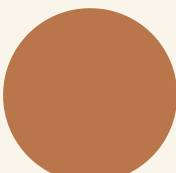
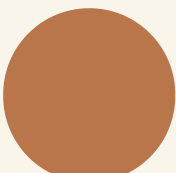


Eggs resting in the warm straw.

- 4** Take four away from ten. Write what is left.

- 5** Draw the counters that are left after taking five away.

TOPIC 5.2 • YOUR TURN • CONTINUED



fewer

6

Ring the take away that leaves more.

9 take away 1

9 take away 7

Hopping back

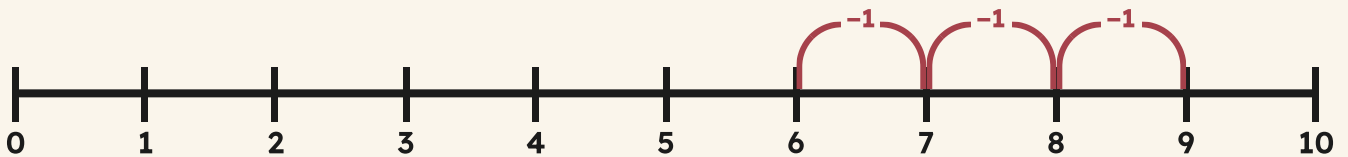
To take away, Pip turns round and hops backwards. One hop back is one less.



fewer

7 Hop back along the number line to take away three.

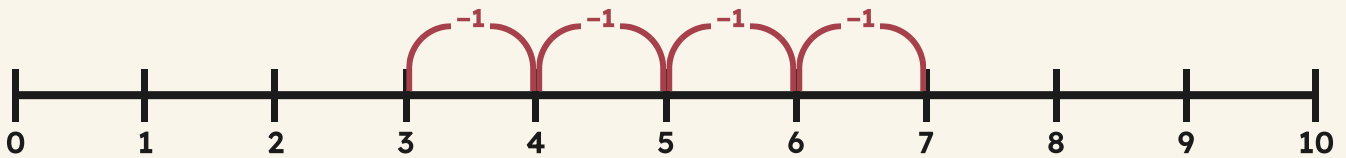
start at nine



TOPIC 5.2 · YOUR TURN · CONTINUED

8 Start at seven. Hop back four. Write where you stop.

start at seven



9 Finish these take away sentences.

$$5 - 5 = \square$$

$$8 - 1 = \square$$

$$10 - 10 = \square$$

$$6 - 3 = \square$$

10 Match each take away sentence to what is left.

$$7 - 2 = \bullet$$

$$\bullet 3$$

$$9 - 6 = \bullet$$

$$\bullet 5$$

$$8 - 4 = \bullet$$

$$\bullet 4$$

- 11** Draw a take away story with eight pebbles.



fewer

- 12** Tell a partner a take away story about the store.

The material you need: ten counters and a small paper bag

🦔 Burrow challenge

Bramble has ten walnuts. He sells some, and four are left. How many did he sell? Use counters to find out.

What can you do now?



☐ Check what you can do

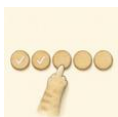
- ☐ 1 Otto has five acorns and picks up four more. Find his total.
- ☐ 2 Draw a plus sentence for seven acorns and two acorns.
- ☐ 3 Bramble sells three of his ten shells. Say what is left.
- ☐ 4 Count back from nine three times. Say where you stop.
- ☐ 5 Explain why adding two sets can never give you fewer things.

- ☐ I can add two sets together and count the total.
- ☐ I can count on from the bigger number to add.
- ☐ I can take away from a set and count what is left.
- ☐ I can write a sentence with a plus sign or a take away sign.

END OF UNIT 5 · CONTINUED

	confident	almost there	needs more work
Adding two sets to find a total	☐	☐	☐
Counting on to add	☐	☐	☐
Taking away and counting what is left	☐	☐	☐

▣ BURROW WORDS



count



total



more



fewer



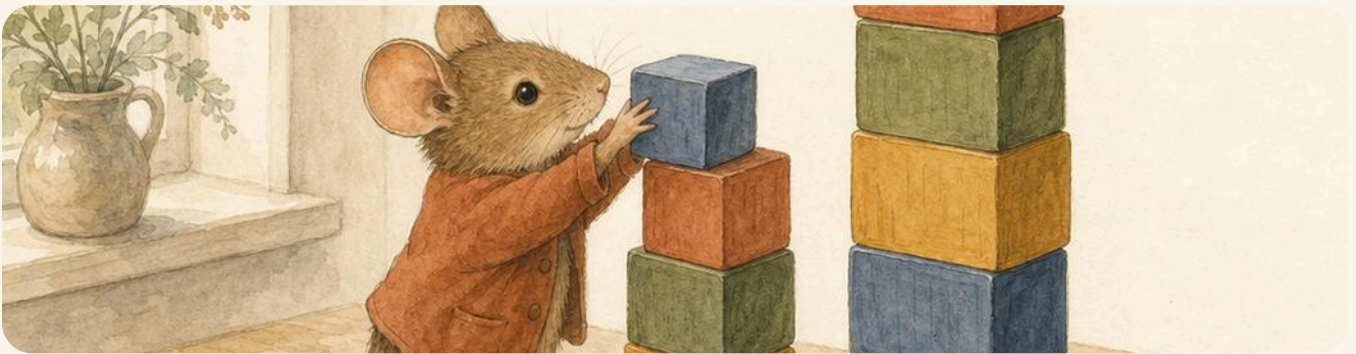
same



equal

Two towers

Milo and Otto have been building. Otto's tower of blocks stands taller than Milo's, and Milo would like them to be exactly the same. He does not want to knock Otto's tower down, so he must build his own one up instead. Milo asks the question he always asks: how many more blocks?



Two towers of wooden blocks, one taller than the other. Milo is putting one more block on the shorter tower.

The material you need

a box of wooden blocks that are all the same size, a partner, and a flat table to build on

- 1 Build a short tower. Ask your partner to build a taller one.
- 2 Stand the two towers side by side on the table.
- 3 Count the blocks in each tower. Write both numbers down.
- 4 Add blocks to the shorter tower until the two towers match.
- 5 Count the blocks you added on. Write that number too.

How do you know?

Tessa says the two towers are equal now. Could you show her that they are, without counting the blocks all over again?

UNIT 6

Where things are

On top, under, next to and between, and finding out who is first in the queue on the woven rug.



Open the door

Before you begin, show what you know already.

1	2	3		5	6		8	9	10
---	---	---	--	---	---	--	---	---	----

Say the numbers along the track, first to last. Write the two numbers that are missing.

☀ In the burrow today

Tessa has unrolled the big woven rug across the schoolroom floor.

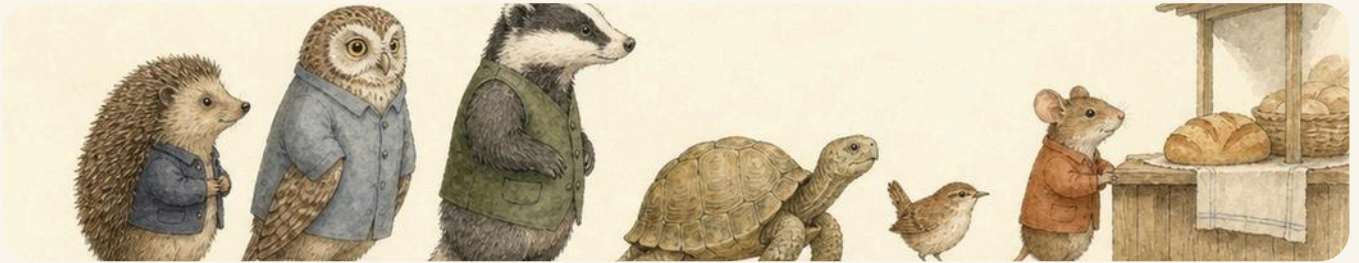
It has wide stripes, and the stripes make a track. Pip hops along them, one hop for one stripe. Otto lands on the rug and Tessa lines five acorns up along the middle stripe.

Milo looks at the row and asks **which one is third?** Nobody can answer until they agree where the row begins. So that is where this unit begins too.

• CONTINUED

✓ **By the end you will**

- use position words such as on top, under, next to and between
- say where something sits on a grid
- say who is first, second and third in a queue
- use the ordinal numbers from first to tenth
- give a partner directions along the woven rug



The woven rug with its wide stripes. Pip stands on one stripe and Tessa waits on the next one along.

Words that tell you where



WAKE UP
YOUR
COUNTING

Hold your pencil on top of your book. Now under it. Now next to it. Say each word out loud as you move.

♣ Where is it?

A position word tells you **where** something is. On top, under, next to, between, in front of, behind. Each one needs two things: the thing you are talking about, and the thing you measure it against.

On a grid you say the **row** first, counting down from the top, and then the **place** along that row, counting from the left.

TOPIC 6.1 · CONTINUED



position



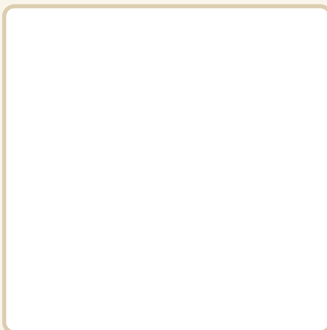
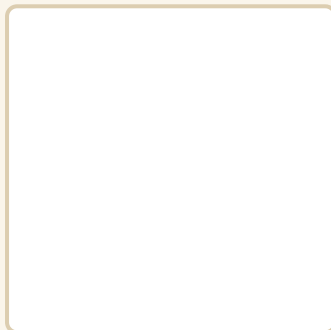
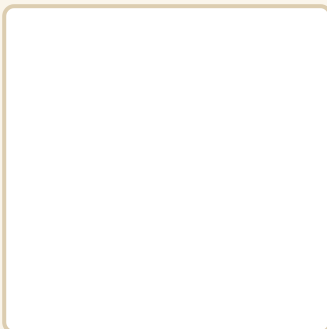
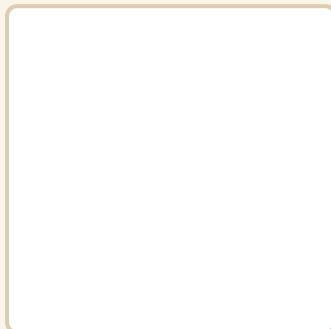
order



line



track



TOPIC 6.1 • CONTINUED

The pebble sits in the top row. The button is below it, in the middle row. The bean is in the bottom row.

Watch me do one 1

Where is the button on the grid?

- 1 Find the row first. Count down from the top: one, two.
 - 2 Now find the place along that row. Count from the left: one, two.
 - 3 Say both parts together, the row and then the place.
-

Answer: the button is in row **2**, place **2**

How do you know?

Otto says the pebble is under the button. Tessa says it is above.
Look again, and say who is right.

On the rug



position

- 1 Draw a shell in the middle box of the top row.

TOPIC 6.1 · YOUR TURN · CONTINUED

2 Ring the word that says where the cat is sitting.



The cat has curled up under the table.

The basket stands next to the cat.

on top

under

next to

3 Draw a bean next to the basket.



TOPIC 6.1 · YOUR TURN · CONTINUED

- 4** Colour the box between the two pebbles.

The material you need: one coloured pencil

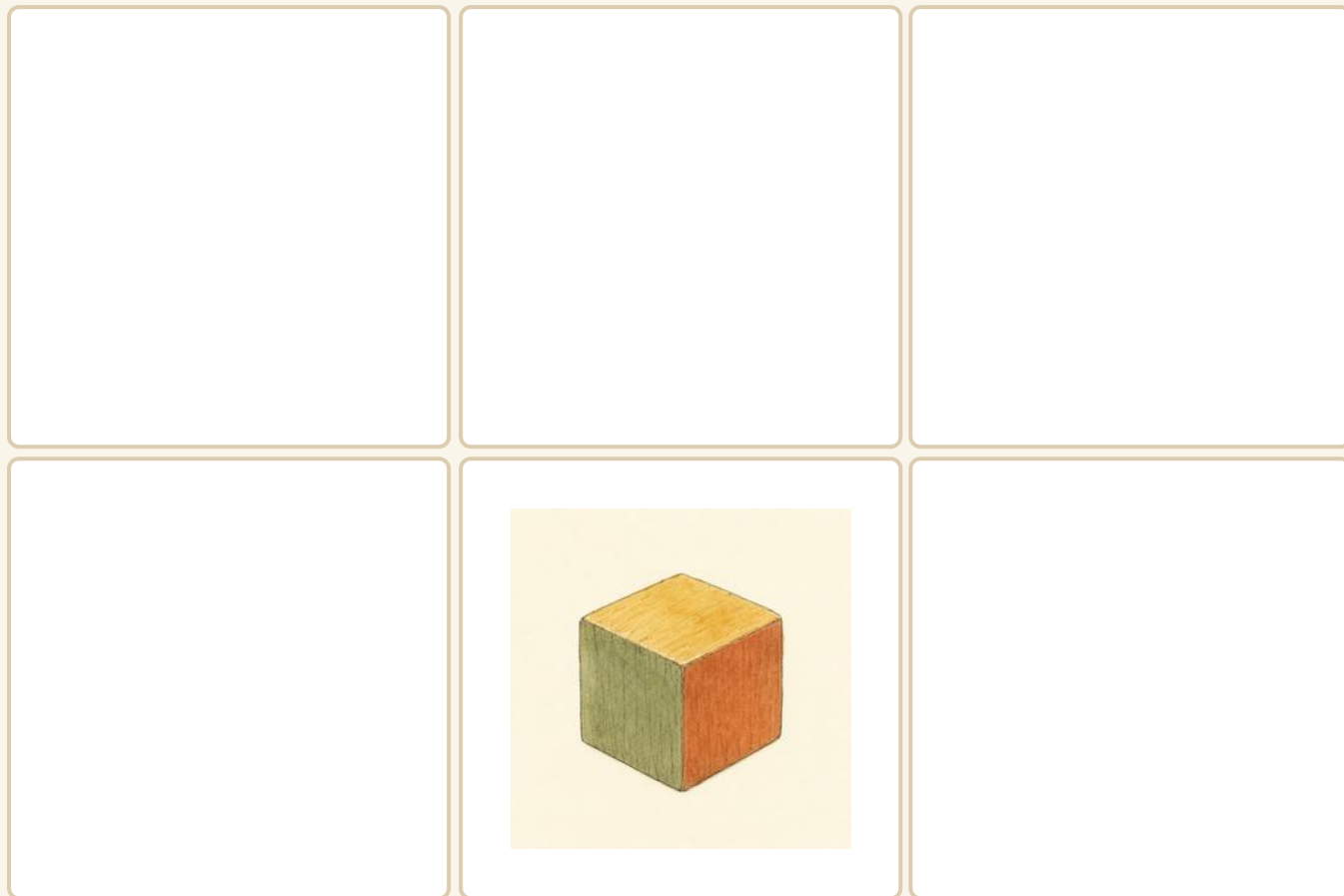


- 5** Tell a partner where your chair is.

Use two position words, such as behind and next to.

- 6** Draw a snail on top of the block.

TOPIC 6.1 · YOUR TURN · CONTINUED



★ Pip's find

Sailors say port and starboard instead of left and right, because left and right swap over when you turn round.

First, second, third

Do you remember?

How many acorns are in a set of five? Now say which acorn is the last one you counted.

♣ Counting numbers and place numbers

One, two, three tell you **how many** there are. First, second, third tell you **which place** something is in.

Everyone in the burrow counts a queue from the front, and keeps going the same way. If you turn round halfway, your answer changes and nobody can trust it.

TOPIC 6.1 · CONTINUED · CONTINUED



hen
1st



cat
2nd



dog
3rd



snail
4th



bird
5th

Watch me do one 1

Which animal is third in the queue?

- 1** Start at the front of the queue, where the hen is standing.
- 2** Count the places and not the animals: first, second, third.
- 3** Stop at the third place and see who is standing there.

Answer: the dog is 3rd in the queue

How do you know?

Pip counts the same queue from the back and says the hen is fifth.
Explain why Pip and Otto disagree.

Which place?



7 Write the place of each animal in the queue.



hen



cat



dog



bird

8 Ring the third pebble from the left.



9 Ring the number where Pip's fourth hop lands.

Pip starts at zero



10 Write the missing numbers on the track.

1	2		4	5		7	8		10
---	---	--	---	---	--	---	---	--	----

11 Draw five acorns in a row. Ring the second one.



position

12 Write these animals in queue order, first to last.

dog

hen

bird

cat

first to last

→ **Try a bigger one**

Line up ten pencils. Ask a partner to point at the seventh one, then at the tenth one.

What can you do now?



☐ Check what you can do

- 1** Draw a walnut under a basket and a bean on top of it.
- 2** Say which animal stands fourth in a queue of six animals.
- 3** Name the place of the last pebble in a row of eight.
- 4** Give a partner two directions for crossing the woven rug.
- 5** Explain why the front of a queue is always the first place.

- ☐ I can use position words to say where something is.
- ☐ I can find a place on a grid by its row and its place.
- ☐ I can say who is first, second, third and last in a queue.
- ☐ I can use the ordinal numbers from first to tenth.

END OF UNIT 6 · CONTINUED

	confident	almost there	needs more work
Using position words on the woven rug	☐	☐	☐
Finding a place on a grid	☐	☐	☐
Saying which place something is in	☐	☐	☐

▣ BURROW WORDS



position



order



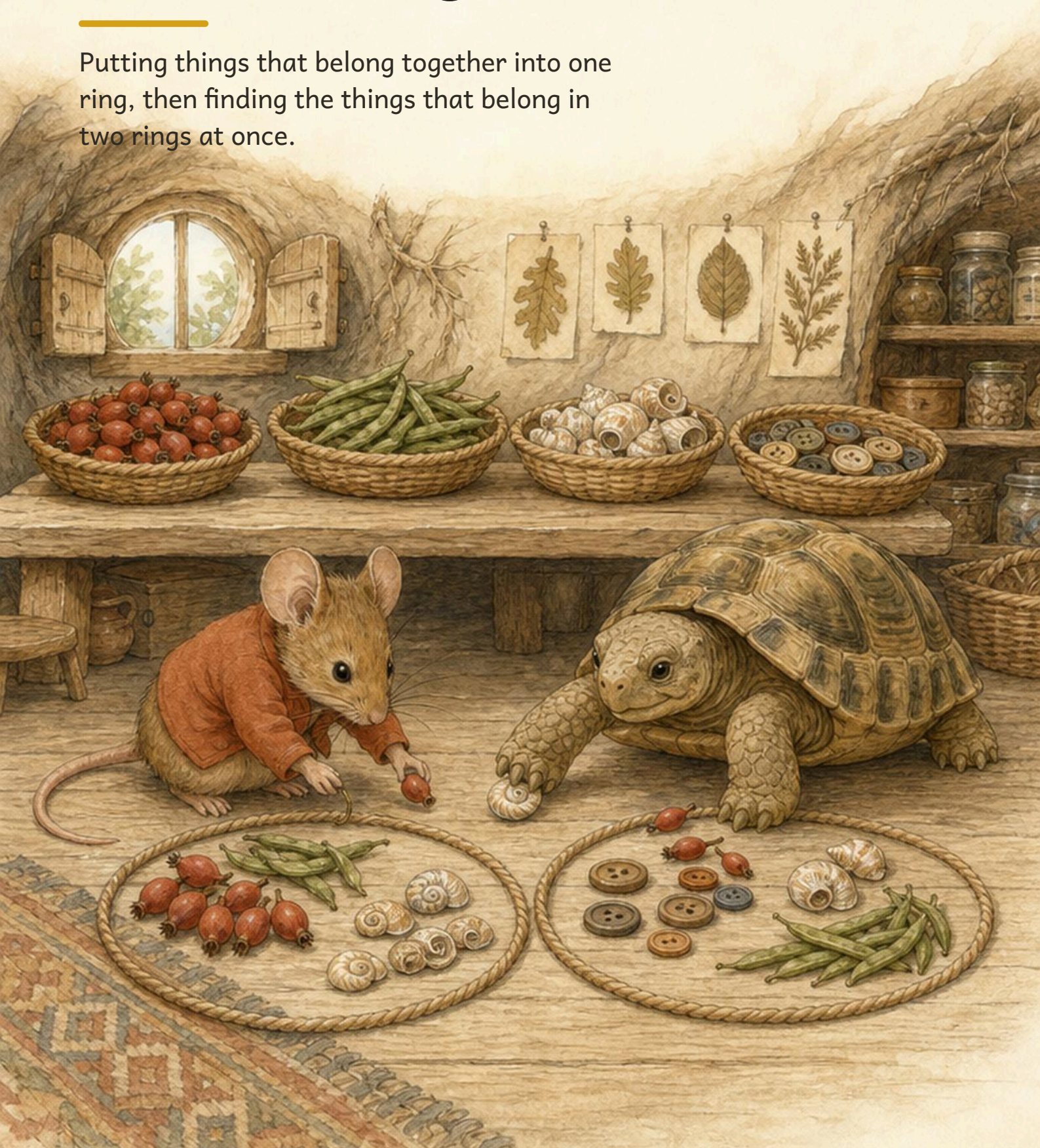
line



track

Sets and rings

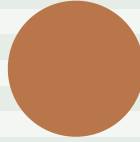
Putting things that belong together into one ring, then finding the things that belong in two rings at once.



Open the door

Before you begin, show what you know already.

Sort these into two groups. Say your rule out loud.



☀ In the burrow today

Bramble has tipped four baskets onto the floor.

Shells, buttons, beans and pebbles, all mixed together. Tessa lays two loops of cord on the floorboards and starts putting things inside them.

Milo asks the useful question: **what is your rule?** A ring is only a set once everybody knows what belongs in it.

✓ **By the end you will**

- put things that belong together into one set
- say the rule that decides what belongs
- count how many are in a set
- use two rings that overlap in the middle
- say what belongs in the overlap



Tessa sorts buttons, shells and beans into two rings of cord. Every ring needs a rule.

Sets

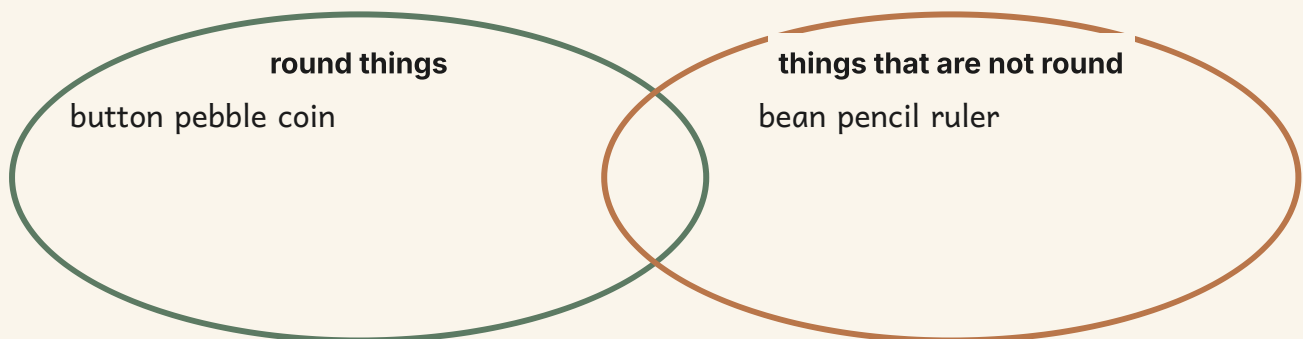
WAKE UP
YOUR
COUNTING

Look at your table. Find three things that belong together. Tell your partner your rule, but do not tell them the things.

✦ A set is a group with a rule

A **set** is a group of things that belong together. The **rule** says what belongs.

If the rule is **round things**, then every round thing goes inside the ring, and everything else stays outside.



 Watch me do one 1

How many shells are in this set?

- 1 Look only inside the ring. Ignore everything outside it.
 - 2 Point at each shell inside and count: one, two, three, four.
 - 3 The last number you said was four.
-

Answer: there are **4** shells in the set

How do you know?

Hattie put a pebble in the **round things** ring and a button outside it. Is she right? Say how you know.

Make a set



set

1 Ring all the buttons.



2 Count the things in the ring. Write how many.

3 Sort these into two baskets. Write your rule.

The material you need: a handful of buttons, shells and beans

my first rule

my second rule

TOPIC 7.1 · YOUR TURN · CONTINUED

4 Draw four things that belong in one set.

5 Write the rule for your set.



6 Ring the one thing that does not belong.



Counting a set



7 Count the shells in the basket. Write how many.



Point at each shell as you count it.

8 Draw a set of six acorns inside the ring.

9 Tick every set that has exactly three things.

●	●	●			●	●	●	●	●	●	●	●		

10 Sort the beans and the pebbles into two rings.

The material you need: six beans and six pebbles

beans

pebbles

11 Write how many things are in your biggest set.



set

12 Say your rule to a partner. Let them add one thing.

★ **Pip's find**

Sorting is the oldest kind of mathematics there is. Long before anybody wrote a number down, people put things into groups.

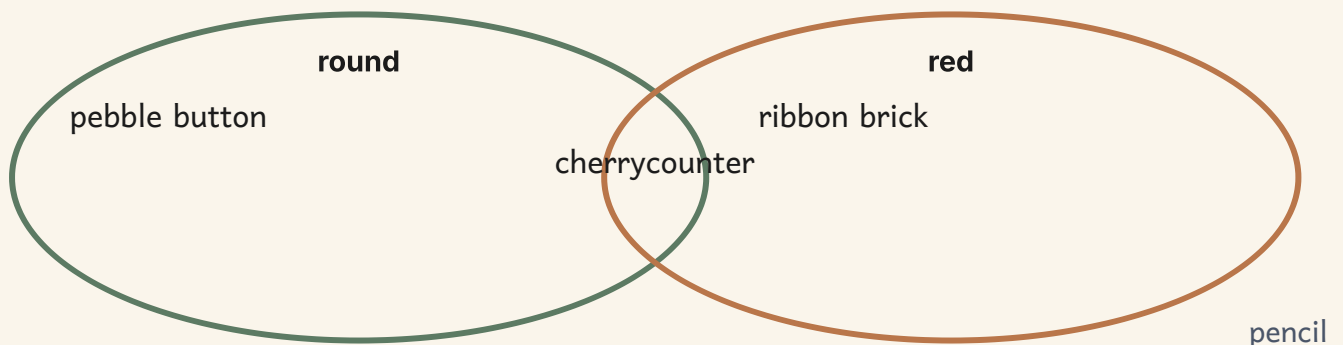
Venn diagrams



♣ Two rings that share a middle

Sometimes a thing belongs in **two** sets at once. Then the two rings overlap, and that thing sits in the middle.

A ring with an overlap is called a **Venn diagram**. Anything that fits neither rule stays outside both rings.



 Watch me do one 1

Where does a red button go?

- 1 Ask the first rule: is it round? Yes, so it belongs in ring one.
 - 2 Ask the second rule: is it red? Yes, so it belongs in ring two.
 - 3 It belongs in both, so it goes in the middle where they overlap.
-

Answer: in the middle, where the two rings overlap

How do you know?

A green bean is not round and not red. Where does it go? Say how you know.

Two rings



group

1 Write each thing in the right part of the rings.

acorn

red apple

round red counter

pencil

round

red

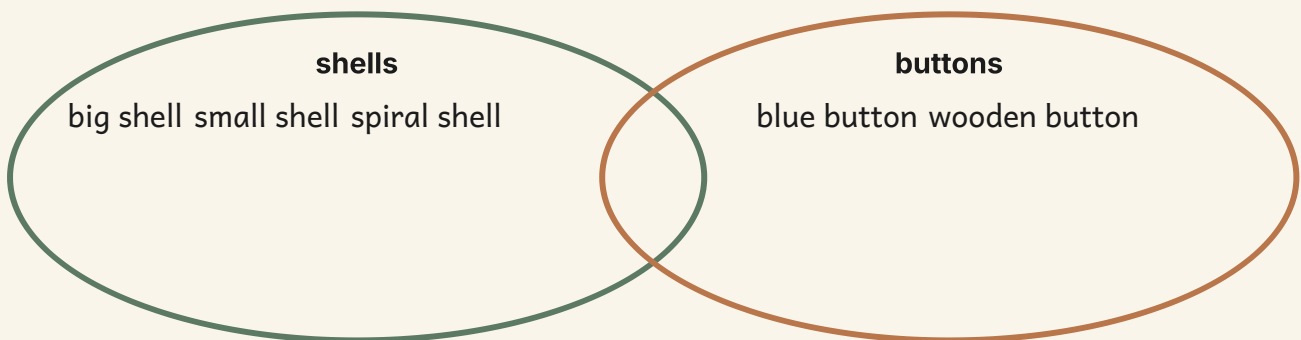
2 Count the things in the middle. Write how many.

TOPIC 7.2 · YOUR TURN · CONTINUED

3 Write one thing that belongs in both rings.

4 Write one thing that belongs in neither ring.

5 Tick the ring that has more things in it.



group

6 Sort your own things into two overlapping rings.

The material you need: two loops of string and ten small objects

Sorting two ways



7 Fill in this table. Tick one box for each thing.

Sort them two ways

	red		not red	
round				
not round				

8 Write how many things you put in each box.

9 Ring the thing that could go in the middle.

a red round counter

a green bean

a wooden ruler

10 Draw one thing for the overlap of these rings.

11 Count everything outside both rings. Write how many.



group

12 Explain your rules to a partner. Swap and check.

→ Try a bigger one

Try three rings that all overlap. How many different places are there for a thing to sit?

The words of this unit

▣ BURROW WORDS



set

things grouped by one rule



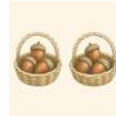
group

things gathered together



label

the name written on a ring



same

no difference at all



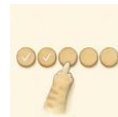
more

the bigger amount



fewer

the smaller amount



count

one number for one thing



data

what we found out

▣ Check what you can do

- 1 Ring all the round things in this picture.
- 2 Write the rule for the set you made.
- 3 Count the things inside the ring and write how many.
- 4 Name an object that fits both rules at once.
- 5 Write one thing that belongs outside both rings.

Now I can

- ☐ put things that belong together into one set
- ☐ say the rule that decides what belongs
- ☐ count how many things are in a set
- ☐ use two rings that overlap in the middle
- ☐ say what belongs in the overlap and what stays outside

	confident	almost there	needs more work
making a set with a clear rule	☐	☐	☐
counting the things in a set	☐	☐	☐
using two overlapping rings	☐	☐	☐
deciding what goes in the middle	☐	☐	☐
sorting things two ways at once	☐	☐	☐



Sorting is how data begins. In Unit 14 these same rings turn into charts.

UNIT 8

Days and o'clock

Naming the seven days in their proper order, then reading the hour that the short hand points to.



Open the door

Before you begin, show what you know already.

Say the days out loud, Monday first. Which two are missing from this strip?

Monday	Tuesday	Wednesday		Friday	Saturday	
--------	---------	-----------	--	--------	----------	--

☀ In the burrow today

Bramble has hung a new week board beside the store door.

Seven wooden cards, one for each day, and a hook to hang them on. Otto points at each card as he counts them, and Pip hops from card to card, one hop for one day.

Above the board hangs the old burrow clock, with one short hand and one long hand. Milo asks the useful question: **how do you know** which hand tells the hour?

• CONTINUED

✓ **By the end you will**

- say the seven days of the week in order
- name yesterday, today and tomorrow
- point to the hour hand and the minute hand
- read an o'clock time on a clock face
- draw hands on a face to show an o'clock time



The week board and the old burrow clock. Both go round again in the same order, every time.

Days of the week



WAKE UP
YOUR
COUNTING

Say the days of the week with your partner, Monday first and Sunday last. Now say them again, quietly, without stopping.

✦ Seven days, always in the same order

A **week** has seven **days**. They always come in the same order, and when Sunday ends the week begins again at Monday.

The day we are in now is **today**. The day that has gone is **yesterday**. The day still to come is **tomorrow**.

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

 Watch me do one 1

Which day comes after Wednesday?

- 1** Say the week in order, starting at Monday.
 - 2** Stop as soon as you reach Wednesday.
 - 3** Say one more day. That day comes after Wednesday.
-

Answer: the day after Wednesday is Thursday

How do you know?

Hattie says Sunday comes before Saturday. Is she right? Say how you know.

*** Pip's find**

Every language in the world counts a week as seven days, and nobody is quite sure who chose seven first.

The clock face

Do you remember?

How many days are there in one week? Which day comes straight before Monday?

♣ The short hand tells the hour

A clock **face** has the numbers one to twelve around its edge, and two **hands** that turn.

The **short** hand is the **hour** hand. The **long** hand is the **minute** hand.

When the long hand points straight up to twelve, we say **o'clock**. Then the short hand tells you which o'clock it is.



2 o'clock



6 o'clock



11 o'clock



 **Watch me do one 2**

What time does the clock above show?

- 1** Look at the long hand first. It points straight up to twelve, so this is an o'clock time.
 - 2** Now look at the short hand. Follow it out to the numbers.
 - 3** The short hand points at nine, so nine is the hour.
-

Answer: the clock shows 9 o'clock

How do you know?

Milo says the long hand tells the hour, because it is bigger. Is he right? Say how you know.

The week board



1 Write the two missing days on this week strip.

Monday	Tuesday		Thursday	Friday		Sunday
--------	---------	--	----------	--------	--	--------

2 Write the day that comes after Tuesday.

3 Write the day that comes before Sunday.

4 Write the days in order, Monday first and Sunday last.

5 Ring the day that comes straight after Friday.

Thursday

Saturday

Monday



6 Ring the word that names a month, not a day.

Monday

April

Friday

7 Lay the day cards out in order, Monday to Sunday.

The material you need: seven cards, one for each day of the week

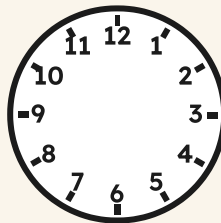
Telling the hour



8 Write the time under each clock face.



9 Draw the two hands to show five o'clock.



10 Ring the clock that shows eight o'clock.



11 Count the numbers around a clock face. Write how many.

12 Match each clock to the words for its time.



6 o'clock



10 o'clock



3 o'clock



clock

- 13** Set a clock to two o'clock. Ask a partner to check.

The material you need: a wooden clock with hands you can turn

- 14** Write the time the schoolroom bell rings each morning.

→ Try a bigger one

Turn the long hand right round once. Watch what the short hand does while it travels.

The words of this unit

▣ BURROW WORDS



clock

it shows us the time



hands

one short, one long



week

seven days together



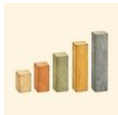
month

many weeks together



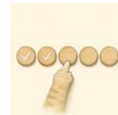
order

what comes first, next,
last



sequence

the same order every
time



count

one number for one
thing



data

what we found out

▣ Check what you can do

- 1 Say the seven days aloud, from Monday through to Sunday.
- 2 Name the day before Thursday and the day after Thursday.
- 3 Point to the short hour hand, then to the long minute hand.
- 4 Read an o'clock time from a clock face.
- 5 Draw hands on a blank face to show six o'clock.

Now I can



- ☐ say the seven days of the week in order
- ☐ name yesterday, today and tomorrow
- ☐ find the hour hand and the minute hand on a clock
- ☐ read an o'clock time on a clock face
- ☐ draw the hands to show an o'clock time

UNIT 8 • HOW DID IT GO • CONTINUED

	confident	almost there	needs more work
saying the days in the right order	☐	☐	☐
naming the day after and the day before	☐	☐	☐
telling the short hand from the long hand	☐	☐	☐
reading an o'clock time	☐	☐	☐
drawing hands to show a given hour	☐	☐	☐



The hour is only the beginning. Half past waits for you, halfway round the same face.

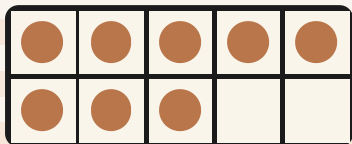
Numbers to 20

Counting past ten to twenty, making a full ten and some ones, saying which set has more, and finding the pattern hiding in the numbers.

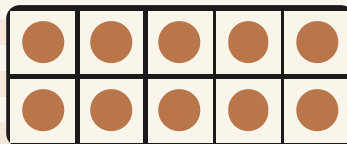


Open the door

You can already count to ten. Show that first.



How many?



How many?

Say each number out loud. Then write it in the box.

--	--

☀ In the burrow today

Bramble has tipped out the whole tin of walnuts.

There are far more than ten. Otto begins to point at them one by one. Hattie looks and says **about twenty**, because that is her estimate, and an estimate comes before the counting.

Milo asks **how do you know?** So Tessa slides ten walnuts into a row and keeps the rest to one side. Now everybody can see it: a full ten, and some ones.

• CONTINUED

✓ **By the end you will**

- count a set of up to twenty things and say how many
- say, read and write every number from ten to twenty
- make a teen number from one full ten and some ones
- say which of two numbers is more and which is less
- estimate first, then count to check
- carry on a number pattern and spot odd and even



Ten in the row, and the rest waiting. Tessa never guesses when she can line it up instead.

Counting past ten

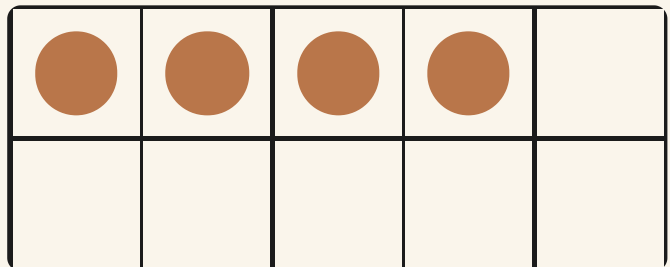
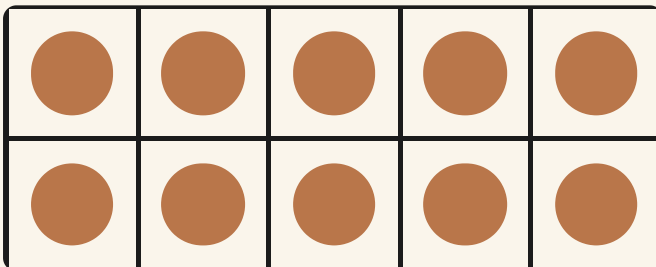
WAKE UP
YOUR
COUNTING

Count to twenty with me. Clap on every number. Now whisper them and count all the way back down to one.

♣ **A teen number is a full ten and some ones**

Fill one frame right up first. That is your ten, and you never have to count it again. The counters left over are the ones.

Ten and four more is fourteen. Say it slowly and you can hear the ten hiding at the end of the word.



Tens and ones

	Tens	Ones
14	1	4
20	2	0

Watch me do one 1

How many walnuts are on the table?

- 1 Slide walnuts into the first frame until it is full. That is a ten.
- 2 Now count the walnuts left over: one, two, three, four.
- 3 A full ten and four more ones is fourteen.

Answer: there are **14** walnuts

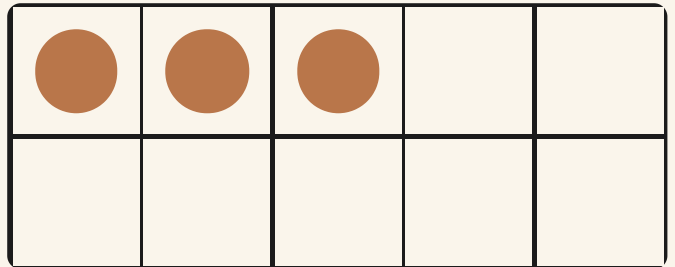
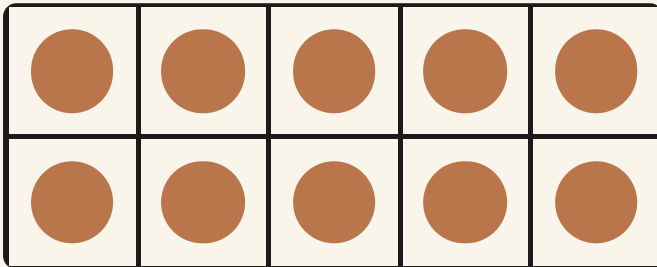
How do you know?

Cover the full frame with your hand. Can you still say how many there are altogether? Tell your partner why the ten does not need counting twice.

Count and write

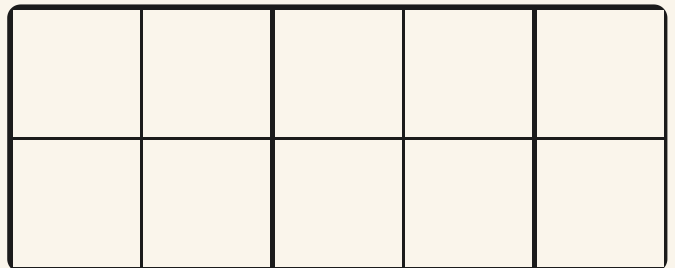
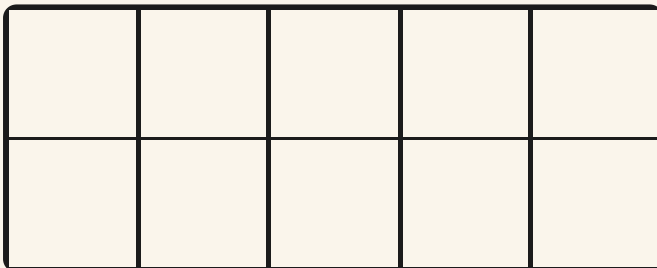


1 Count the counters in both frames. Write how many.



2 Draw counters in these frames to show sixteen.

Fill the first frame right up before you start the second one.



TOPIC 9.1 · YOUR TURN · CONTINUED

3 Fill in the missing numbers on this number track.

11	12		14	15		17	18		20
----	----	--	----	----	--	----	----	--	----

4 Write this number in figures: nineteen.

5 Write **12** in words.



ten

6 Trace this numeral three times. Say it as you write.

15 15 15 15

Tens and ones



7 Write the tens and the ones for each number.

	Tens	Ones
12		
15		
18		

8 Match each number to its tens and ones.

15 ●

12 ●

18 ●

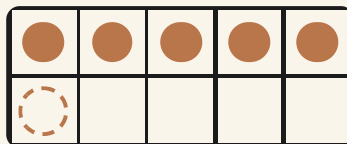
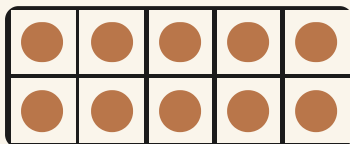
● 1 ten and 2 ones

● 1 ten and 8 ones

● 1 ten and 5 ones

TOPIC 9.1 • YOUR TURN • CONTINUED

- 9 Draw one more counter. Write the new number.



- 10 Count out seventeen pebbles. Make a full ten first.

The material you need: a pot of pebbles and a small tray

- 11 Count aloud from ten to twenty. Then count back down.



ten

- 12 Ring the number that comes just after nineteen.

18

20

17

* Pip's find

Fourteen once meant **four and ten**. The word kept the old order, which is why we say the four part first.



TOPIC 9.2

More, less, and about how many

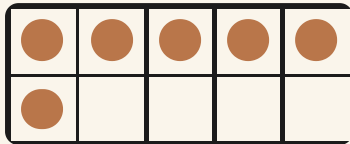
WAKE UP
YOUR
COUNTING

Show me six fingers. Now show me one more. Now one fewer. Say each number as your fingers change.

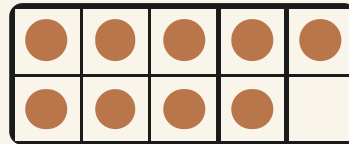
✦ Line them up, then you can see

Set one row under the other, one for one. The row that pokes out the end has **more**. The shorter row has **fewer**.

Hattie always estimates first. An estimate is a sensible guess you make before you count, and counting is how you check it.



Otto's counters



Pip's counters

Watch me do one 1

Which is more, thirteen or seventeen?

- 1** Find both numbers on the number track and point at them.
- 2** Say the numbers along the track. Seventeen comes later than thirteen.
- 3** The number you say later is always the bigger one.

Answer: **17** is more than **13**

How do you know?

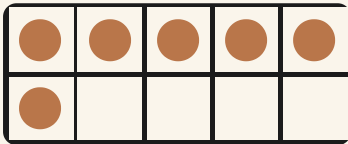
Hattie says twenty is more than nineteen because it sounds bigger. Is that a good reason? Give her a better one.

More, fewer, in order

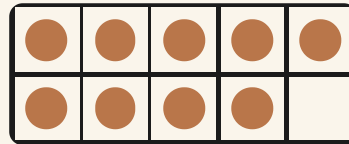


estimate

1 Ring the frame with more counters.

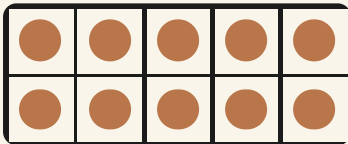


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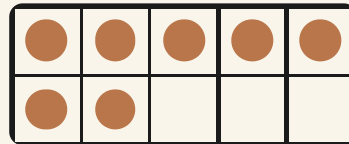


B

2 Ring the frame with fewer counters.



C



D

TOPIC 9.2 · YOUR TURN · CONTINUED

3 Ring the bigger number in each pair.

11 and **15**

18 and **13**

20 and **17**

12 and **19**

4 Write these numbers from smallest to largest.

17

11

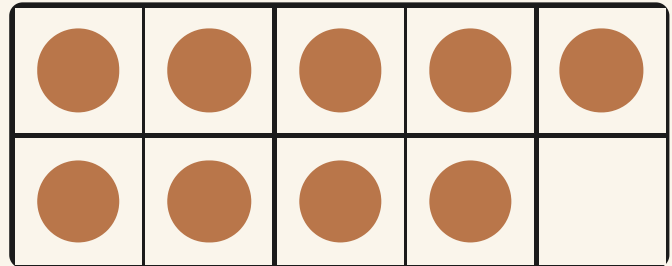
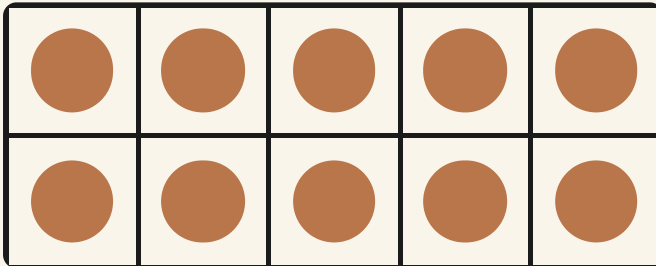
20

14

--	--	--	--

5 Estimate how many counters. Then count to check.

Say your estimate out loud before you point at anything.



Is it true?



6 Ring TRUE or NOT TRUE for each sentence.

14 is more than 11	TRUE / NOT TRUE
16 is less than 13	TRUE / NOT TRUE
20 is more than 19	TRUE / NOT TRUE

7 Fill the gaps on this track. Then read it backwards.

11

13

14

16

17

19

20

8 Write the number that is one less than eighteen.

9 Ring the number that comes between twelve and fourteen.

11

13

15

10 Build two towers of cubes. Say which tower has more.

The material you need: two handfuls of cubes



estimate

11 Write these four numbers from largest to smallest.

13

20

16

11

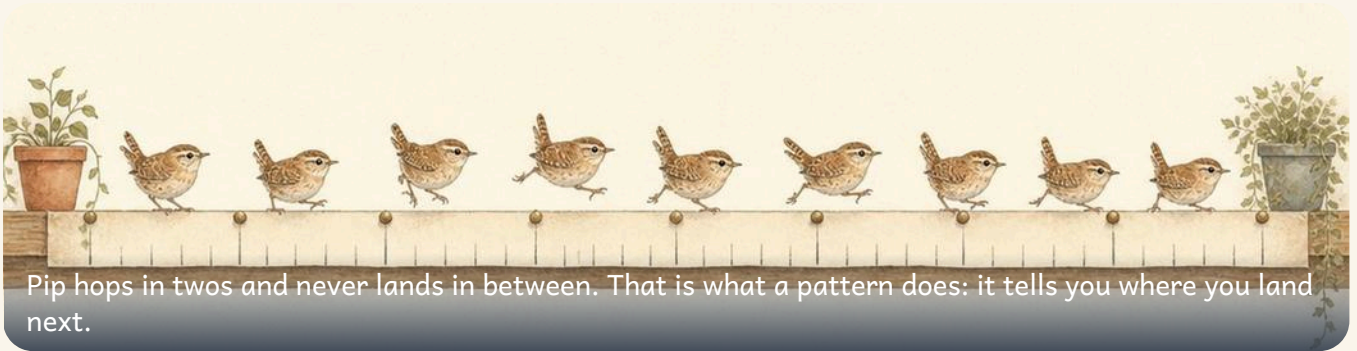
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12 Choose two numbers. Ask a friend which one is more.

--	--

🦔 Burrow challenge

Hattie estimates twenty acorns and counts eighteen. Was her estimate a good one? Say why, and make a better one for the next handful.



Pip hops in twos and never lands in between. That is what a pattern does: it tells you where you land next.

TOPIC 9.3

Patterns in the numbers

WAKE UP
YOUR
COUNTING

Stamp and clap: stamp, clap, stamp, clap. What comes next? Now do it with two claps every time.

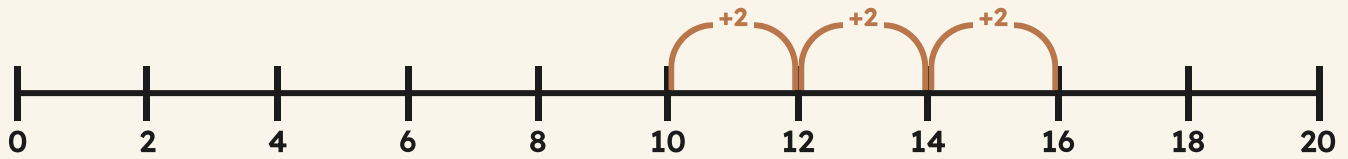
✿ A pattern tells you what comes next

Pip the wren hops along the counting track, and one hop is always the same size. Two at a time, he lands on two, four, six, eight. He never lands in between.

Numbers you land on when you hop in twos are **even**. They always split into pairs with none left over. The others are **odd**, and odd numbers always have one left on its own.

TOPIC 9.3 · CONTINUED

hopping in twos



 **Watch me do one 1**

What comes next: two, four, six, eight?

- 1** Look at the hops. Every hop is the same size, two.
- 2** Count the hops with Pip: two, four, six, eight.
- 3** One more hop of two from eight lands on ten.

Answer: the next number is **10**

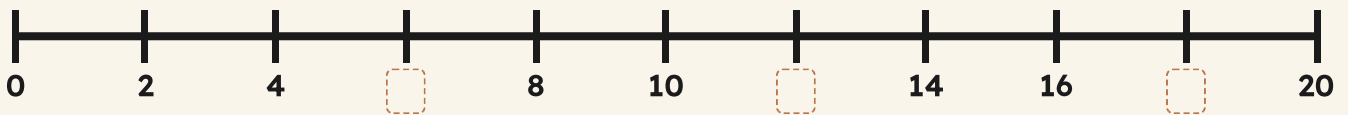
How do you know?

Pip hops in twos, starting at two. Will he ever land on seven? Try it with your finger, then say how you know.

Odd, even and what comes next



1 Write the missing numbers on this number line.



2 Ring every even number on this track.



TOPIC 9.3 · YOUR TURN · CONTINUED

3 Ring every odd number in the bank.

11

14

15

18

19

20

4 Draw nine dots in pairs. Is nine odd or even?

5 Carry on this pattern. Draw the next two cards.



6 Say what comes next. Then draw it in the empty cards.



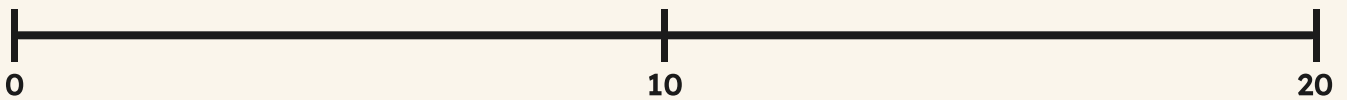
Hopping in twos, fives and tens



sequence

7 Draw hops of ten on this number line.

hops of ten



8 Write the number that is one more than nineteen.

9 Fill the gaps. Count in fives to twenty.



10 Sort these numbers into the odd tray and the even tray.

12

13

16

17

19

20

odd

even

11 Make your own pattern with two kinds of counter.

The material you need: a pot of counters of two kinds



sequence

12 Count in twos to twenty with a friend. Take turns.

★ **Pip's find**

Even numbers can always be shared out fairly between two. That is why Bramble counts the store in twos.

What you can do now



☐ Check what you can do

- 1** Say every number from ten to twenty, then back again.
- 2** Show sixteen as one full ten and some ones.
- 3** Name a number that is more than fourteen.
- 4** Name a number that sits between eleven and fifteen.
- 5** Keep counting in twos: two, four, six, and on.
- 6** Tell a friend one odd number and one even number.

- ☐ count a set of up to twenty things and say how many
- ☐ read and write the numbers from ten to twenty
- ☐ make a teen number from a full ten and some ones
- ☐ say which number is more and which is less
- ☐ estimate first, and then count to check
- ☐ carry on a pattern of twos, fives or tens

END OF UNIT 9 • CONTINUED

	confident	almost there	needs more work
Counting to twenty and back again	☐	☐	☐
Tens and ones inside a teen number	☐	☐	☐
Comparing, ordering and estimating	☐	☐	☐
Number patterns, odd and even	☐	☐	☐

▣ BURROW WORDS



digit



estimate



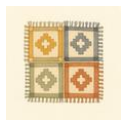
more



fewer



order



pattern



odd



even

How many minnows?



A shallow tray of small silver minnows on wet leaves, waiting to be counted.

Hattie has carried a tray in from the stream, and the whole class wants to know how many minnows are in it. Hattie looks for one moment and says **about fifteen**. Milo folds his arms and asks how she knows. A guess made with your eyes open has a proper name: it is an **estimate**. An estimate is not a wild guess, and it is not the answer either. You still have to count.

& The material you need

a tray of counters or paper fish, a pencil, paper and a partner to check with

- 1 Look at the tray while your partner claps ten times.
- 2 Write your estimate down before you count anything.
- 3 Now count the minnows. Move each one as you say its number.
- 4 Write down how many there really were.
- 5 Look at your two numbers. How close was your estimate?

How do you know?

Do it again with a fuller tray. Was your second estimate better than your first? Say what helped you.

Shapes again

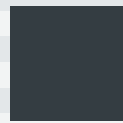
Faces, edges, sides and corners: what rolls, what stacks, and what happens when you turn a shape round.



Open the door

Before you begin, show what you know already.

Name each flat shape. Say one thing about each one.



☀ In the burrow today

Bramble is loading the store shelves before the rain comes.

The crates go up in a neat tower. The tins will not stay put: they roll off the shelf and away across the floorboards. Otto points at each face of a crate as he counts them.

Tessa draws round the bottom of a block to be sure of its shape, and Milo asks the useful question: **how do you know** a tin will roll and a crate will not?

• CONTINUED

✓ **By the end you will**

- count the faces and edges of a solid shape
- say which solids roll and which solids stack
- count the sides and corners of a flat shape
- name the flat shapes up to six sides
- say what stays the same when a shape is turned



Crates stack and tins roll. Bramble sorts the shelf by what each solid will do.

Solid shapes

WAKE UP
YOUR
COUNTING

Pick up something solid from your table. Run one finger along an edge, then lay your whole hand flat on a face.

♣ Faces you can rest, edges you can feel

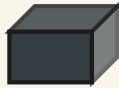
A solid shape has **faces**, the parts you can lay a hand on, and **edges**, the lines where two faces meet.

A solid with only **flat** faces will **stack** and stay put. A solid with a **curved** face will **roll**.

A cylinder does both: it stacks on its flat ends, and it rolls on its curved side.



cube



cuboid



cylinder



sphere

 Watch me do one 1

How many faces has a cube?

- 1** Stand the cube on the table and count the face on the top.
- 2** Turn it round and count each face at the side: four of them.
- 3** Now count the face resting on the table. Add them all up.

Answer: a cube has **6** faces

How do you know?

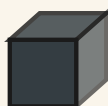
Hattie says a sphere will stack on top of a sphere. Is she right? Say how you know.

Rolls and stacks



edge

1 Count the faces on this cube. Write how many.



2 Count the edges on this cuboid. Write how many.



3 Tick every solid here that will stack and stay put.



4 Ring the solid that has no flat face at all.



5 Sort these solids into stacks and rolls.

The material you need: a wooden brick, a tin, a ball and a small box

TOPIC 10.1 · YOUR TURN · CONTINUED

these stack

these roll



edge

6 Roll a tin and a ball. Say which one rolls straight.

The material you need: one tin and one ball from the store shelf

7 Draw one solid that stacks. Draw one that rolls.

A large dashed rectangular box for drawing.

Counting faces



8 Write how many edges this pyramid has.



9 Count the flat faces on this cylinder. Write how many.



10 Match each solid to the number of faces it has.

TOPIC 10.1 • YOUR TURN • CONTINUED



2



6



1

11 Draw round one face of a block. Name the flat shape.

The material you need: one wooden block and a sharp pencil

12 Write the name of a solid you can see on the shelf.



edge

13 Say how a cube and a cuboid are different.

Use the words face and edge.

🔨 Burrow challenge

Choose a solid but keep it secret. Let a partner ask three questions about its faces, then guess it.

Flat shapes



Tessa lines up her card shapes on the floorboards, then turns one of them right round to be sure.

♣ Sides, corners, and turning a shape

A flat shape has **sides**, its straight edges, and **corners**, the points where two sides meet.

A circle is the odd one out: it has one **curved** side and no corners at all.

If you **turn** a shape round, it is still the same shape. It has the same sides and the same corners, only they now point a different way.

TOPIC 10.2 · CONTINUED



triangle



square



pentagon



hexagon



 Watch me do one 1

How many sides has this hexagon?

- 1** Put your pencil tip on one corner of the hexagon.
- 2** Move along each side in turn, counting as you go.
- 3** Stop when your pencil is back where it began.

Answer: a hexagon has **6** sides

How do you know?

Milo turns a square halfway round and says it became a new shape. Is he right? Say how you know.

Sides and corners



1 Count the sides of each shape. Write how many.



2 Count the corners of this hexagon. Write how many.



TOPIC 10.2 · YOUR TURN · CONTINUED

3 Tick every shape that has three straight sides.



4 Ring the shape that has no corners at all.

circle

square

pentagon

5 Turn this shape halfway round. Draw where it lands.



TOPIC 10.2 · YOUR TURN · CONTINUED



6 Sort your card shapes into curved sides and straight sides.

The material you need: a set of card shapes and two paper trays

curved sides

Three horizontal dashed lines for writing.

straight sides

Three horizontal dashed lines for writing.

7 Draw a shape with six sides. Write its name.

A large dashed rectangular box for drawing and writing.

Turning a shape



8 Ring the two shapes that match after one turn.



9 Match each shape to its number of corners.



3



8



4

10 Write the name of this shape on the line.



A horizontal dashed line for writing the name of the shape.

11 Find two flat shapes in the schoolroom. Draw them.

A large horizontal dashed rectangle for drawing two flat shapes.

12 Make a repeating pattern with a square and a triangle.

The material you need: six square tiles and six triangle tiles



13 Say how you know a shape has been turned, not swapped.

→ **Try a bigger one**

Turn a rectangle a quarter turn. Does it still fit the same hole?
Try it with card and scissors.

The words of this unit

▣ BURROW WORDS



face

a part you can lay a hand on



edge

where two faces meet



side

one straight edge of a flat shape



curved

bent, with no straight part



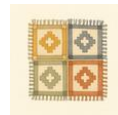
straight

not bent at all



turn

the shape stays, it points elsewhere



pattern

the same thing happening again



same

no difference at all

▣ Check what you can do

- 1 Count the faces of a cube and write the number.
- 2 Say which of these solids will roll, and why.
- 3 Count the corners of a pentagon and write the number.
- 4 Name a flat shape that has no straight sides.
- 5 Turn a card triangle round and say what stayed the same.

Now I can



- ☐ count the faces and the edges of a solid shape
- ☐ say which solids roll and which solids stack
- ☐ count the sides and corners of a flat shape
- ☐ name the flat shapes up to six sides
- ☐ say what stays the same when a shape is turned

UNIT 10 · HOW DID IT GO · CONTINUED

	confident	almost there	needs more work
counting faces and edges	☐	☐	☐
deciding what rolls and what stacks	☐	☐	☐
counting sides and corners	☐	☐	☐
naming a flat shape from its sides	☐	☐	☐
turning a shape and saying what stayed the same	☐	☐	☐



Shapes never stop. Later you will fold them flat, and open them out again.

Which one does not belong?

Tessa has cut five shapes out of blue paper and laid them on the table: three squares, one triangle and one circle. She asks which one does not belong with the others. Otto points at one shape. Hattie points at a different shape. They both have a good reason, and this is the surprising part: in this project they can both be right.



Five flat paper shapes on the table: three squares, a triangle and a circle.

& The material you need

five shapes cut from card: three squares, one triangle and one circle, plus a pencil

- 1 Lay your five shapes out in a row on the table.
- 2 Point at the shape you think does not belong.
- 3 Tell your partner the reason for your choice.
- 4 Count the straight sides of every shape.
- 5 Now find a different shape that could be the odd one out.

How do you know?

Otto and Hattie chose different shapes and both gave a good reason. How can two different answers both be right?

Halves of sets

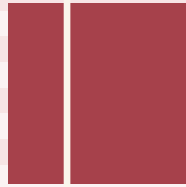
Sharing a group of things into two equal parts, and knowing which numbers will share fairly and which will not.



Open the door

Before you begin, show what you know already.

Tick the shape that is cut into two equal parts.



☀ In the burrow today

Bramble has six ripe apples on the counter and two paper bags.

Otto points at each apple and counts them, one, two, three, four, five, six. Hattie looks and says three will go in each bag. Tessa lays the apples out in two lines to be sure.

Milo asks the useful question: **how do you know the two bags are the same?** A half is only a half when the two parts match exactly.

• CONTINUED

✓ By the end you will

- find half of a set of objects
- share a quantity equally between two
- check that the two parts really are equal
- tell half of a whole shape from half of a group
- say which numbers halve into two equal whole groups



Bramble shares six apples between two paper bags. Otto counts one bag to find half.

Halves of sets

WAKE UP
YOUR
COUNTING

Count out eight acorns. Deal them one at a time, one to Otto, one to Hattie, one to Otto, one to Hattie, until none are left.

✦ Half of a set is one of two equal parts

To find **half** of a group, share it into **two equal parts**. One of those parts is one half.

Half of a whole shape is one of two equal pieces. Half of a group is one of two equal piles. The word is the same because the idea is the same: two parts, exactly equal, nothing left over.



Six apples dealt out fairly. Count one side only.

Watch me do one 1

What is half of six apples?

- 1** Count all six apples. Point at each one as you count it.
- 2** Deal them out one at a time, one to this side, one to that side, until none are left.
- 3** Count one side only. Check the other side has the same number.

Answer: half of 6 apples is 3 apples

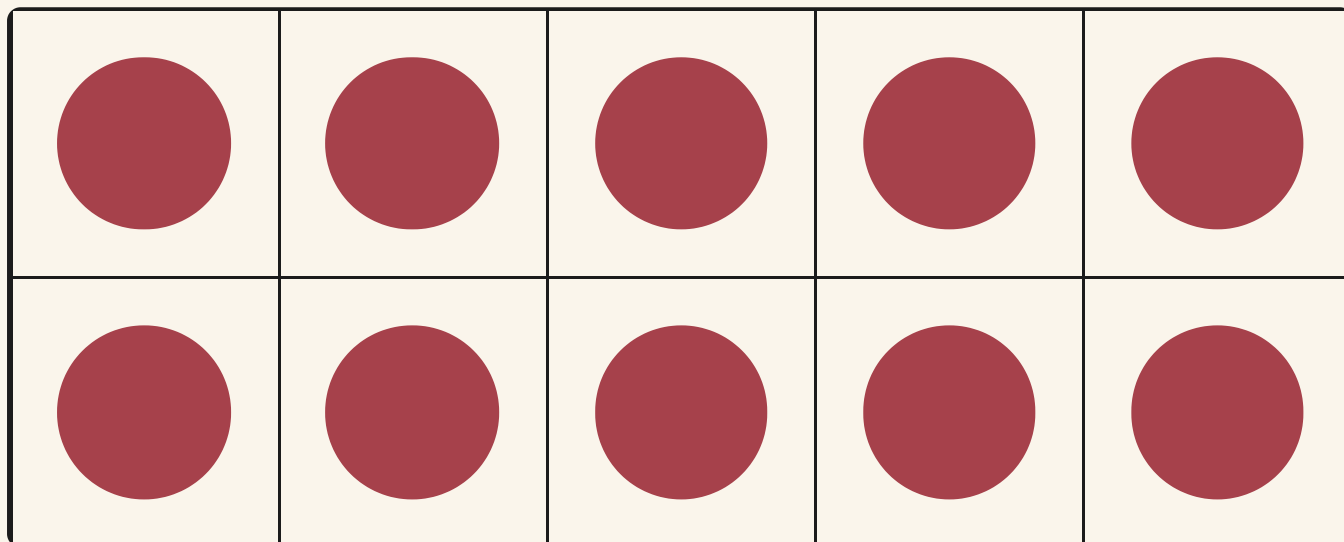
How do you know?

Hattie put four cherries on one plate and two on the other. Is that half? Say how you know.

Half of a group



1 Ring half of these ten counters.



ten counters

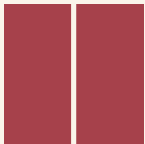
2 Write half of ten.

3 Share eight acorns equally between two bowls.

The material you need: eight acorns and two small bowls

4 Draw your two equal groups here.

5 Tick the shape cut into two equal halves.



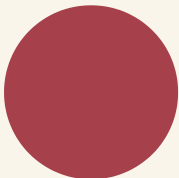
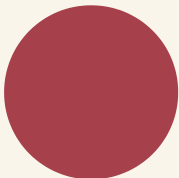
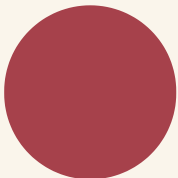
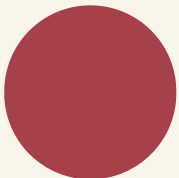
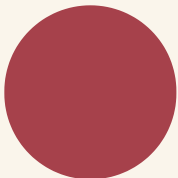
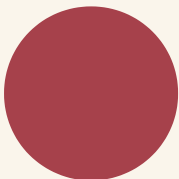
TOPIC 11.1 • YOUR TURN • CONTINUED



share

6 Write half of four.

7 Ring half of these six counters.

Which numbers halve



8 Tick every set that shares into two equal groups.

●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
●										●	●	●		

9 Cross out the number that will not halve fairly.

4	7	10
---	---	----

TOPIC 11.1 • YOUR TURN • CONTINUED

10 Match each number to half of it.

four



three

six



five

ten



two

11 Write half of eight, then half of two.

--	--

12 Sort ten counters into two equal trays.

The material you need: ten counters and two trays

one half

the other half

TOPIC 11.1 • YOUR TURN • CONTINUED



share

13 Write how many counters are in one tray.

14 Halve twelve beans with a partner. Say how you know.

The material you need: twelve dried beans and two saucers

→ **Try a bigger one**

Try halving twenty acorns. Then try halving nineteen. What happens, and why?

The words of this unit

▣ BURROW WORDS



half

one of two equal parts



whole

all of it, nothing missing



equal

exactly the same amount



share

give out fairly



pair

two that go together



double

two of the same amount



group

things gathered together



same

no difference at all

▣ Check what you can do

- 1 Ring half of the acorns in this picture.
- 2 Write half of twelve.
- 3 Colour one half of this rectangle.
- 4 Draw a set of eight, then split it fairly.
- 5 Explain to Otto why five will not halve fairly.

Now I can



- ☐ find half of a set of objects
- ☐ share a quantity equally between two
- ☐ check that my two parts really are equal
- ☐ tell half of a whole shape from half of a group
- ☐ say which numbers halve into two equal whole groups

UNIT 11 • HOW DID IT GO • CONTINUED

	confident	almost there	needs more work
sharing a group into two equal parts	☐	☐	☐
counting one half to find the answer	☐	☐	☐
checking that both parts are equal	☐	☐	☐
halving a whole shape	☐	☐	☐
saying which numbers will not halve fairly	☐	☐	☐



Halving is sharing you can check. In Unit 12 the balance scales do the checking for you.

Eight blackberries



Bramble shares a plate of eight blackberries between Otto and Milo, four moving to each side.

Bramble has picked eight blackberries from the hedge by the meadow, and Otto and Milo are both waiting by the table. Bramble wants the two plates to be **fair**, so that nobody can grumble afterwards. Fair does not mean nearly the same. It means exactly the same, with none left over in the middle.

& The material you need

eight blackberries or eight counters, two small plates and a partner to share with

- 1 Lay the eight blackberries out in a row on the table.
- 2 Put one on the first plate, then one on the second plate.
- 3 Keep going, one plate then the other, until the row is empty.
- 4 Count what is sitting on each plate now.
- 5 Draw the two plates and what you put on them.

How do you know?

Both plates hold the same amount. Say how you know the sharing was fair, without counting the plates again.

Heavy, light, full, empty

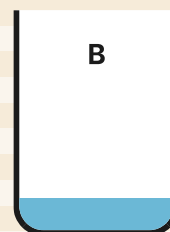
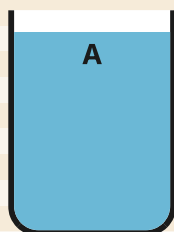
Weighing, filling and comparing: heavier and lighter, full and empty, and choosing the tool that fits the question.



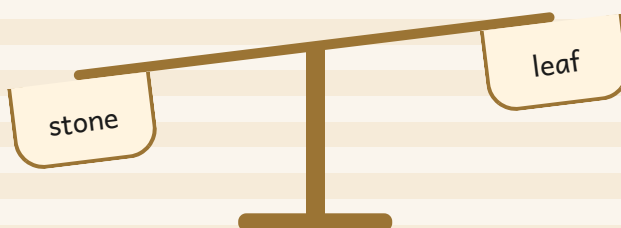
Open the door

Before you begin, show what you know already.

Tick the fuller jug.



Tick the pan that went down.



☀ In the burrow today

Bramble has set the brass balance scales on the store counter.

A basket of potatoes sits in one pan and a basket of dry leaves sits in the other. The potato pan swings low and stays there. Otto points at each pan in turn. Tessa lines the baskets up so nothing can roll away.

Beside the scales stand a tall jug and a small cup. Hattie guesses the jug holds more, then Bramble pours to check. Milo asks: **how do you know?**

• CONTINUED

✓ By the end you will

- say which of two things is heavier and which is lighter
- read balance scales and name the heavier pan
- say whether a container is full, part full or empty
- say which container holds more
- choose the right tool for measuring mass, length or capacity



The potato basket pulls its pan low while the leaves rise. The lower pan holds the heavier load.

Heavier, lighter, full, empty

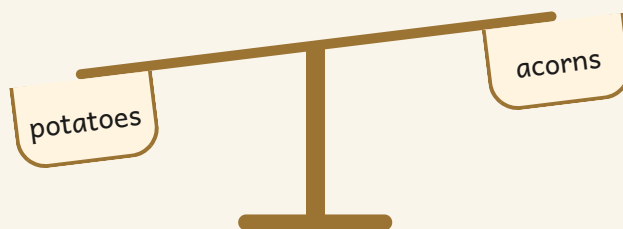
WAKE UP
YOUR
COUNTING

Hold a stone in one hand and a dry leaf in the other. Close your eyes. Which hand can feel the pull?

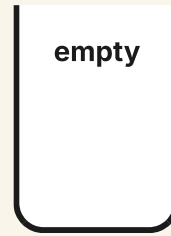
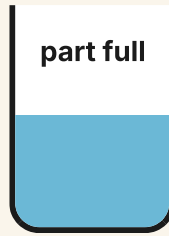
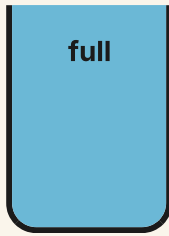
♣ The lower pan holds the heavier thing

Balance scales have two pans. Put one thing in each pan. The **heavier** thing pulls its pan down. The **lighter** thing lets its pan rise. If the pans sit level, the two things weigh the same.

A container that will take no more is **full**. A container with nothing in it is **empty**. The container that **holds more** is the one that takes more cups to fill.



TOPIC 12.1 • CONTINUED



Watch me do one 1

Which basket is heavier, the potatoes or the acorns?

- 1 Put one basket in each pan of the balance scales.
- 2 Let the pans settle. Do not touch them while they move.
- 3 Look for the pan that sits low. That pan holds the heavier basket.

Answer: the first basket, the potatoes, because its pan went down

How do you know?

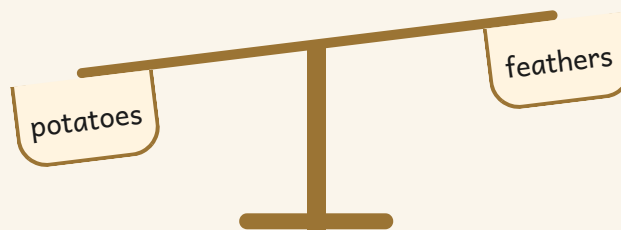
Milo says the bigger box must be the heavier one. Is he right? Say how you know.

Heavier and lighter



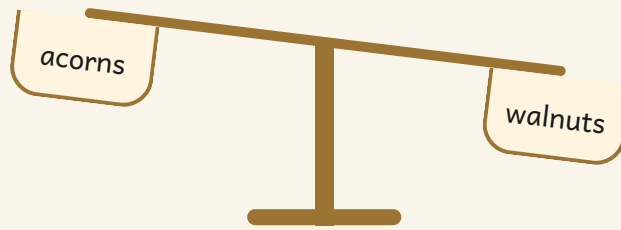
balance

1 Tick the heavier basket.

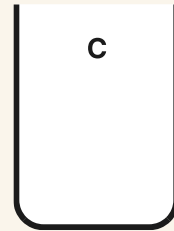
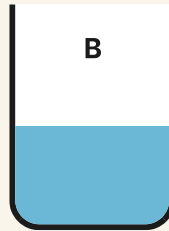
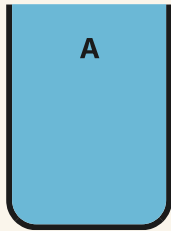


2 Tick the lighter jar.

TOPIC 12.1 · YOUR TURN · CONTINUED



3 Tick the full jug. Cross the empty jug.



4 Write the letter of the jug holding the most water.

A dashed rectangular box for writing the answer.

5 Hold a stone and a leaf. Ring the heavier one.

The material you need: a small stone and a dry leaf

TOPIC 12.1 · YOUR TURN · CONTINUED

the stone

the leaf

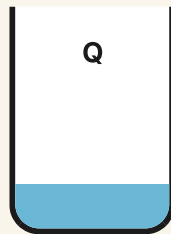
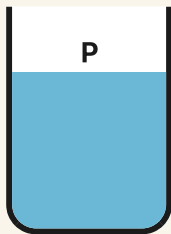


balance

6 Pour one cup from a full jug. Draw what is left.

The material you need: a jug of water, a cup and a tray

7 Order these jugs from emptiest to fullest.



Weighing and filling



balance

8 Order these baskets from lightest to heaviest.

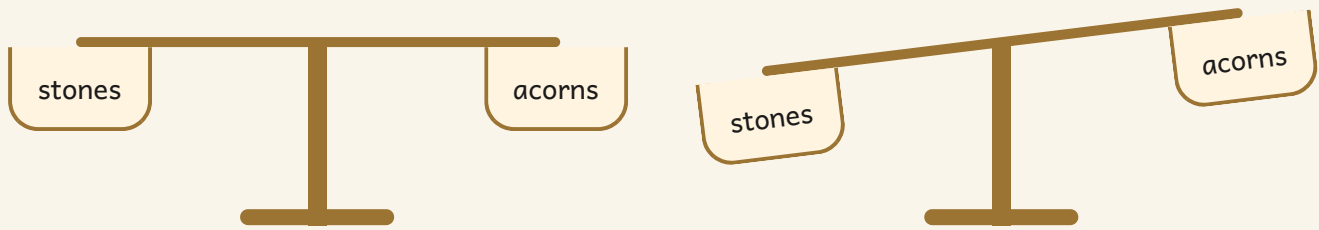
a basket of feathers

a basket of acorns

a basket of potatoes

9 Tick the pans that are level.

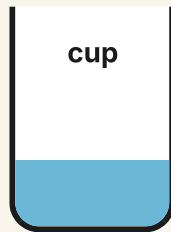
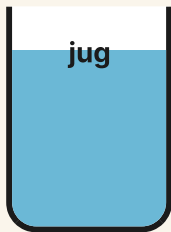
TOPIC 12.1 · YOUR TURN · CONTINUED



10 Balance a pencil against acorns. Write how many acorns.

The material you need: a pencil, ten acorns and balance scales

11 Tick the container that holds the most water.



- 12** Pour one full cup into the jug. Mark the water level.

The material you need: a cup, a clear jug and a wax pencil



balance

- 13** Find something heavier than your shoe. Draw it.

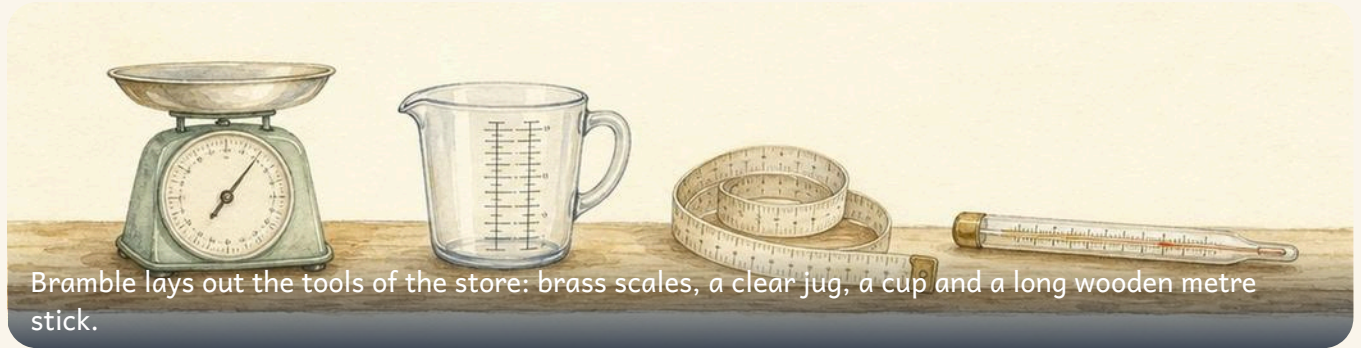
The material you need: balance scales, your shoe and things from the shelf

- 14** Tell Milo how you know which jar is heavier.

★ **Pip's find**

A balance never tells you how many grams a thing weighs. It only tells you which side is heavier, and that is often all you need.

How do we measure?



Bramble lays out the tools of the store: brass scales, a clear jug, a cup and a long wooden metre stick.

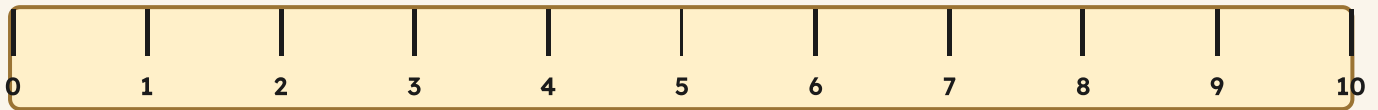
♣ Choose the tool that fits the question

Every question has its own tool. To find out which is heavier, use **balance scales**. To find out how long, use a **ruler**. To find out how much a jug holds, fill it with **cups**.

Hands and cups work, but my hand is not the same size as yours, so our numbers disagree. That is why everybody uses the same units: the **centimetre** and the **metre** for length, the **litre** for how much a jug holds, and the **kilogram** for mass.

ribbon

a ruler in centimetres



 **Watch me do one 1**

How long is this ribbon?

- 1 Lay the ribbon along the ruler. Line its end up with zero.
- 2 Keep it straight and flat. A curled ribbon reads too short.
- 3 Read the number the far end reaches, then say the unit.

Answer: the ribbon is **6** centimetres long

How do you know?

Tessa measured a ribbon starting at one, not at zero. Say how you know her answer is wrong.

Choosing the tool



full

1 Ring the tool for weighing a bag of walnuts.

balance scales

a ruler

a jug

2 Ring the tool for measuring a ribbon.

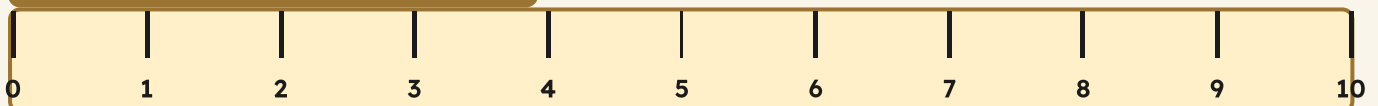
a jug

a ruler

balance scales

3 Measure this bar. Write how many centimetres.

bar



4 Measure your table with your hands. Write how many hands.

The material you need: your two hands and a table

--	--

5 Measure the same table with a metre stick.

The material you need: a wooden metre stick

--



full

6 Write why two people count different numbers of hands.

--

7 Fill the jug using cups. Write how many cups.

The material you need: a cup, a clear jug and water

--	--

Units everyone shares



full

8 Order these ribbons from shortest to longest.

ribbon A

ribbon B

ribbon C

9 Match each thing to the unit you would use.

a ribbon



a jug of water



a bag of flour



litres



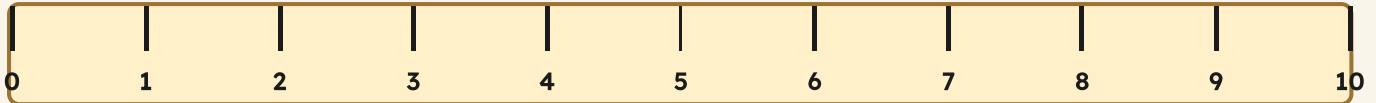
kilograms



centimetres

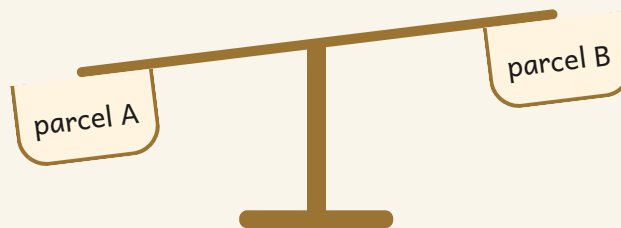
TOPIC 12.2 · YOUR TURN · CONTINUED

10 Draw a line exactly five centimetres long.



A dashed rectangular box for drawing a line.

11 Tick the heavier parcel on these scales.



12 Weigh two potatoes on Bramble's scales. Write the mass.

The material you need: two potatoes and a kitchen scale marked in kilograms

Two dashed rectangular boxes for writing the mass of the potatoes.

TOPIC 12.2 · YOUR TURN · CONTINUED



full

13 Measure a shelf in centimetres, then check with a partner.

The material you need: a tape measure and a shelf

14 Explain to Milo why everybody should use the same unit.

→ **Try a bigger one**

Measure the same shelf in hands, in pencils and in centimetres.
Which number could you post to a friend?

The words of this unit

▣ BURROW WORDS



heavy

hard to lift, pulls the pan down



light

easy to lift, the pan rises



full

it will take no more



empty

nothing inside at all



balance

two pans that sit level



long

it reaches a long way



short

it does not reach far



order

put in place, smallest first

▣ Check what you can do

- 1 Tick the basket whose pan went down.
- 2 Ring the emptiest of these three jugs.
- 3 Name the tool for weighing an apple.
- 4 Write the length of this line in centimetres.
- 5 Say why a metre stick beats counting hands.

Now I can



- ☐ say which of two things is heavier and which is lighter
- ☐ read balance scales and name the lower pan
- ☐ say whether a container is full, part full or empty
- ☐ say which container holds more
- ☐ choose the right tool, and measure in centimetres, litres and kilograms

UNIT 12 • HOW DID IT GO • CONTINUED

	confident	almost there	needs more work
comparing two things by mass	☐	☐	☐
reading balance scales	☐	☐	☐
using the words full and empty	☐	☐	☐
choosing the right measuring tool	☐	☐	☐
measuring in centimetres and litres	☐	☐	☐



Measuring always begins by comparing two things and saying which is more. The units come after.

Working with numbers to 20

Counting on, counting back, hopping along the number line with Pip, and paying for things at Bramble's store.



Open the door

Before you begin, show what you know already.

●	●	●	●	●	●	●	●	●	
●	●	●	●	●					

Write how many. Then write one more.

☀ In the burrow today

Bramble has opened the store for the morning.

There are fourteen acorns in the tray and a jar of coins on the counter. Otto wants three more acorns. Hattie wants to know what is left if she takes five away.

Pip lands on the long number strip along the wall. Every hop Pip makes is worth exactly one, forwards or backwards.

• CONTINUED

✓ By the end you will

- add by counting on from the bigger number
- subtract by counting back
- use a number line to add and to take away
- name the euro coins and count a small total
- say how you know your answer is right



Hattie counts on along the strip while Pip hops forwards. One hop is one.

Addition by counting on

WAKE UP
YOUR
COUNTING

Start at eleven and count on three, out loud: twelve, thirteen, fourteen. Now start at fifteen and count on two.

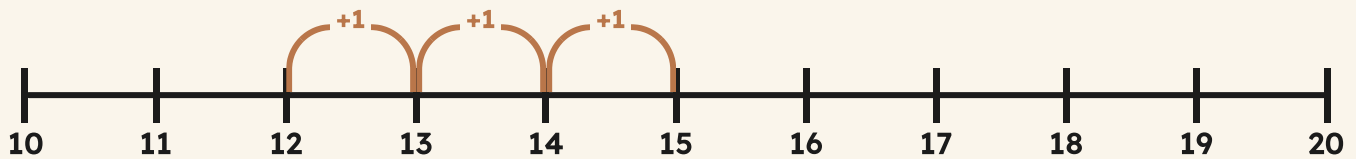
✦ Start at the bigger number and count on

You do not have to start at one. To add, hold the **bigger** number in your head, then count on the smaller one.

Adding means hopping **forwards** along the line above. Each hop is one.

TOPIC 13.1 • CONTINUED

start at 12, count on 3



Watch me do one 1

What is 12 add 3?

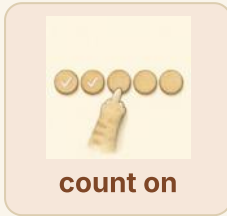
- 1** Hold twelve in your head. Do not count up to it.
- 2** Hop forwards three times: thirteen, fourteen, fifteen.
- 3** You landed on fifteen, so twelve add three is fifteen.

Answer: $12 + 3 = 15$

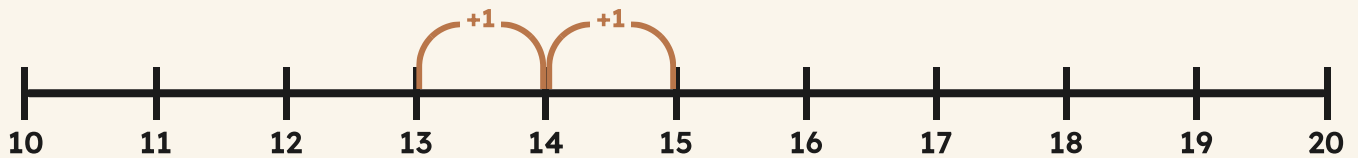
How do you know?

Otto counted on from the smaller number instead and got the same answer. Why does that happen?

Count on to add

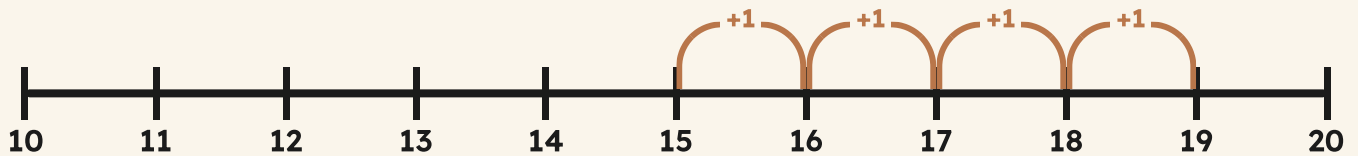


1 Count on along the line. Write the answer.



$$13 + 2 = \boxed{}$$

2 Count on along the line. Write the answer.



$$15 + 4 = \boxed{}$$

3 Write the answers. Count on from the bigger number.

TOPIC 13.1 • YOUR TURN • CONTINUED

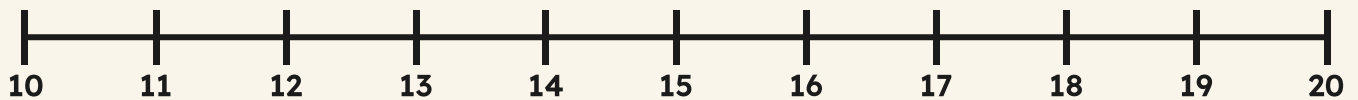
$11 + 3 = \boxed{}$

$14 + 2 = \boxed{}$

$16 + 3 = \boxed{}$

$12 + 5 = \boxed{}$

4 Draw the hops on the line, then write the answer.



$17 + 2 = \boxed{}$

5 Use counters to add. Write both answers.

The material you need: twenty counters and a ten frame

$13 + 4 = \boxed{}$

$15 + 5 = \boxed{}$



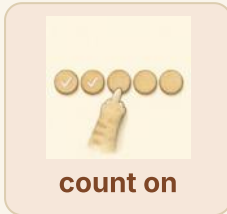
count on

6 Ring the number you would hold in your head first.

3

16

Adding at the store



7 Count the acorns. Add three more. Write the total.

●	●	●	●	●	●	●			
●	●	●	●	●					

8 Write the missing number in each sentence.

$$10 + 6 = \boxed{}$$

$$11 + 8 = \boxed{}$$

9 Add two to every number on this track.

TOPIC 13.1 • YOUR TURN • CONTINUED

10

11

12

13

14

15

16

10 Write two sentences that both make eighteen.

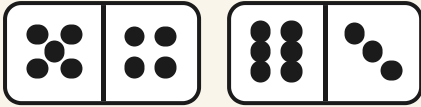
11 Count on five from thirteen. Write where you land.



count on

12 Add the two dice. Write the total.

TOPIC 13.1 • YOUR TURN • CONTINUED



✱ **Pip's find**

Counting on works because a number already holds everything below it. Twelve is ten and two, so you never need to start again.

Subtraction by counting back



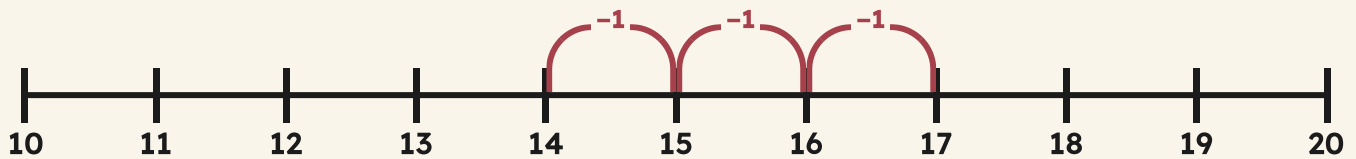
✦ Taking away means hopping backwards

To take away, start at the number you have and count **back**. Each hop backwards is one less.

The number you land on is what is left.

TOPIC 13.2 · CONTINUED

start at 17, count back 3



 **Watch me do one 1**

What is 17 take away 3?

- 1** Start at seventeen on the line.
- 2** Hop backwards three times: sixteen, fifteen, fourteen.
- 3** You landed on fourteen, so seventeen take away three is fourteen.

Answer: $17 - 3 = 14$

How do you know?

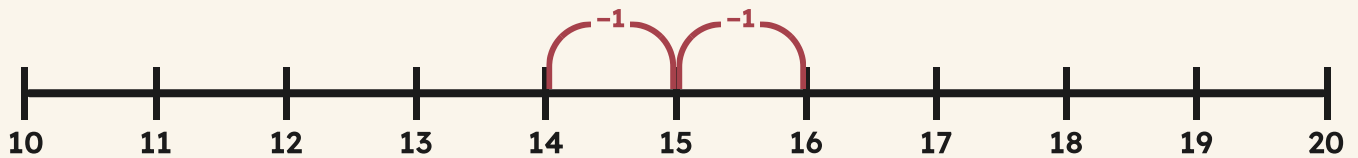
Milo says counting back is just counting on the wrong way. Is he right? Show him on the line.

Count back to take away



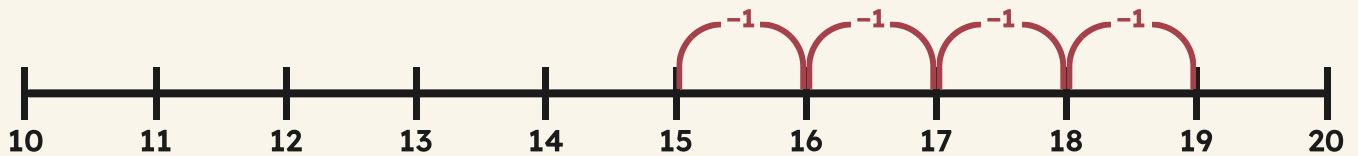
track

1 Count back along the line. Write the answer.



$$16 - 2 = \boxed{}$$

2 Count back along the line. Write the answer.



$$19 - 4 = \boxed{}$$

3 Write the answers. Count back each time.

$$15 - 3 = \boxed{}$$

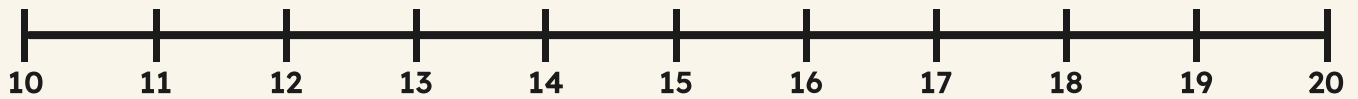
$$18 - 2 = \boxed{}$$

$$14 - 4 = \boxed{}$$

$$20 - 5 = \boxed{}$$

TOPIC 13.2 · YOUR TURN · CONTINUED

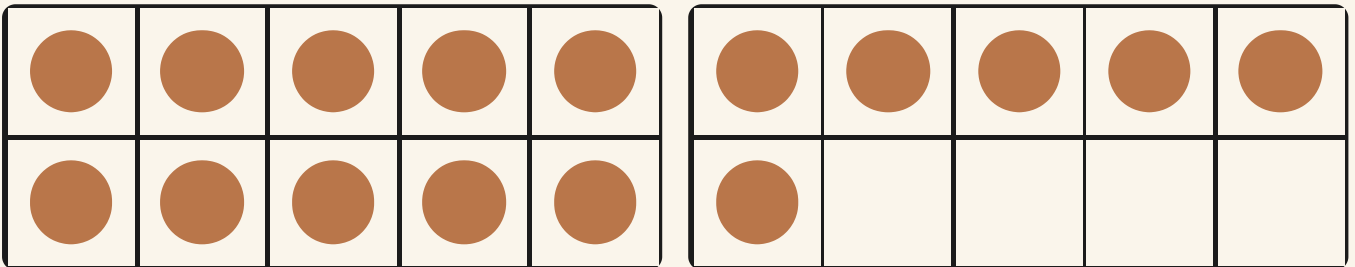
4 Draw the hops backwards, then write the answer.



$13 - 3 = \boxed{}$

5 Take three acorns away. Write what is left.

The material you need: twenty counters



track

6 Ring the number you start on when you count back.

4

18

What is left?



track

7 Write the missing number in each sentence.

$$17 - 6 = \boxed{}$$

$$20 - 8 = \boxed{}$$

8 Take two away from every number on this track.

12

13

14

15

16

17

18

9 Write two sentences that both leave eleven.

10 Count back six from eighteen. Write where you land.

11 Tick the sentence that leaves the most.

$19 - 2 = \square$

$19 - 7 = \square$



track

12 Explain counting back to a partner. Use the line.

→ Try a bigger one

Start at twenty and count back in twos. Which numbers do you land on? Are they odd or even?

Using the number line

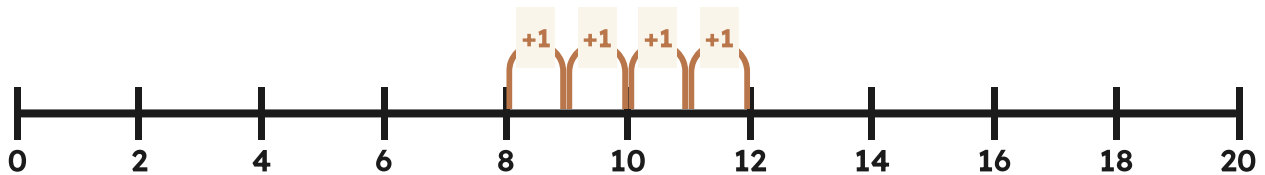


♣ One line, both jobs

The **number line** does adding and taking away with the same picture. Forwards is more. Backwards is less.

Always say where you started and how many hops you made.

8 and 4 more



 Watch me do one 1

Otto is on nine and hops forwards five. Where does he land?

- 1** Find nine on the line and put your finger on it.
 - 2** Hop forwards five times, counting each hop out loud.
 - 3** Your finger is on fourteen.
-

Answer: Otto lands on **14**

How do you know?

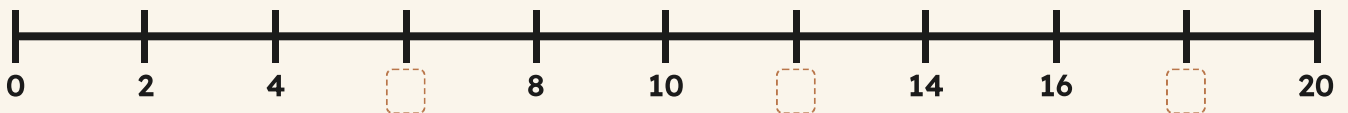
Two children start on different numbers and both hop three forwards. Do they land on the same number? Say why.

Hopping on the line

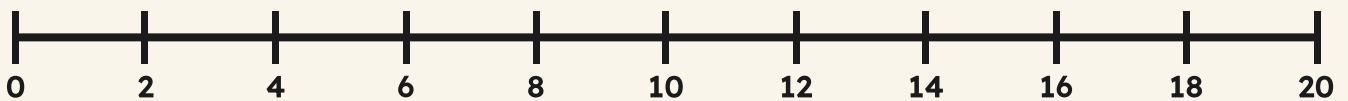


track

1 Write the missing numbers on the number line.

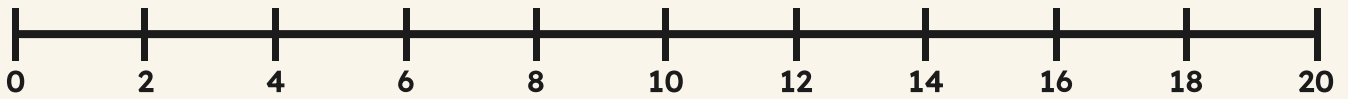


2 Start on seven. Hop forwards four. Write where you land.



3 Start on fifteen. Hop back six. Write where you land.

TOPIC 13.3 · YOUR TURN · CONTINUED

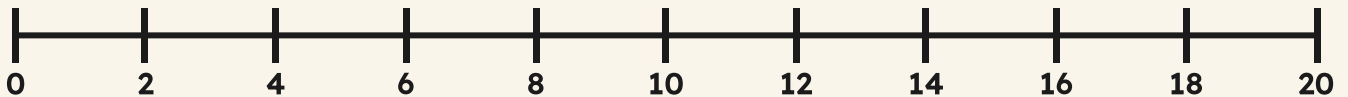


4 Write the answers using the number line.

$9 + 6 = \boxed{}$

$16 - 7 = \boxed{}$

5 Draw a hop that lands exactly on twenty.



track

6 Ring the direction you hop when you add.

forwards

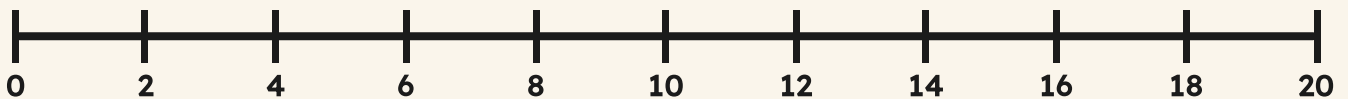
backwards

Reading the line



track

7 Write which number is halfway between zero and twenty.



8 Ring the bigger number on the line.

11

17

9 Order these numbers from smallest to largest.

16

9

20

13

smallest to largest

--	--	--	--

10 Write the number that is two more than eighteen.

--

11 Write the number that is three less than fifteen.

--



track

12 Make up a hop for a partner to solve on the line.

The material you need: a counting line marked from zero to twenty

Burrow challenge

Pip made four hops and landed on twelve. Where could Pip have started? Find two different answers.

Money



♣ A coin says how much it is worth

Coins are not all worth the same. The biggest coin is not always worth the most, so you must read the number on it.

To find a total, count the coins on and add them up.



TOPIC 13.4 · CONTINUED

ten, then fifteen, then seventeen



acorn
8c



apple
12c



leaf
5c

Watch me do one 1

How much are these three coins worth?

- 1 Start with the biggest coin: ten cents.
- 2 Count on the five: that makes fifteen cents.
- 3 Count on the two: that makes seventeen cents.

Answer: 17c altogether

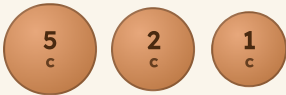
How do you know?

Hattie says a five cent coin must be worth more than a ten because it is bigger. Show her why she is wrong.

Counting coins



1 Count these coins. Write the total.



2 Count these coins. Write the total.



3 Ring the coin that is worth the most.

TOPIC 13.4 · YOUR TURN · CONTINUED



4 Write which two coins make twelve cents.

5 Tick the price you could pay with one coin.



ribbon
20c



basket
13c



coin

6 Sort the coins into copper and gold.

The material you need: a handful of one, two, five, ten and twenty cent coins

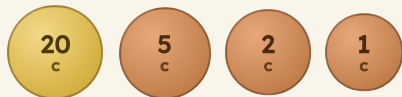
copper

gold

Paying at the store



7 Write the total of these coins.



8 Order these prices from cheapest to dearest.

jug
14c

button
6c

cup
11c

cheapest to dearest

TOPIC 13.4 · YOUR TURN · CONTINUED

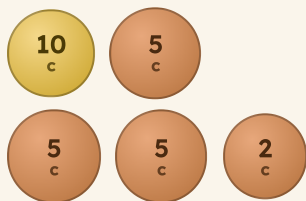
9 Draw the coins you would use to pay seven cents.



10 Write how much two ten cent coins are worth.



11 Tick the set of coins worth more.



TOPIC 13.4 · YOUR TURN · CONTINUED



coin

12 Play shopkeeper. Take turns paying and checking.

The material you need: coins and five priced objects

★ Pip's find

Every euro coin has the same value in every country that uses it, but each country draws its own picture on the back.

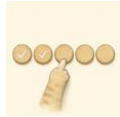
The words of this unit

▣ BURROW WORDS



total

how many altogether



count on

start big, then hop forwards



number line

a line with the numbers in order



track

numbered squares in a row



coin

money you can hold



note

paper money



price

what a thing costs



double

two of the same amount

▣ Check what you can do

- 1 Count on four from thirteen. Write where you land.
- 2 Count back five from nineteen. Write what is left.
- 3 Fill the gaps along this counting line.
- 4 Add up this handful of coins.
- 5 Write which coin is worth the most.

Now I can



- ☐ add by counting on from the bigger number
- ☐ subtract by counting back
- ☐ use a number line for adding and taking away
- ☐ name the euro coins and count a small total
- ☐ say how I know my answer is right

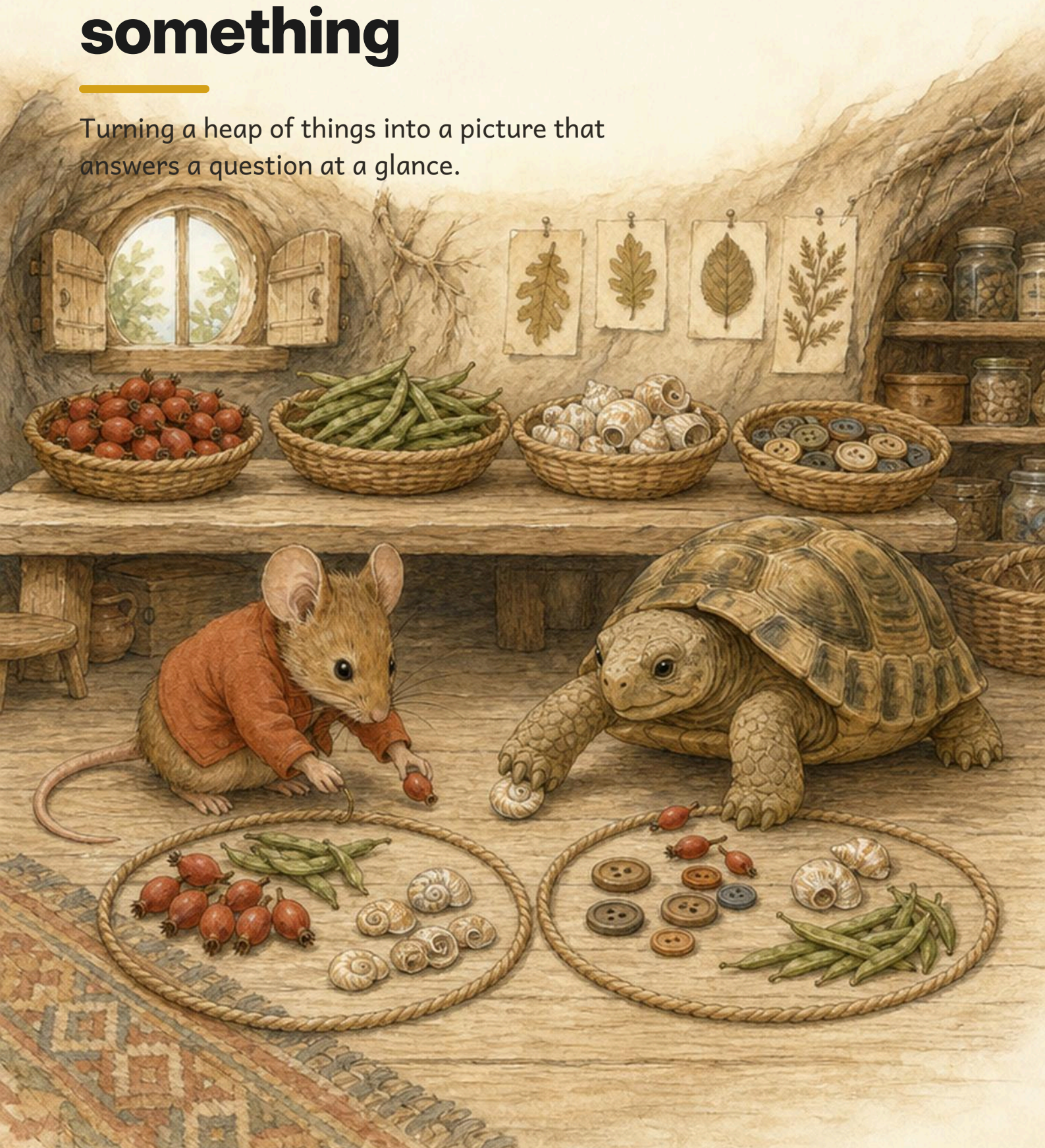
	confident	almost there	needs more work
counting on to add	☐	☐	☐
counting back to take away	☐	☐	☐
reading and using the number line	☐	☐	☐
counting a total of coins	☐	☐	☐
explaining how I know	☐	☐	☐



One line does both jobs. Forwards is more, backwards is less.

Charts that tell you something

Turning a heap of things into a picture that answers a question at a glance.



Open the door

Before you begin, show what you know already.

Count each kind and write how many.

What we found	Tally	How many
acorns		
shells		
buttons		
pebbles		

☀ In the burrow today

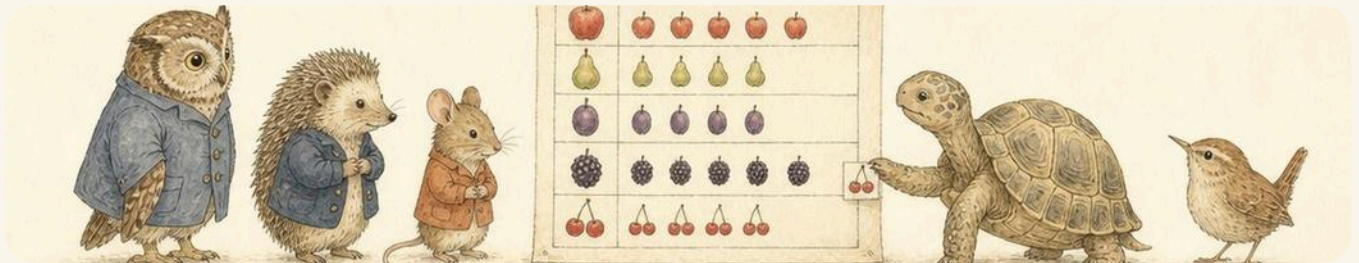
Tessa has emptied the whole sorting shelf onto the table.

Acorns, shells, buttons and pebbles, all in one heap. Otto asks a simple question: **which did we find most of?**

Counting the heap again and again is slow. So Tessa draws a picture of the counting instead, and the answer is there to see.

✓ **By the end you will**

- make a tally while you count
- read a pictogram where one picture means one
- read and draw a block chart
- sort into two rings and into a table
- answer a question by looking at a chart



Tessa adds one more picture to the chart. Each picture stands for one thing she found.

Rings, tables and pictograms

WAKE UP
YOUR
COUNTING

Count how many children in your group are wearing something green. Hold up that many fingers all together.

✦ A picture that answers a question

A **pictogram** uses one picture for one thing. The longest row is the one there was most of.

You do not have to count again. The picture already did.

What we found on the shelf

acorns



shells



buttons



one picture means one thing found

 Watch me do one 1

Which did Tessa find most of?

- 1 Look at the rows and find the longest one.
 - 2 The buttons row is longest.
 - 3 Check by counting that row: eight buttons.
-

Answer: buttons, and there were **8** of them

How do you know?

Milo says the shells row must be biggest because shells are bigger.
Show him why the row length is what matters.

Reading the picture

acorns



shells



buttons



 one picture means one



data

1 Write how many shells there were.

2 Ring the row that has the fewest.

acorns

shells

buttons

3 Write how many acorns and shells there were altogether.

4 Draw one more shell on the chart. Write the new count.

5 Sort these two ways. Tick one box for each thing.

	wooden	not wooden
round		
not round		



data

6 Write one thing that is round and wooden.

Make your own tally



data

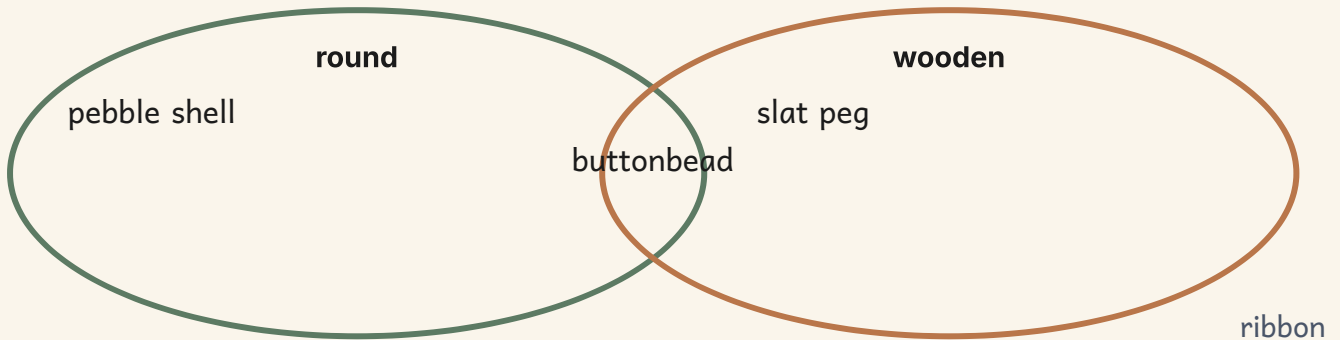
7 Make a tally as you count. Then write how many.
The material you need: a tray of mixed buttons, shells and pebbles

What we found	Tally	How many
buttons		
shells		
pebbles		

8 Write which kind you found most of.

9 Write which kind you found fewest of.

10 Sort your things into two overlapping rings.



11 Write how many things sat in the middle.



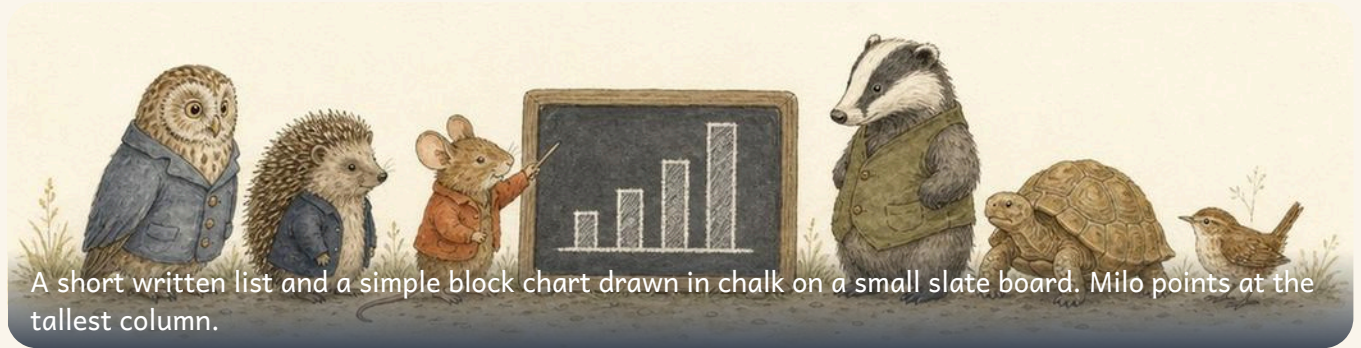
data

12 Ask a partner one question about your tally.

★ **Pip's find**

Tally marks are grouped in fives because a hand has five fingers. It is the oldest way of counting without writing numbers.

Lists, tables and block charts

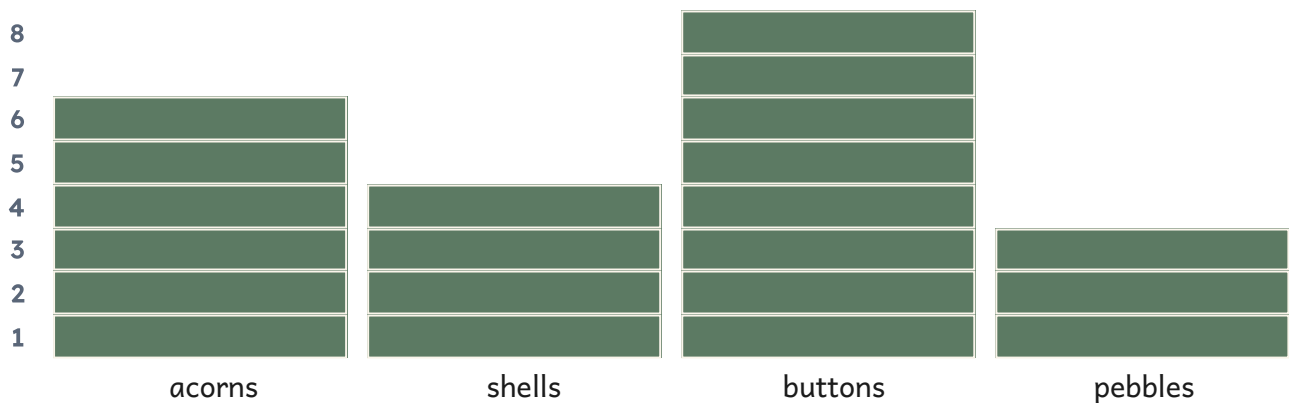


♣ A block chart shows the answer by height

A **block chart** uses one block for one thing, so the **tallest** column is the one there was most of.

A **list** writes things one under another. A **table** puts them in rows and columns so you can read across.

What we found on the shelf



 Watch me do one 1

How many more buttons than pebbles were there?

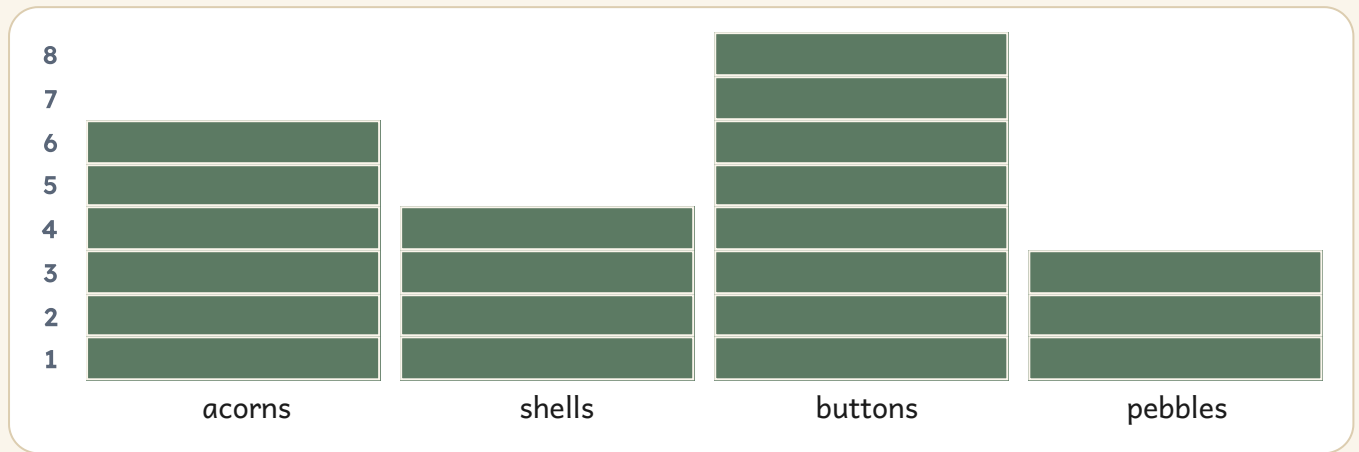
- 1 Read the buttons column: it is eight blocks tall.
 - 2 Read the pebbles column: it is three blocks tall.
 - 3 Count up from three to eight: that is five more.
-

Answer: 5 more buttons

How do you know?

Two columns are exactly the same height. What does that tell you without counting anything?

Reading a block chart



1 Write how many buttons there were.

2 Ring the tallest column.

- acorns
- shells
- buttons
- pebbles

3 Write how many pebbles and shells there were altogether.

4 Write how many fewer pebbles than acorns there were.

5 Order these from fewest to most.

shells

buttons

pebbles

fewest to most



table

6 Write the total of everything on the chart.

Lists and tables



table

7 Fill in this table. Write one number in each box.

What we found	How many
acorns	
shells	
buttons	

8 Write a list of four things on your table.

9 Draw a block chart for your own tally.
The material you need: squared paper and a pencil

10 Write one question your chart can answer.

11 Write the answer to your own question.



table

12 Swap charts with a partner. Answer their question.

→ Try a bigger one

Collect the same data twice, on two different days. Did your chart change? Why might that happen?

The words of this unit

▣ BURROW WORDS



data

what we found out



list

things written one under another



table

rows and columns to read across



count

one number for one thing



total

how many altogether



more

the bigger amount



fewer

the smaller amount



set

things grouped by one rule

▣ Check what you can do

- 1 Make a tally for this tray and write how many.
- 2 Write which kind there was most of.
- 3 Write how many there were altogether.
- 4 Ring the tallest column on the block chart.
- 5 Think of something your chart could tell a friend.

Now I can

- ☐ make a tally while I count
- ☐ read a pictogram where one picture means one
- ☐ read and draw a block chart
- ☐ put my data into a list and a table
- ☐ answer a question by looking at a chart

	confident	almost there	needs more work
making a tally	☐	☐	☐
reading a pictogram	☐	☐	☐
reading a block chart	☐	☐	☐
drawing my own chart	☐	☐	☐
answering a question from a chart	☐	☐	☐



The tallest column answers the question without any counting at all.

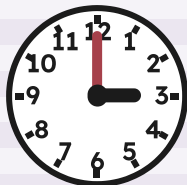
Half past and the year

The two hands at half past, the seven days of the week, and the twelve months that turn one whole year.



Open the door

Before you begin, show what you know already.



Say the time on each clock out loud. Both of these are o'clock times, so the long hand points straight up.

☀ In the burrow today

Bramble winds the old cuckoo clock above the schoolroom door.

It ticks slowly, and the long hand creeps round while nobody is watching. Otto points at each numeral and counts round the face, one to twelve.

Then the long hand slides to the bottom of the clock and Milo asks the useful question: **how do you know** what to call this time? Half of the hour has gone, so the burrow calls it half past.

• CONTINUED

✓ **By the end you will**

- say where the two hands point at half past
- read a clock that shows half past
- draw the hands for a half past time
- name the seven days of the week in order
- name the twelve months of one year and the four seasons



The pale cuckoo clock face above the schoolroom door. Otto points at each numeral as he counts round.

Half past



At half past, the long hand points straight down and the short hand sits halfway between two numerals.

WAKE UP
YOUR
COUNTING

Stand up and hold both arms straight up for twelve o'clock. Now keep one arm up and point the other one straight down.

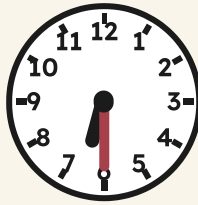
♣ Half past is half the way round

The **long hand** counts the minutes. It begins at the top of the clock and travels all the way round in one hour.

When the long hand has gone **half the way round** it points straight down. The short hand has moved too, so it sits halfway between two numerals. We say **half past** and then the number the short hand has already passed.



half past 2



half past 6



half past 9

 Watch me do one 1

What time does the middle clock show?

- 1** Look at the long hand first. It points straight down, so this is a half past time.
- 2** Now look at the short hand. It sits between two numerals, past six but not yet at seven.
- 3** Say the two parts together: half past, then the number the short hand has passed.

Answer: the middle clock shows half past 6

How do you know?

Hattie says the short hand must point straight at a numeral at half past. Tessa says it never does. Say who is right.

Reading half past

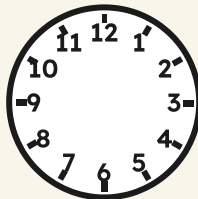


month

1 Write the time under each clock.



2 Draw the two hands to show half past three.



TOPIC 15.1 • YOUR TURN • CONTINUED

- 3 Ring the clock that shows half past ten.



- 4 Ring the two times that are half past times.

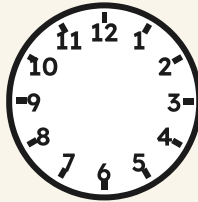
half past four

five o'clock

half past nine

- 5 Draw the hands for the time your lessons begin.

The material you need: a sharp pencil and a small ruler



The seven days

✦ A week is seven days long

A **week** has seven days, and they always come in the same order. After Sunday the week begins again with Monday, and it never stops going round.



month

6 Write the two days that are missing.

Monday

Tuesday

Thursday

Friday

Sunday

7 Write the day that comes after Thursday.

8 Write the day that comes just before Sunday.

9 Say the seven days out loud, first to last.
Begin at Monday and keep going the same way round.

10 Draw one thing you do on Saturday.

The twelve months

✦ Twelve months turn one year

One **year** has twelve **months**. January is the first month and December is the last one.

A year also turns through four **seasons**: spring, summer, autumn and winter. Then the oak begins again.



month

11 Write the three months that are missing.

1 January	2 February	3	4 April
5 May	6 June	7	8 August
9 September	10	11 November	12 December

12 Write the month that comes after August.

13 Ring the season when the oak leaves turn brown.

spring

summer

autumn

winter

14 Write the last month of the year.

15 Tell a partner which month your birthday is in.

Say whether it comes before or after this month.

★ **Pip's find**

The cuckoo in a cuckoo clock calls once for every hour, so at twelve o'clock you have to wait a long while for quiet.

The words of this unit

▣ BURROW WORDS



clock

it shows us the time



hands

the two pointers on a clock



week

seven days together



month

one of the twelve parts of a year



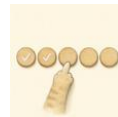
order

which one comes next



sequence

things in a settled order



count

one number for one thing



same

no difference at all

▣ Check what you can do

- 1 Show half past eight on a clock face with two hands.
- 2 Say what the long hand does at every half past time.
- 3 Name the day that follows Tuesday.
- 4 Say the twelve months of one year, first to last.
- 5 Name the season that follows summer.

Now I can



- ☐ I can say where the two hands point at half past.
- ☐ I can read a clock that shows a half past time.
- ☐ I can draw the hands for a half past time.
- ☐ I can name the seven days of the week in order.
- ☐ I can name the twelve months and the four seasons.

UNIT 15 • HOW DID IT GO • CONTINUED

	confident	almost there	needs more work
Reading half past on a clock face	☐	☐	☐
Drawing the two hands myself	☐	☐	☐
Naming the days of the week in order	☐	☐	☐
Naming the twelve months of one year	☐	☐	☐
Naming the four seasons	☐	☐	☐



One year under the old oak: frost, then blossom, then long grass, then falling leaves.

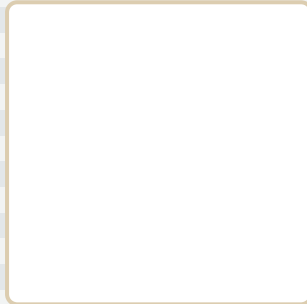
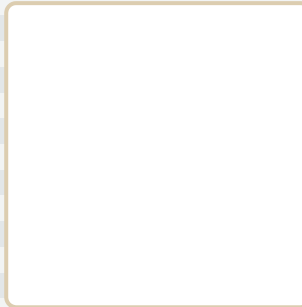
Position, direction and patterns

Above, below, beside and between, who is first along the runner, and what comes next when a pattern repeats.



Open the door

Before you begin, show what you know already.



Say where the leaf is. Now say where the basket is. Use one word that tells somebody where to look.

☀ **In the burrow today**

Tessa has unrolled the long woven runner along the schoolroom floor.

She lines things up on it so she can be sure: a leaf, a paper star, a block, then a leaf, a star and a block again. Pip hops along the runner, one hop for one thing.

Otto points at each one and counts. Milo looks at the end of the row and asks **what comes next?** Nobody can answer until they find the part that repeats.

✓ **By the end you will**

- use position words such as above, below, beside and between
- say whether a thing is inside or outside
- use the ordinal numbers from first to tenth
- make a whole turn and a half turn, and use left and right
- carry on a repeating pattern and say what comes next



The long woven runner laid along the schoolroom floor. Tessa lines her things up on it to be sure.

Where a thing sits



A leaf, a paper star and a block laid out along the woven runner. Every one of them is somewhere.

WAKE UP
YOUR
COUNTING

Hold your pencil above your book. Now below it. Now beside it. Say each word out loud as you move it.

✦ Words that tell you where

A position word tells you **where** something is: **above**, **below**, **beside**, **between**, **inside** and **outside**. Each one needs two things, the thing you are talking about and the thing you measure it against.

Between needs **two** other things, one on each side. On the boxes below, count down from the top to find the row, then count from the left to find the place along it.

TOPIC 16.1 • CONTINUED



position



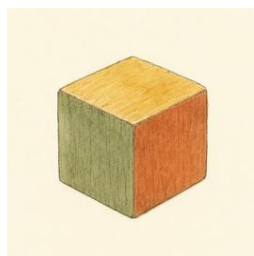
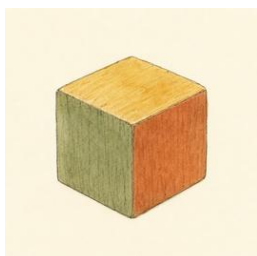
side



line



track



TOPIC 16.1 • CONTINUED

The star sits in the top row. The leaf is below the star. The basket space is between the two blocks.

Watch me do one 1

Where is the leaf on the boxes?

- 1 Find the row first. Count down from the top: one, two.
 - 2 Now count along that row from the left: one, two.
 - 3 The star is in the row above it, so the leaf is below the star.
-

Answer: the leaf is in row **2**, place **2**, below the star

How do you know?

Otto says the star is beside the leaf. Hattie says it is above the leaf. Look again, and say who is right.

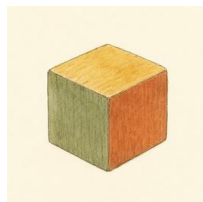
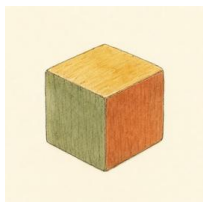
Above, below, between



1 Draw a leaf in the box above the basket.

2 Ring the leaf that sits between the two blocks.



3 Colour the box below the star.

The material you need: one coloured pencil

TOPIC 16.1 • YOUR TURN • CONTINUED

- 4** Draw a crate. Draw one leaf inside it and one outside.

- 5** Tell a partner where your book is now.
Use two position words, such as beside and below.

Which place, which way

✦ Place numbers and turns

One, two, three tell you **how many**. First, second, third tell you **which place** a thing is in. The tenth place is written 10th, and everybody counts from the same end.

A **whole turn** brings you back facing the way you began. A **half turn** leaves you facing the other way.

6 Write the place of each thing along the runner.



leaf



star



snail



bird

7 Ring the third basket from the left.



8 Stand up and make one whole turn.
Stop facing exactly the way you began.

9 Ring the hand you write with.

left hand

right hand

10 Draw an arrow to show a half turn.



What comes next

✦ A pattern has a part that repeats

A **repeating pattern** is built from one small part said again and again. Find that part, say it under your breath, and you always know what comes next.

🔑 Watch me do one 2

What comes next in this pattern?

- 1 Read the row from the left and listen to it: circle, triangle, circle, triangle.
- 2 The part that repeats is two shapes long, a circle and then a triangle.
- 3 The row ends with a triangle, so the repeating part has to begin again.

Answer: a circle comes next, and then a triangle

TOPIC 16.1 · YOUR TURN · CONTINUED

carry it on



pattern

11 Draw the next two shapes in this pattern.



12 Ring the part of this pattern that repeats.



13 Colour these boxes to make a repeating pattern.

The material you need: two coloured pencils

TOPIC 16.1 • YOUR TURN • CONTINUED

--	--	--	--	--	--	--	--

14 Draw the shape that is missing from this pattern.



15 Make your own repeating pattern for a partner.

The material you need: ten buttons and ten beans

→ **Try a bigger one**

Make a pattern with three things in the repeating part, then ask a partner to say the part out loud.

The words of this unit

▣ BURROW WORDS



position

where a thing sits



side

one edge of a shape



line

a straight path



track

boxes to hop along



order

which one comes next



sequence

things in a settled order



pattern

the part that repeats



rotate

turn it round

▣ Check what you can do

- 1 Place a leaf above a basket and a bean beside it.
- 2 Name the place of the fifth thing in a line of eight.
- 3 Make a half turn, then say which way you now face.
- 4 Carry on a repeating pattern of two colours for six boxes.
- 5 Explain how you know which shape comes next.

Now I can



- ☐ I can use *above*, *below*, *beside* and *between* to say where a thing is.
- ☐ I can say whether something is *inside* or *outside*.
- ☐ I can use the ordinal numbers from *first* to *tenth*.
- ☐ I can make a *whole turn* and a *half turn*, and use *left* and *right*.
- ☐ I can carry on a repeating pattern and say what comes next.

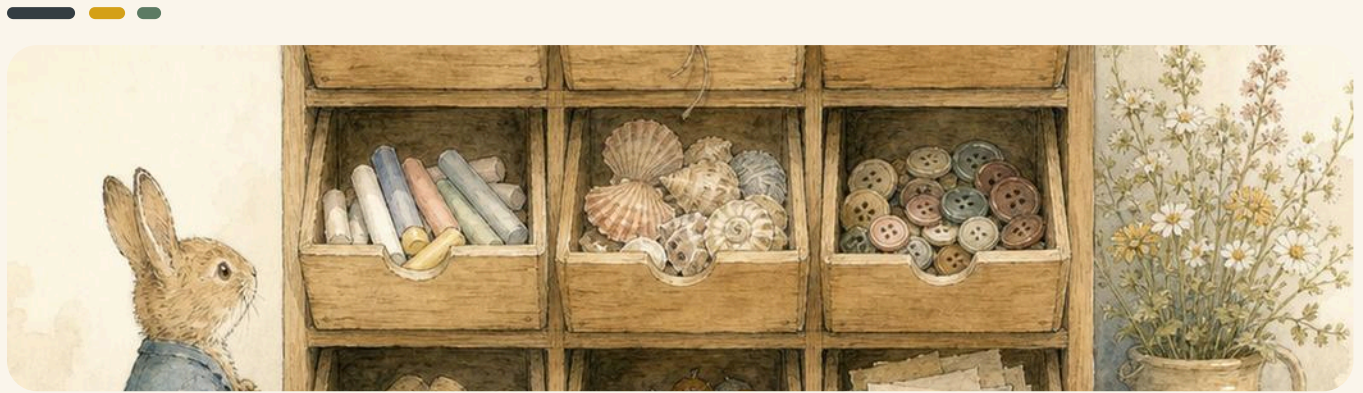
UNIT 16 · HOW DID IT GO · CONTINUED

	confident	almost there	needs more work
Using position words on the woven runner	☐	☐	☐
Saying which place a thing is in	☐	☐	☐
Making a whole turn and a half turn	☐	☐	☐
Finding the part of a pattern that repeats	☐	☐	☐
Saying what comes next	☐	☐	☐



A pattern you can say out loud is a pattern you can carry on. Every shape unit builds on this one.

Nine drawers



Bramble's cabinet: nine open drawers in three rows of three, each one holding something different.

Bramble keeps the store tidy in a tall cabinet of nine drawers, three rows of three. There is ribbon in one, string in another, then pegs, chalk, shells, buttons, corks, acorns and paper. Pip can never remember which drawer the pegs live in, and pointing is no help at all when Pip is up on the shelf and Bramble is down on the floor. They need words.

& The material you need

the cabinet in the picture, a pencil, paper and a partner to talk to

- 1 Find the drawer right in the middle of the cabinet.
- 2 Say out loud what is in the drawer above it.
- 3 Tell your partner where the shells are kept.
- 4 Now describe one drawer without pointing at it.
- 5 Ask your partner to find the drawer you described.

How do you know?

Your partner found the right drawer from your words alone. Which words helped the most? Which ones were no use?

Run your own counting fair



The whole burrow at work in Fern Meadow: Otto weighs the acorns, Hattie takes the coins, Tessa keeps the chart and Milo calls out to the visitors.

☀ In the burrow today

The trestle table is out in the meadow, and today the stall is yours.

Bramble has lent you the scales, the coin tray and a basket of acorns. Every visitor who stops at your stall has to be counted, every acorn has to be weighed, and every coin has to be added up at the end of the day. This is the last project in the book, and it uses everything you have learned all year.

The material you need

a table, a basket of acorns or counters, balance scales, a tray of coins, paper, a pencil and coloured pencils

- 1 Choose four things to sell. Give each one a price.
- 2 Write your prices on paper labels. Lay them on the table.
- 3 Ask one friend to count the visitors, one to take the coins.
- 4 Open your stall. Fill in the recording sheet as you go.
- 5 At the end, count the coins in the tray together.

What happened at your stall

Fill this sheet in while the stall is open. Say every number out loud before you write it down.

1 · One tally mark for every visitor who stops at your stall.

When they came	Tally	How many
Before break		
After break		
Home time		
Altogether		

2 · Colour one square for each thing you sold. Start at the bottom.

What the visitors bought

5
4
3
2
1

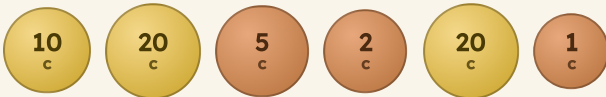
acorns

ribbons

pebbles

shells

3 · Count the coins in the tray. Write the total in the box.



The coins in Hattie's tray at the end of the day.

How do you know?

Which hour was the busiest at your stall? Say how your tally chart tells you that, without counting it all again.

add to digit

All the new words of the year, in the order of the alphabet, said the way you would say them out loud.

add	put two sets together to find how many there are	
altogether	how many you have when every set is put together	
balance	scales that tip down on the heavier side	
coin	money you can hold, with its value written on it	
count on	start at the bigger number, then hop forwards	
data	what you found out when you counted or sorted	
block chart	a chart of coloured squares, one square for one thing	
capacity	how much a jug or a cup holds when it is full	
Carroll diagram	a table of four boxes that asks two questions	
centre	the middle of a shape, the same way round from every side	
chart	a picture of what you found out, so you can read it quickly	
clock	the round face with hands that tells you the time	
compare	look at two amounts and say which is more	
count	you say one number for one thing, and point as you go	
curved	bent all the way along, like the side of a ball	
digit	one of the figures 0 to 9 that numbers are written with	

double to hands



double	two of the same amount, added together	
edge	the line where two faces of a solid shape meet	
empty	with nothing inside it at all	
equal	exactly the same amount, with nothing left over	
estimate	a careful guess at how many, made before you count	
even	when everything pairs up, with nobody left on their own	
face	one of the flat parts of a solid shape	
fewer	the smaller amount when you compare two sets	
flat shape	a shape you can draw, like a circle or a square	
full	with no room left inside for any more	
half	one of two equal parts of a whole thing	
group	things gathered together in one place	
half past	half an hour after the hour, the long hand pointing down	
heavy	hard to lift, because it has a lot of mass	
how many	the question you answer by counting a set	
hands	the two pointers that go round a clock face	

hour hand to numeral



hour hand	the short hand on a clock, which tells you the hour	
label	the name you write on a set or a ring	
largest	the biggest one of them all	
length	how long a thing is, from one end to the other end	
light	easy to lift, because it has only a little mass	
list	things written down one under the other, in order	
line	a straight path with no bends in it	
longer	more long than the thing you are comparing it with	
long	reaching a long way from end to end	
month	one of the twelve parts of a year	
mass	how heavy a thing is when you weigh it on scales	
minute hand	the long hand on a clock, which counts the minutes	
more	the bigger amount when you compare two sets	
number line	a line with the numbers in order, for hopping along	
number word	a number written in letters, like four or ten	
numeral	a number written in figures, like 4 or 10	

note to short



note paper money, worth more than a coin



o'clock the time when the long hand points straight up

odd when you pair them up, one is left over

order put things in their places, from smallest to largest

pair two that go together, like two shoes

pattern shapes or colours that come round again and again

pictogram a chart where one small picture stands for one thing

position where a thing is: on top, behind, beside or under

price how much money a thing costs to buy

repeat do the very same thing again, in the same order

sequence things that follow one after another in a settled order



same no difference at all between two amounts

set some things grouped together, like a set of five acorns

share give out fairly, so that everybody gets an equal amount

shorter less long, or less tall, than the other one

short not reaching far from end to end



side to whole



side	one of the straight lines around a flat shape	
smallest	the littlest one of them all	
solid shape	a shape you can pick up and hold, like a cube	
sort	put things into sets that belong together	
straight	not bent and not curved anywhere along it	
subtract	take a part away, then count what is left	
table	rows and columns that hold what you found out	
tally	one mark for one thing, counted in groups of five	
taller	more tall than the other one	
total	the number you get when you add everything up	
rotate	turn a shape round without changing it	
turn	move round on the spot, like a half turn or a whole turn	
track	numbered boxes in a row that you hop along	
Venn diagram	rings that show which things belong in which set	
week	seven days, from Monday all the way to Sunday	
whole	all of it, with no part missing	

Now I can



Read this list with a partner. Tick every box you can say yes to, and show your partner how you do it.

- ☐ count a set of things and say how many there are
- ☐ say, read and write every number from zero to twenty
- ☐ add two sets together, and take one part away
- ☐ find half of a whole shape, and half of a set of things
- ☐ name flat shapes and solid shapes, and sort them
- ☐ say where a thing is, and describe a turn
- ☐ compare how long, how tall, how heavy and how full
- ☐ read the clock at o'clock and at half past
- ☐ make a tally, a pictogram and a block chart
- ☐ say how I know, in my own words, every single time

THE WHOLE YEAR · HOW DID IT GO · CONTINUED



One year in Fern Meadow: frost, blossom, long grass and brown leaves.

★ Pip's find

You have counted, compared, halved, weighed, sorted and told the time. In Year 2 the same four steps carry you past one hundred: count it, check it, then say how you know.

A note for the teacher



The desk at the end of the day: the register, the pencils and the pile of workbooks.

 For teachers

There are no printed answers in this book. The single exception is the **Watch me do one** panel, which opens every topic with one example solved in full, step by step, before pupils try one of their own. Everywhere else the page is left for the pupil's own work, and the answer belongs in the discussion, not in the margin.

This book is meant to be written in. A dashed line, a dashed box or a dotted frame is always the pupil's space: to write a numeral, draw counters, ring a choice, join a pair or colour a square. Practical work comes first, and the material to fetch is listed on the page under **The material you need**.

Every topic asks How do you know? at least once, and that question carries the mathematics of Year 1. Accept the explanation in the pupil's own words: pointing at each thing in turn, counting again out loud from the other end, pairing counters up, lining two sets side by side or moving the objects across the table are all good answers. Ask it again in the corridor, at the sand tray and over the fruit bowl, until saying how you know becomes a habit.

Each unit closes with **Now I can** and a self-assessment table. Its three columns are worded **confident**, **almost there** and **needs more work**. Please keep that wording when you talk the table through: it names what a pupil is going to do next, which a colour cannot, and no pupil has to be told that a colour means they were wrong. Pupils tick their own boxes.

The six **Along the way** projects and the final **Counting fair** need a table, a partner and about a lesson each. They are the places where the year's mathematics leaves the page, so they are worth the room they take.

PRIME SCHOOL PRESS

Mathematics

Year 1 · Prime School Press · Student Manual

Count it. Check it. Then say how you know.

Down under the roots of the old oak there is a schoolroom. Otto the owl counts acorns, Hattie the hedgehog guesses and then checks, and Milo the mouse asks the question that matters most: how do you know?

Sixteen units of real mathematics, six projects along the way and a counting fair of your own at the end. Every page is meant to be written in.

INSIDE THIS BOOK

- Sixteen units, six projects, a counting fair to finish
- Every page meant to be written in
- Count it, check it, say how you know
- A second method whenever there is one
- Original watercolour illustration throughout

Prime School Press · Mathematics

Ages 5–6 · Lower Primary

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